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Chapter 11

RADIOACTIVE WASTE MANAGEMENT

11.1 SOURCE TERMS

The italicized information is historical and was provided to support the application for an operating license.

Radioactive material sources (activation products and fission product release from fuel) have been evaluated in operating boiling water reactors (BWRs) over the past decade. These source terms are reviewed and periodically revised to incorporate up-to-date information. Release of radioactive material from operating BWRs has resulted in doses to offsite persons which have been only a small fraction of 10 CFR 20 permissible or natural background doses.

The information provided in this section defines the design-basis radioactive material levels in the reactor water, steam, and offgas. The various radioisotopes listed have been grouped as coolant activation products, noncoolant activation products, and fission products. The fission product levels are based on measurements of BWR reactor water and offgas at several stations through mid-1971. Emphasis was placed on observations made at Kernkraftwerk RWE-Bayernwerk GmbH (KRB) and Dresden-2. The design-basis radioactive material levels do not necessarily include all the radioisotopes observed or predicted theoretically to be present. The radioisotopes included are considered significant to one or more of the following criteria:

a. Plant equipment design,
b. Shielding design,
c. Understanding system operation and performance,
d. Measurement practicability, and
e. Evaluating radioactive material releases to the environment.

The inventory of radionuclides used to determine shielding requirements of system components are discussed in Chapter 12.

11.1.1 FISSION PRODUCTS

11.1.1.1 Noble Radiogas Fission Products

The noble gas radionuclide fission product source terms observed in operating BWRs are generally complex mixtures whose sources vary from miniscule defects in cladding to "tramp" uranium on external cladding surfaces. The relative concentrations or amounts of noble radiogas can be described as follows.

Equilibrium: $R_{g,i} = K_1 Y_1$ *(11.1-1)*

Recoil: $R_{g,i} = K_2 Y_1 \lambda_i$ *(11.1-2)*

The nomenclature in Section 11.1.1.4 defines the terms in these and succeeding equations. The constants K_1 and K_2 describe the fractions of the total fissions that are involved in each of the releases. The equilibrium and recoil mixtures are the two extremes of the mixture spectrum that are physically possible. When a sufficient time delay occurs between the fission event and the time of release of the radiogases from the fuel to the coolant, the radiogases approach equilibrium levels in the fuel and the equilibrium mixture results. Where there is no delay between the fission event and the release of the radiogases, the recoil mixture is observed. Prior to Vallecitos Boiling Water Reactor (VBWR) and Dresden-1 experience, it was assumed that noble radiogas leakage from the fuel would be the equilibrium mixture of the noble radiogases present in the fuel.

The VBWR and early Dresden-1 experience (Reference 11.1-1) indicated that the actual mixture most often observed approached a distribution which was intermediate in character to the two extremes. This intermediate decay mixture was termed the "diffusion" mixture. It must be emphasized that this "diffusion" mixture is merely one possible point on the mixture spectrum ranging from the equilibrium to the recoil mixture and does not have the absolute mathematical and mechanistic basis for the calculational methods possible for equilibrium and recoil mixtures. However, the "diffusion" distribution pattern which has been described is as follows:

Diffusion: $R_{g,i} = K_3 Y_i \lambda_i^{0.5}$ *(11.1-3)*

The constant K_3 describes the fraction of total fissions that are involved in the release. The value of the exponent of the decay constant, λ_i, is midway between the values for the equilibrium case, 0, and recoil case 1. The "diffusion" pattern value of 0.5 was originally derived from diffusion theory.

Although the previously described "diffusion" mixture was used by GE as a basis for design since 1963, the design-basis release magnitude used has varied from 0.5 Ci/sec to 0.1 Ci/sec as measured after 30-minute decay (t=30 minutes). Since about 1967, the design-basis release magnitude used (including the 1971 source terms) was established at an annual average of 0.1 Ci/sec (t = 30 minutes). This design basis is considered as an annual average with some time above and some time below this value. This design value was selected on the basis of operating experience rather than predictive assumptions. Several judgment factors,*

**The noble radiogas source term rate after 30-minute decay has been used as a conventional measure of the design-basis fuel leakage rate since it is conveniently measurable and was consistent with the nominal design-basis 30-minute offgas holdup system used on a number of previous plants.*

including the significance of environmental release, reactor water radioisotope concentrations, liquid waste handling and effluent disposal criteria, building air contamination, shielding design, and turbine and other component contamination affecting maintenance, have been considered in establishing this level.

Noble radiogas source terms from fuel above 0.1 Ci/sec (t = 30 minutes) can be tolerated for reasonable periods of time. Continual assessment of these values is made on the basis of actual operating experience in BWRs (References 11.1-2 and 11.1-3).

While the noble radiogas source-term magnitude was established at 0.1 Ci/sec (t = 30 minutes), it was recognized that there may be a more statistically applicable distribution for the noble radiogas mixture. Sufficient data were available from KRB operations from 1967 to mid-1971 along with Dresden-2 data from operation in 1970 and several months in 1971 to more accurately characterize the noble radiogas mixture pattern for an operating BWR.

The basic equation for each radioisotope used to analyze the collected data is:

$$R_{g,i} = K_g Y_i \lambda_i{}^m \left(1 - e^{-\lambda_i T}\right)\left(e^{-\lambda_i t}\right) \qquad (11.1\text{-}4)$$

With the exception of Kr-85 with a half-life of 10.74 years, the noble radiogas fission products in the fuel are essentially at an equilibrium condition after an irradiation period of several months (rate of formation is equal to the rate of decay). So for practical purposes the term $(1 - e^{-\lambda_i T})$ approaches 1 and can be neglected when the reactor has been operating steady state for long periods of time. The term $(1 - e^{-\lambda_i T})$ is used to adjust the releases from the fuel (t = 0) to the decay time for which values are needed. Historically, t = 30 minutes has been used. When discussing long steady-state operation and leakage from the fuel (t = 0), the following simplified form of Equation 11.1-4 can be used to describe the leakage of each noble radiogas:

$$R_{g,i} = K_g Y_i \lambda_i{}^m \qquad (11.1\text{-}5)$$

The constant, K_g, describes the magnitude of leakage from the fuel. The relative rates of leakage of the different noble radiogas isotopes is accounted for by the variable, m, the exponent of the decay constant, λ_i.

Dividing both sides of Equation 11.1-5 by y_i, the fission yield, and taking the logarithm of both sides results in the following equation:

$$\log\left(R_{g,i} / Y_i\right) = m \log\left(\lambda_i\right) + \log\left(K_g\right) \qquad (11.1\text{-}6)$$

Equation 11.1-6 represents a straight line when log ($R_{g,i}/y_i$) is plotted vs. log (λ_i); m is the slope of the line. This straight line is obtained by plotting ($R_{g,i}/y_i$) vs. (λ_i) on logarithmic graph paper. By fitting actual data from KRB and Dresden-2 (using least squares techniques) to the equation, the slope, m, can be obtained. This can be estimated on the plotted graph. With radiogas leakage at KRB over the nearly 5-year period varying from 0.001 to 0.056 Ci/sec (t = 30 minutes) and with radiogas leakage at Dresden-2 varying from 0.001 to 0.169 Ci/sec (t = 30 minutes), the average value of m was determined. The value for m is 0.4 with a standard deviation of ±0.07. This is illustrated in Figure 11.1-1 as a frequency histogram. As can be seen from this figure, variations in m were observed in the range of m = 0.1 to m = 0.6. After establishing the value of m = 0.4, the value of K_g can be calculated by selecting a value for R_g, or as has been done historically, the design basis is set by the total design-basis source-term magnitude at t = 30 minutes. With SR_g at 30 minutes = 100,000 mCi/sec, K_g can be calculated as being 2.6 x 10^7 and Equation 11.1-4 becomes:

$$R_{g,i} = 2.6 \times 10^{7}\ Y_i \lambda_i^{\ 0.4} \left(1 - e^{-\lambda_i t}\right)\left(e^{-\lambda_i t}\right) \qquad (11.1\text{-}7)$$

This updated noble radiogas source-term mixture has been termed the "1971 mixture" to differentiate it from the "diffusion" mixture. The noble gas source term for each radioisotope can be calculated from Equation 11.1-7. The resultant source terms are presented in Table 11.1-1 as leakage from fuel (t = 0) and after 30-minute decay. While Kr-85 can be calculated using Equation 11.1-7, the number of confirming experimental observations was limited by the difficulty of measuring very low release rates of this isotope. Therefore, the table provides an estimated range for ^{85}Kr based on a few actual measurements.

11.1.1.2 Radiohalogen Fission Products

Historically, the radiohalogen design-basis source term was established by the same equation as that used for noble radiogases. In a fashion similar to that used with gases, a simplified equation can be shown to describe the release of each radiohalogen:

$$R_{h,i} = K_h Y_i \lambda_i^{\ n} \qquad (11.1\text{-}8)$$

The constant, K_h, describes the magnitude of leakage from fuel. The relative rates of radiohalogen leakage is expressed in terms of n, the exponent of the decay constant, λ_i. As was done with the noble radiogases, the average value was determined for n. The value for n is 0.5 with a standard deviation of ±0.19. This is illustrated in Figure 11.1-2 as a frequency histogram. As can be seen from this figure, variations in n were observed in the range of n = 0.1 to n = 0.9.

It appeared that the use of the previous method of calculating radiohalogen leakage from fuel was overly conservative. Figure 11.1-3 relates KRB and Dresden-2 noble radiogas versus I-131 leakage. While it can be seen from Dresden-2 data, during the period August 1970 to January 1971, that there is a relationship between noble radiogas and I-131 leakage under one fuel condition, there was no simple relationship for all fuel conditions experienced. Also, it can be seen that during this period, high radiogas leakages were not accompanied by high radioiodine leakage from the fuel. Except for one KRB datum point, all steady-state I-131 leakages observed at KRB or Dresden-2 were equal to or less than 505 μCi/sec. Even at Dresden-1 in March 1965, when severe defects were experienced in stainless steel clad fuel, I-131 leakages greater than 500 μCi/sec were not experienced. Figure 11.1-3 shows that these higher radioiodine leakages from the fuel were related to noble radiogas source terms of less than the design-basis value of 0.1 Ci/sec (t = 30 minutes). This may be partially explained by inherent limitations due to internal plant operational problems that caused plant derating.

In general, it would not be anticipated that operation at full power would continue for any significant time period with fuel cladding defects, which would be indicated by ^{131}I leakage from the fuel in excess of 700 μCi/sec. When high radiohalogen leakages are observed, other fission products will be present in greater amounts.

Using these judgment factors and experience to date, the design-basis radiohalogen source terms from fuel were established based on ^{131}I leakage of 700 μCi/sec. This value, as seen in Figure 11.1-3, accommodates the experience data and the design-basis noble radiogas source term of 0.1 Ci/sec (t = 30 minutes). With the I-131 design-basis source-term established, K_h can be calculated as being 2.4 x 10^7 and radiohalogen release can be expressed by the following equation:

$$R_{h,i} = 2.4 \times 10^{7}\ Y_i \lambda_i^{0.5} \left(1 - e^{-\lambda_i T}\right)\left(e^{-\lambda_i t}\right) \qquad (11.1\text{-}9)$$

Concentrations of radiohalogens in reactor water can be calculated using the following equation:

$$C_{h,i} = \frac{R_{h,i}}{(\lambda_i + \beta + \gamma)M} \qquad (11.1\text{-}10)$$

The observed "carryover" for radiohalogens has varied from 0.1% to about 2% on newer plants. The average of observed radiohalogen carryover measurements has been 1.2% by weight of reactor water in steam with a standard deviation of ±0.9%. In the present source term definition, a radiohalogen carryover of 2% (0.02 fraction) was used.

The radiohalogen release rate from the fuel was calculated from equation 11.1-9.

Concentrations in reactor water were calculated from equation 11.1-10.

The resultant concentrations are presented in Table 11.1-2. Radiohalogens with half-lives less than 3 minutes were omitted.

11.1.1.3 Other Fission Products

The observations of other fission products (and transuranic nuclides, including ^{239}Np) in operating BWRs are not adequately correlated by simple equations. For these radioisotopes, design-basis concentrations in reactor water have been estimated conservatively from experience data (Reference 11.1-8) and are presented in Table 11.1-3. Radioisotopes with half-lives less than 10 minutes were not considered. Carryover of these radioisotopes from the reactor water to the steam is estimated to be ≤0.1% (≤ 0.001 fraction) (Reference 11.1-8). In addition to carryover, however, decay of noble radiogases in the steam leaving the reactor will result in production of noble gas daughter radioisotopes in the steam and condensate systems.

Some daughter radioisotopes (e.g., yttrium and lanthanum) were not listed as being in reactor water. Their independent leakage to the coolant is negligible; however, these radioisotopes may be observed in some samples in equilibrium or approaching equilibrium with the parent radioisotope.

Except for ^{239}Np, trace concentrations of transuranic isotopes have been observed in only a few samples where extensive and complex analyses were carried out. The predominant alpha emitter present in reactor water is ^{242}Cm at an estimated concentration of 10^{-6} μCi/g or less, which is below the maximum permissible concentration in drinking water applicable to continuous use by the general public. The concentration of alpha-emitting plutonium radioisotopes is more than one order of magnitude lower than that of ^{242}Cm.

Plutonium-241 (a beta emitter) may also be present in concentrations comparable to the ^{242}Cm level.

11.1.1.4 Nomenclature

The following list defines the terms used in equations for source term calculations:

$R_{g,i}$ = *leakage rate of noble gas radioisotope i (μCi/sec)*

$R_{h,i}$ = *leakage rate of a halogen radioisotope i (μCi/sec)*

Y_i = *fission yield of a radioisotope i (atoms/fission)*

λ_I = *decay constant of a radioisotope i (sec^{-1})*

T = *fuel irradiation time (sec)*

t = *decay time following leakage from fuel (sec)*

m = *noble radiogas decay constant exponent (dimensionless)*

n = *radiohalogen decay constant exponent (dimensionless)*

K_g = *a constant establishing the level of noble radiogas leakage from fuel*

K_h = *a constant establishing the level of radiohalogen leakage from fuel*

$C_{h,i}$ = *concentration of a radiohalogen i in reactor water (μCi/g)*

M = *mass of water in the operating reactor (g)*

β = *cleanup system removal constant (sec^{-1})*

$$\beta = \frac{\text{cleanup system flow rate (g/sec)}}{M(g)}$$

g = *grams mass*

γ = *halogen steam carryover removal constant (sec^{-1})*

$$\gamma = \frac{\text{concentration of radiohalogen isotope in steam } (\mu Ci / g)}{C_{h,i}\,(\mu Ci / g)} \; x \; \frac{\text{steam flow } (g / sec)}{M\,(g)}$$

11.1.2 ACTIVATION PRODUCTS

11.1.2.1 Coolant Activation Products

The coolant activation products are not adequately correlated by simple equations. Design basis concentrations in reactor water and steam have been estimated conservatively from experience data. The resultant concentrations are presented in Table 11.1-4.

11.1.2.2 Noncoolant Activation Products

The activation products formed by activation of impurities in the coolant or by corrosion of irradiated system materials are not adequately correlated by simple equations. The design-basis source terms of noncoolant activation products have been estimated conservatively

from experience data (Reference 11.1-8). The resultant concentrations are presented in Table 11.1-5. Carryover of these isotopes from the reactor water to the steam is estimated to be <0.1% (<0.001 fraction) (Reference 11.1-8).

11.1.2.3 Steam and Power Conversion System N-16 Inventory

The main steam and reactor feedwater systems sources are discussed in Section 12.2.1.2.2.7. This section discusses the N-16 source strength in the moisture separators and reheaters, main condenser and hotwell, feedwater heaters, and associated piping.

11.1.3 TRITIUM

In a BWR, tritium is produced by three principal methods:

a. Activation of naturally occurring deuterium in the primary coolant,
b. Nuclear fission of UO_2 fuel, and
c. Neutron reactions with boron used in reactivity control rods.

The tritium formed in control rods, which may be released from a BWR in liquid or gaseous effluents, is believed to be negligible. The prime source of tritium available for release from a BWR is that produced from activation of deuterium in the primary coolant. Some fission product tritium may also transfer from fuel to primary coolant. This discussion is limited to the uncertainties associated with estimating the amounts of tritium generated in a BWR which are available for release. All of the tritium produced by activation of deuterium in the primary coolant is available for release in liquid or gaseous effluents. The tritium formed in a BWR from deuterium activation is calculated using the equation

$$R_{act} = \frac{\Sigma \phi V \lambda}{3.7 \times 10^4 P} \qquad (11.1\text{-}11)$$

where:

R_{act} = *tritium formation rate by deuterium activation act (μCi/sec/MWt)*

Σ = *macroscopic thermal neutron cross section (cm^{-1})*

ϕ = *thermal neutron flux (neutrons/cm^2-sec)*

V = *coolant volume in core (cm^3)*

λ = *tritium radioactive decay constant (1.78×10^{-9} sec^{-1})*

P = *reactor power level (MWt)*

For recent BWR designs, R_{act} is calculated to be (1.3 ± 0.4) x 10^{-4} μCi/sec/MWt. The uncertainty indicated is derived from the estimated errors in selecting values for the coolant volume in the core, coolant density in the core, abundance of deuterium in light water [some additional deuterium will be present because of the H(n,γ) D reaction], thermal neutron flux, and microscopic cross section for deuterium.

The fraction of tritium produced by fission which may transfer from the fuel to the coolant (which will then be available for release in liquid and gaseous effluents) is more difficult to estimate. However, since zircaloy-clad fuel rods are used in BWRs, essentially all fission product tritium will remain in the fuel rods unless defects are present in the cladding material (Reference 11.1-4).

The study made at Dresden-1 in 1968 by the U.S. Public Health Service (USPHS) suggests that essentially all of the tritium released from the plant could be accounted for by the deuterium activation source (Reference 11.1-3). For purposes of estimating the leakage of tritium from defective fuel, it is assumed that it leaks in a manner similar to the leakage of noble radiogases. Thus, use is made of the empirical relationship described as the "diffusion" mixture used for predicting the source term of individual noble radiogas isotopes as a function of the total noble gas source term. The equation which describes this relationship is:

$$R_{dif} = Ky\lambda^{0.5} \qquad (11.1\text{-}12)$$

where:

R_{dif} = *leakage rate of tritium from fuel (μCi/sec)*

y = *fission yield fraction (atoms/fission)*

λ = *radioactive decay constant (sec^{-1})*

K = *a constant related to total tritium leakage rate.*

When the total noble radiogas source term is 100,000 μCi/sec after 30-minutes decay, leakage from fuel is calculated to be about 0.24 μCi/sec of tritium. To place this value in perspective, in the USPHS study (Reference 11.1-3), the observed rate of ^{85}Kr (which has a half-life similar to that of tritium) was 0.06 to 0.4 times that calculated using the "diffusion" mixture relationship. This would suggest that the actual tritium leakage rate might range from 0.015 to 0.10 μCi/sec. Since the annual average noble radiogas leakage from a BWR is expected to

be less than 100,000 μCi/sec (t = 30-minutes), the annual average tritium release rate from the fission source is conservatively estimated at 0.12 ± 0.12 μCi/sec, or 0.0 to 0.24 μCi/sec. Based on this approach, the estimated total tritium appearance rate in reactor coolant and release rate in the effluent is about 19 Ci/yr.

Tritium formed in the reactor is present as tritiated oxide (HTO) and to a lesser degree as tritiated gas (HT). Tritium concentration on a weight basis in the steam formed in the reactor is the same as in the reactor water at any given time. This tritium concentration is also present in condensate and feedwater. Since radioactive effluents generally originate from the reactor and power cycle equipment, radioactive effluents also have this tritium concentration. The condensate storage tanks receive treated water from the liquid waste management systems and rejected water from the condensate system. Thus, all plant process water approaches a common tritium concentration.

Offgases released from the plant contain tritium, which is present as tritiated gas (HT) resulting from reactor water radiolysis as well as tritiated water vapor (HTO). In addition, water vapor from the turbine gland seal steam packing exhauster and some water vapor present in ventilation air due to process steam leaks and evaporation from sumps and tanks also contain tritium. The remainder of the tritium leaves the plant in liquid effluents or with solid wastes.

Recombination of radiolysis gases in the offgas system forms water, which is condensed and returned to the main condenser. This tends to reduce the amount of tritium leaving in gaseous effluents. Reducing the gaseous tritium release results in a slightly higher tritium concentration in the plant process water. Reducing the amount of liquid effluent discharged will also result in a higher process coolant tritium concentration.

Essentially all tritium in the primary coolant is eventually released to the environs, either as water vapor and gas to the atmosphere or as liquid effluent to the plant discharge or as solid waste. Reduction due to radioactive decay is negligible due to the 12.3-year half-life of tritium.

The USPHS study at Dresden-1 estimated that approximately 90% of the tritium release was observed in liquid effluent, with the remaining 10% leaving as gaseous effluent (Reference 11.1-5).

Efforts to reduce the volume of liquid effluent discharges may change this distribution so that a greater amount of tritium will leave as gaseous effluent. From a practical standpoint, the fraction of tritium leaving as liquid effluent may vary between 60% and 90% with the remainder leaving in gaseous effluent.

The amount of tritium released to the environment in liquid and gaseous effluents is based on the draft Regulatory Guide 1.CC, BWR-GALE code analyses. This is discussed in Sections 11.2 and 11.3 respectively.

11.1.4 FUEL FISSION PRODUCT INVENTORY AND FUEL EXPERIENCE

11.1.4.1 Fuel Fission Product Inventory

Fuel fission product inventory information is used in establishing fission product source terms for accident analysis and is, therefore, discussed in Chapter 15.

11.1.4.2 Fuel Experience

A discussion of BWR fuel experienced including fuel failure, burn-up and thermal conditions under which the experience was gained, is presented in References 11.1-2, 11.1-3, and 11.1-6.

11.1.5 RADIOACTIVITY LEAKAGE AND EFFLUENT SOURCES

Process leakage results in potential release paths for noble gases and other volatile fission products via ventilation systems. Liquids from process leaks are collected and routed to radioactive equipment and floor drain systems. Radioisotope releases via ventilation paths are at extremely low levels and have been insignificant compared to process offgas from operating BWR plants. However, because the implementation of improved process offgas treatment systems makes the ventilation release comparatively significant, measurements have been conducted to identify and quantify these low-level release paths. In addition an awareness of measurements by the Electric Power Research Institute, other organizations, and routine measurements by other utilities with operating BWRs has been maintained. Design basis estimates of the various liquid, gaseous, and solids effluents are discussed in Sections 11.2, 11.3, and 11.4 which follow.

Concurrently, analytical and mathematical model studies are being performed to provide a description of the transport, residence, and release of various radionuclides in and from an operating BWR.

Process leakage measurement, detection, and control methods are further discussed in Sections 5.2.5, 11.2.2, 11.3.2, and 12.1.2.

The effect of process leakage sources on the in-plant airborne radionuclide concentrations and the adequacy of plant ventilation systems is discussed in Section 12.2.2. Liquid radioactive sources are discussed in Section 11.2 and gaseous radioactive sources are discussed in Section 11.3.

11.1.6 REFERENCES

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11.1-2 *Williamson, H. E., Ditmore, D. C., "Experience with BWR Fuel Through September 1971," NEDO-10505, May 1972. (Update)*

11.1-3 *Elkins, R. B., "Experience with BWR Fuel Through September 1974," NEDO-20922, June 1975.*

11.1-4 *Ray, J. W., "Tritium in Power Reactors," Reactor and Fuel Processing Technology, 12 (1), pp. 19-26, Winter 1968-1969.*

11.1-5 *Kahn, B., et al, "Radiological Surveillance Studies at a Boiling Water Nuclear Power Reactor," BRH/DER 70-1, March 1970.*

11.1-6 *Williamson, H. E., Ditmore, D. C., "Current State of Knowledge of High Performance BWR Zircaloy Clad UO Fuel," NEDO-10173, May 1970.*

11.1-7 *Marrero, T. R., "Airborne Releases From BWRs for Environmental Impact Evaluation," NEDO-21159, March 1976.*

11.1-8 *Gilbert, R. S., Skarpelos, J. M., "Technical Derivation of 1971 BWR Design Basis Radioactive Material Source Terms," NEDO-10871, March 1973.*

Table 11.1-1

Noble Radiogas Source Terms

Isotope	*Half-Life*	*Source Term t = 0 (μCi/sec)*	*Source Term t = 30 minutes (μCi/sec)*
^{83m}Kr	*1.86 hr*	3.4×10^{3}	2.9×10^{3}
^{85m}Kr	*4.4 hr*	6.1×10^{3}	5.6×10^{3}
^{85}Kr	*10.74 years*	*10 to 20*[a]	*10 to 20*[a]
^{87}Kr	*76 minutes*	2.0×10^{4}	1.5×10^{4}
^{88}Kr	*2.79 hr*	2.0×10^{4}	1.8×10^{4}
^{89}Kr	*3.18 minutes*	1.3×10^{5}	1.8×10^{2}
^{90}Kr	*32.3 sec*	2.8×10^{5}	---
^{91}Kr	*8.6 sec*	3.3×10^{5}	---
^{92}Kr	*1.84 sec*	3.3×10^{5}	---
^{93}Kr	*1.29 sec*	9.9×10^{4}	---
^{94}Kr	*1.0 sec*	2.3×10^{4}	---
^{95}Kr	*0.5 sec*	2.1×10^{3}	---
^{97}Kr	*1.0 sec*	1.4×10^{1}	---
^{131m}Xe	*11.96 days*	1.5×10^{1}	1.5×10^{1}
^{133m}Xe	*2.26 days*	2.9×10^{2}	2.8×10^{2}
^{133}Xe	*5.27 days*	8.2×10^{3}	8.2×10^{3}
^{135m}Xe	*15.7 minutes*	2.6×10^{4}	6.9×10^{3}
^{135}Xe	*9.16 hr*	2.2×10^{4}	2.2×10^{4}
^{137}Xe	*3.82 minutes*	1.5×10^{5}	6.7×10^{2}

Table 11.1-1

Noble Radiogas Source Terms (Continued)

Isotope	*Half-Life*	*Source Term t = 0 (μCi/sec)*	*Source Term t = 30 minutes (μCi/sec)*
^{138}Xe	*14.2 minutes*	8.9×10^{4}	2.1×10^{4}
^{139}Xe	*40 sec*	2.8×10^{5}	---
^{140}Xe	*13.6 sec*	3.0×10^{5}	---
^{141}Xe	*1.72 sec*	2.4×10^{5}	---
^{142}Xe	*1.22 sec*	7.3×10^{4}	---
^{143}Xe	*0.96 sec*	1.2×10^{4}	---
^{144}Xe	*9.0 sec*	5.6×10^{2}	---
	Totals	$\sim 2.5 \times 10^{6}$	$\sim 1.0 \times 10^{5}$

[a] *Estimated from experimental observations.*

Table 11.1-2

Halogen Radioisotopes in Reactor Water

Isotope	*Half-Life*	*Concentration (μCi/g)*
^{83}Br	*2.40 hr*	1.5×10^{-2}
^{84}Br	*31.8 minutes*	2.7×10^{-2}
^{85}Br	*3.0 minutes*	1.7×10^{-2}
^{131}I	*8.07 days*	1.3×10^{-2}
^{132}I	*2.28 hr*	1.2×10^{-1}
^{133}I	*20.8 hr*	8.8×10^{-2}
^{134}I	*52.3 minutes*	2.4×10^{-1}
^{135}I	*6.7 hr*	1.3×10^{-1}

Table 11.1-3

Other Fission Product Radioisotopes in Reactor Water

Isotope	*Half-Life*	*Concentration (μCi/g)*
^{89}Sr	*50.8 days*	3.1×10^{-3}
^{90}Sr	*28.9 years*	2.3×10^{-4}
^{91}Sr	*9.67 hr*	6.9×10^{-2}
^{92}Sr	*2.69 hr*	1.1×10^{-1}
^{95}Zr	*65.5 days*	4.0×10^{-5}
^{97}Zr	*16.8 hr*	3.2×10^{-5}
^{95}Nb	*35.1 days*	4.2×10^{-5}
^{99}Mo	*66.6 hr*	2.2×10^{-2}
^{99m}Tc	*6.007 hr*	2.8×10^{-1}
^{101}Tc	*14.2 minutes*	1.4×10^{-1}
^{103}Ru	*39.8 days*	1.9×10^{-5}
^{106}Ru	*368 days*	2.6×10^{-6}
^{129m}Te	*34.1 days*	4.0×10^{-5}
^{132}Te	*78.0 hr*	4.9×10^{-2}
^{134}Cs	*2.06 years*	$1.6 \times - 10^{-4}$
^{136}Cs	*13.0 days*	1.1×10^{-4}
^{137}Cs	*30.2 years*	2.4×10^{-4}
^{138}Cs	*32.3 minutes*	1.9×10^{-1}
^{139}Ba	*83.2 minutes*	1.6×10^{-1}

Table 11.1-3

Other Fission Product Radioisotopes in Reactor Water (Continued)

Isotope	*Half-Life*	*Concentration (μCi/g)*
^{140}Ba	*12.8 days*	9.0×10^{-3}
^{141}Ba	*18.3 minutes*	1.7×10^{-1}
^{142}Ba	*10.7 minutes*	1.7×10^{-1}
^{141}Ce	*32.53 days*	3.9×10^{-5}
^{143}Ce	*33.0 hr*	3.5×10^{-5}
^{144}Ce	*384.4 days*	3.5×10^{-5}
^{143}Pr	*13.58 days*	3.8×10^{-5}
^{147}Nd	*11.06 days*	1.4×10^{-5}
^{239}Np	*2.35 days*	2.4×10^{-1}

Table 11.1-4

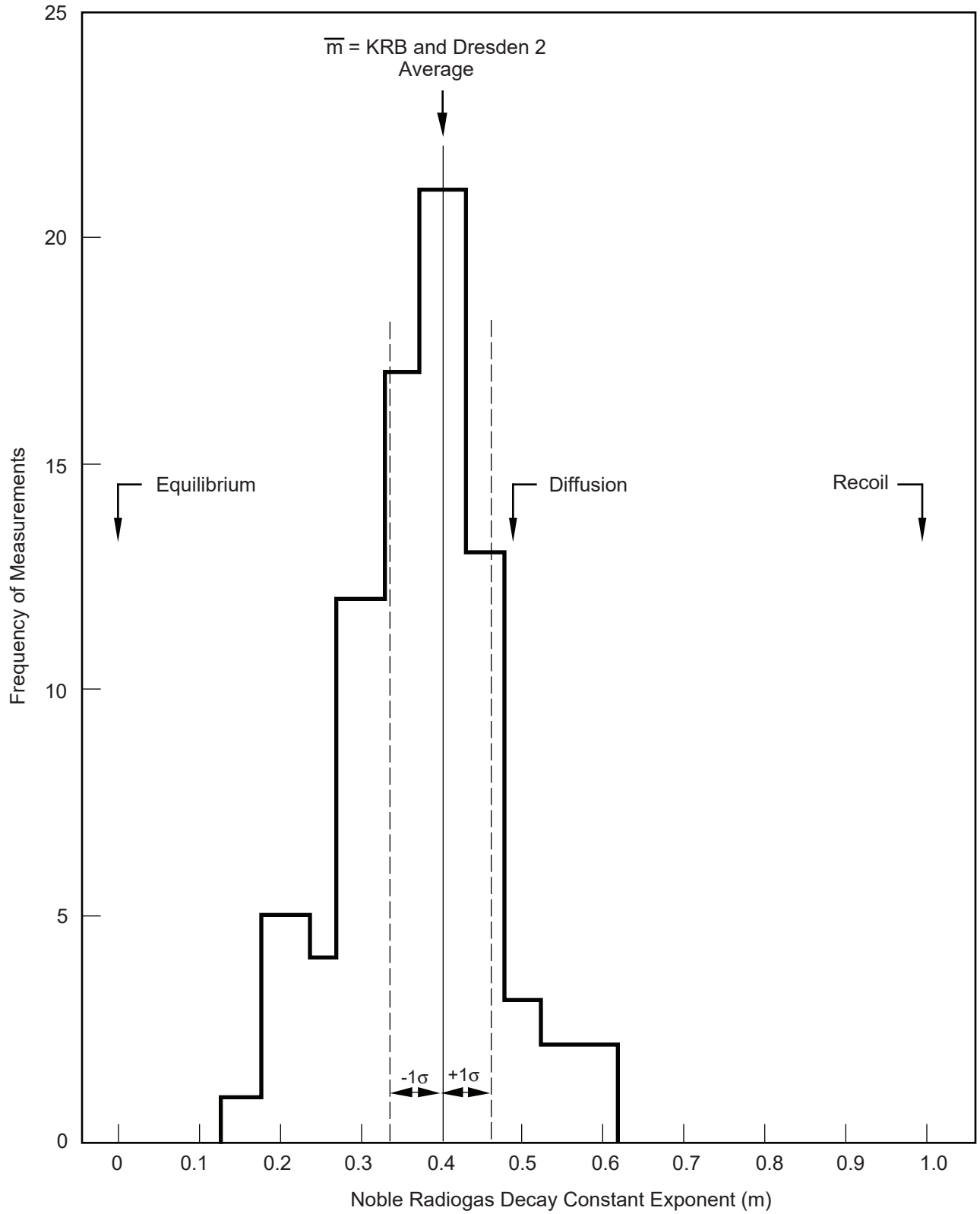
Coolant Activation Products in Reactor Water and Steam

Isotope	*Half-Life*	*Steam Concentration (μCi/g)*	*Reactor Water Concentration (μCi/g)*
^{13}N	*9.99 minutes*	7×10^{-3}	4×10^{-2}
^{16}N	*7.13 sec*	5×10^{1}	4×10^{1}
^{17}N	*4.14 sec*	2×10^{-2}	6×10^{-3}
^{19}O	*26.8 sec*	8×10^{-1}	7×10^{-1}
^{18}F	*109.8 minutes*	4×10^{-3}	4×10^{-3}

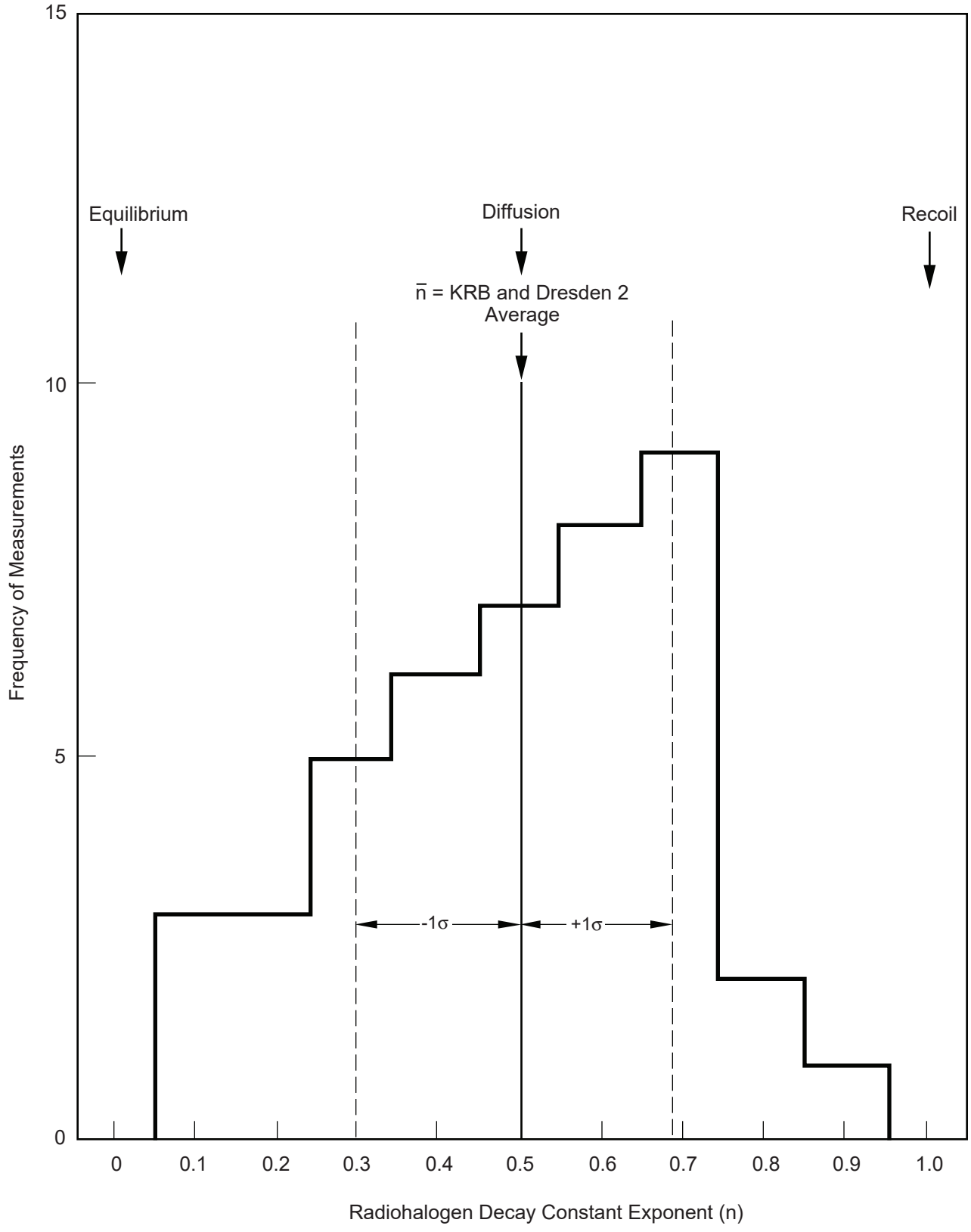
Table 11.1-5

Noncoolant Activation Products in Reactor Water

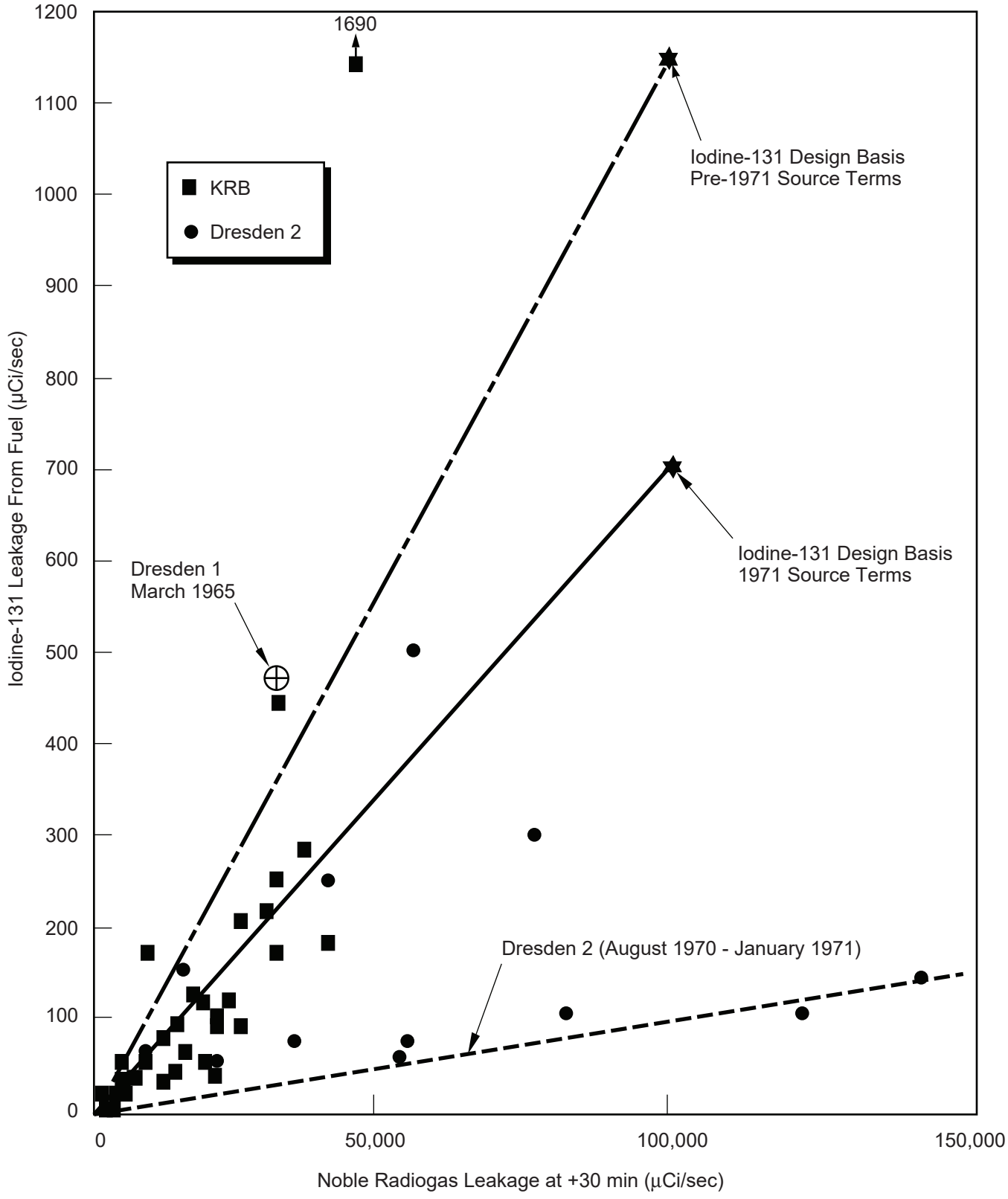
Isotope	*Half-Life*	*Concentration (μCi/g)*
^{24}Na	*15 hr*	2×10^{-3}
^{32}P	*14.31 days*	2×10^{-5}
^{51}Cr	*27.8 days*	5×10^{-4}
^{54}Mn	*313 days*	4×10^{-5}
^{56}Mn	*2.58 hr*	5×10^{-2}
^{58}Co	*71.4 days*	5×10^{-3}
^{60}Co	*5.25 years*	5×10^{-4}
^{59}Fe	*45 days*	8×10^{-5}
^{65}Ni	*2.55 hr*	3×10^{-4}
^{65}Zn	*243.7 days*	2×10^{-6}
^{69m}Zn	*13.7 hr*	3×10^{-5}
^{110m}Ag	*253 days*	6×10^{-5}
^{187}W	*23.9 hr*	3×10^{-3}



Columbia Generating Station Final Safety Analysis Report	**Noble Radiogas Decay Constant Exponent Frequency Histogram**		
	Draw. No. **900547.45**	Rev.	Figure **11.1-1**



Columbia Generating Station Final Safety Analysis Report	**Radiohalogen Decay Constant Exponent Frequency Histogram**		
	Draw. No. **900547.46**	Rev.	Figure **11.1-2**



Columbia Generating Station Final Safety Analysis Report	Noble Radiogas Leakage Versus Iodine-131 Leakage		
	Draw. No. 900547.47	Rev.	Figure 11.1-3

11.2 LIQUID WASTE MANAGEMENT SYSTEM

11.2.1 DESIGN BASIS

The liquid waste management system is designed to collect, segregate, store, and process potentially radioactive liquids generated during normal plant operation and anticipated operational occurrences. The design objective is to keep the radiation dose in unrestricted areas as low as is reasonably achievable (ALARA) within the guidelines of Appendix I to 10 CFR 50. The design incorporates the objectives of maximum recycle and minimum release of radioactive liquids without limiting plant operations or availability.

The criteria considered in the design of this system include volume, radioactivity, operational exposure, and required quality for recycle of the processed liquid. Radioisotopic inventories of the components used for the design are listed in Tables 11.2-1 through 11.2-5. These values are based on the reactor water source term associated with the design-basis fuel leakage rate. Allowance is made for concentration, decay and daughter product buildup in filters, demineralizers, and tanks. Equipment locations and arrangements are shown in Section 1.2.

The system is designed to treat process liquids with radioisotope concentrations associated with the design-basis fuel leakage and produce a quality of water which allows its recycle for plant reuse. Water inventory will occasionally require the discharge of processed liquids to the environs, in which case concentrations of radioisotopes in the effluent (Table 11.2-6) will be significantly less than the values specified in 10 CFR 20 and within the release limits established in the Technical Specifications. Radiation exposure to persons in unrestricted areas resulting from liquid waste discharged during normal operation and anticipated operational occurrences is less than the guidelines specified in 10 CFR 50, Appendix I.

Tanks that hold radioactive liquid, including the condensate storage tanks, are monitored for level and alarm primarily in the radwaste control room. Table 11.2-7 lists the design features of the tanks in the radwaste building used to prevent uncontrolled releases due to spillage and shows overflow alarms and drainage paths. The radwaste systems are operated from the radwaste control room; hence, additional local alarms are not required for radwaste tank levels.

All liquid waste management system tank overflows and drains are routed to the radwaste building equipment and floor drain sumps (see Figure 9.3-11). Radioactive liquid samples are primarily routed to sampling sinks. Those samples, which are local, drain into floor drain trenches, equipment or floor drain funnels, and pump beds. The above sample receivers are routed to various radioactive sumps, all of which are processed by the liquid waste management system.

Indoor tanks that hold radioactive liquid are not enclosed by individual curbs or elevated thresholds. The portion of the radwaste building that houses radioactive liquid tanks is Seismic

Category I, as discussed in Section 3.8.4.1.2. The radwaste building can retain the normal operating capacity of all liquid radwaste tanks, which are nonseismic. Analysis of a postulated radioactive release due to a liquid radwaste tank failure is presented in Section 15.7.3.

The radioactive and nonradioactive equipment and floor drains within the plant are segregated. Equipment and floor drains within the reactor building and radwaste building are routed to the liquid waste management system. The turbine building equipment and floor drains are segregated for radioactivity by component source and area (see Section 9.3.3). Although three of the turbine building sumps are designated as collectors of nonradioactive waste water, there is a possibility of low level contamination of the effluent water. One contamination source is steam leaks inside the building which condense on interior surfaces and are routed to floor drains. Because of this possibility, the discharge of these three sumps is routed to the radwaste system for processing.

The design of the system was accomplished prior to the issuance of Regulatory Guide 8.8. However, the system does incorporate substantially the guidance provided in this regulatory guide.

Like the nonradioactive sumps, the storm water drainage system (see Section 9.3.3.2.3.1) is not intended to collect radioactive materials. Nonetheless, radionuclides have been detected in the pond water and sediments. These concentrations are attributed to unanticipated and unavoidable occurrences. For example, special system draindowns during maintenance activities may have contributed minor amounts of tritium and corrosion products. Tritium that leaves the plant as a vapor can condense on building roofs and exterior walls and be carried to the ponds in storm drainage. Also, water treatment filter backwashes that are routed to the ponds can contribute radionuclides which were withdrawn from the river.

The degree of compliance with Regulatory Guide 1.143 is described in Section 1.8.

The liquid waste management system is designed to the requirements of General Design Criteria, Appendix A to 10 CFR 50, as follows:

General Design Criterion 60

The system capacity as required by General Design Criterion 60 is sufficient for the volume of liquid waste expected from normal operation and anticipated operational occurrences such as condenser leakage, maintenance activities, and process equipment down time. Flow rates are listed in Tables 11.2-8 through 11.2-10.

General Design Criterion 64

Radioactivity monitoring in the sample tanks and in the effluent discharge path ensures that excess liquid discharged to the environs does not exceed the limits of 10 CFR 20. Sampled

fluids exceeding these limits are returned to an appropriate collector tank for reprocessing. The radwaste effluent radiation monitoring system is described in Section 11.5.2.2.2.3.

11.2.2 SYSTEM DESCRIPTION

11.2.2.1 Process Description

Radioactive liquid wastes are collected and segregated into three categories: high purity waste, low purity waste, and chemical waste. Wastes thus classified are treated in subsystems designated as equipment drain, floor drain, and chemical waste, respectively.

High purity wastes, treated in the equipment drain subsystem, have low conductivity and relatively high radioactivity concentrations. Radioactive material is removed from these wastes by filtration and ion exchange. Following treatment and batch sampling, the processed waste is normally returned to condensate storage for reuse in plant.

Low purity wastes, treated in the floor drain subsystem, have moderate conductivity and generally low radioactivity. As with high purity wastes, radioactive material is removed by filtration and ion exchange. Following treatment and batch sampling, the processed waste is normally returned to condensate storage for reuse in the plant.

The high conductivity and organic content of chemical wastes preclude normal treatment in the system demineralizers by ion exchange. These wastes can be neutralized by adding appropriate neutralizing agent and thorough mixing. If necessary, following neutralization, these wastes are routed to a backwash tank or phase separator where unexpended ion-exchange capacity of the resins is used to remove contaminants from the chemical waste. After thorough mixing and a period of quiescence to allow the ion-exchange process to proceed, the excess liquid is decanted to the floor drain system for further processing. The spent resins in the separators are then processed as described below.

The installed chemical waste concentrators are currently not used. Their preoperational testing and use has been deferred until a need is identified. There are currently no plans to activate the concentrators.

The installed detergent drain tanks are not normally used. Any wastes requiring disposal are handled on a case-by-case basis and routed for processing and disposal in accordance with plant procedures and regulatory requirements and guidelines.

All liquid radwaste process streams terminate in one of the sample or distillate tanks. Since the liquid waste management system is operated on a batch basis, this arrangement allows each treated batch to be sampled to ensure that the treatment was effective. If the sample indicates that the processed liquid is still above acceptable radioactivity limits or substandard in purity, equipment is provided to either recycle the batch through the same treatment or through a

subsystem providing a higher degree of treatment. If the sample indicates that the level of activity is within limits required to discharge and the water is in excess of inventory capacity or is substandard in purity, the processed liquid may be discharged. The actual release of effluents from any processed and sampled batch tank requires the opening of a key-locked valve in accordance with written operating procedures. All required information regarding the batch release must be documented. These procedures are established to prevent inadvertent release of liquids that have not been suitably processed and analyzed.

Expended ion exchange resins are removed by backwashing to the spent resin tank and phase separators. Excess backwash water is removed from the phase separators by decantation and routed to either the floor drain or equipment drain collector tank for treatment. The powdered resin sludge is accumulated for radioactive decay. Following accumulation of successive layers in the phase separator, the sludge is then transferred to the radwaste processing system for dewatering or solidification. The deep bed resins in the spent resin tank, after a decay period, are also transferred to the solid radwaste processing system. Water separated from the wastes is returned via the waste sludge phase separator to the floor drain collector tank for treatment. The solid waste management system is described in Section 11.4.

Noncondensable gases from the liquid waste processing vessels are vented through the radwaste building exhaust system. This system is described in Section 9.4.3.2.

A process flow diagram, Figure 11.2-1, together with process flow diagram data, Table 11.2-11, show the volumes, flow rates, and radioactivity concentration used in the design of the liquid and solid waste management system.

The radionuclide distribution of liquid waste management system influents is based on the reactor coolant concentrations, as discussed in Section 11.1, taking into consideration mixing and dilution sources.

Decontamination factors which were used for evaluations of the system are those values specified in draft Regulatory Guide 1.CC. Table 11.2-12 shows the process equipment design basis decontamination factors which were used to generate the radioactivity values in Table 11.2-11.

11.2.2.2 Subsystems Description

11.2.2.2.1 Equipment Drain Subsystem

The equipment drain subsystem consists of a waste collector tank, waste surge tank, pressure precoat filter, deep bed demineralizer, two waste sample tanks, and auxiliary equipment necessary to operate the subsystem. Sizes and capacities of the equipment are listed in Table 11.2-13. The waste surge tank principally serves as a receptacle for reactor hydrotest and thermal expansion water and residual heat removal (RHR) system flush water during

startup and testing of these systems. The waste surge tank also serves as backup during equipment downtime. The piping and instrumentation drawing for this subsystem is shown in Figure 11.2-2.

High purity (low solids content) liquid wastes are collected in the waste collector tank from the following sources:

a. Drywell equipment drain sump,
b. Reactor building equipment drain sump,
c. Radwaste building equipment drain sump,
d. Turbine building equipment drain sump,
e. Reactor water cleanup system,
f. RHR system,
g. Cleanup phase separators (decant water),
h. Condensate phase separators (decant water), and
i. Fuel pool seal rupture drains (to waste surge tank).

The quantities of these wastes are summarized in Table 11.2-8. Since these wastes can contain a high percentage of primary reactor water, the radioactive concentration could be relatively high (on the order of 2.4 μCi/ml).

In the event of a component malfunction within the equipment drain subsystem, sufficient cross ties are provided to the floor drain subsystem to permit continued processing of the wastes. Sufficient capacity and treatment capability are provided to handle such conditions.

Normally, the equipment drain subsystem treated effluents are recycled to the condensate storage tanks for reuse within the plant. When condensate storage capacity is not available or the water is substandard in purity, the purified liquid from this subsystem may be sampled, analyzed and, if acceptable for release as described previously, routed to the blowdown line for discharge. Liquid waste that is unacceptable for discharge is reprocessed.

11.2.2.2.2 Floor Drain Subsystem

The floor drain subsystem consists of a floor drain collector tank, pressure precoat filter, deep bed demineralizer, sample tank, and auxiliary equipment necessary to operate the subsystem. Sizes and capacities of the equipment are listed in Table 11.2-13. The flow diagram for this subsystem is shown in Figure 11.2-3.

Intermediate purity liquid wastes are collected in the floor drain collector tank from the following sources:

a. Drywell floor drain sump,
b. Reactor building floor drain sumps,

c. Radwaste building floor drain sumps,
d. Turbine building floor drain sump, and
e. Waste sludge phase separator (decant water).

The quantities of these wastes are summarized in Table 11.2-9. These wastes are normally of intermediate purity (50 mho/cm and higher) and have radioactive concentrations on the order of 0.1 μCi/ml.

Similar to the equipment drain subsystem, the floor drain subsystem normally functions as an independent process stream. Equipment redundancy and intersystem cross ties have been provided to allow substitution for any failed component.

High purity effluent is routed to condensate storage for reuse in the plant. When condensate storage capacity is exceeded or the processed liquid is substandard in purity, it may be discharged via the blowdown line, if it meets acceptable limits.

11.2.2.2.3 Chemical Waste Subsystem

The chemical waste subsystem consists of two of each of the following: detergent drain tanks, chemical waste tanks, decontamination solution concentrators, decontamination solution concentrated waste tanks, and distillate tanks. It contains also a polishing (deep bed) demineralizer and auxiliary equipment necessary to operate the subsystem. Sizes and capacities of the equipment are listed in Table 11.2-13. The flow diagram for this subsystem is shown in Figure 11.2-4. The decontamination solution concentrators, concentrated waste tanks, distillate tanks, and polishing demineralizer have been installed but will not be used until plant operating experience indicates a need and system testing is accomplished. Chemical wastes are currently processed by routing to a backwash tank or phase separator for use of unexpended ion-exchange capacity of the resins to clean the water prior to decanting to the floor drain subsystem for further processing.

Chemical wastes collected in the chemical waste tank are from the following sources:

a. Detergent drains,
b. Shop decontamination solutions,
c. Reactor and turbine building decontamination drains,
d. Low purity wastes from either the equipment or floor drain subsystems,
e. Filter demineralizer element chemical cleaning solutions,
f. Battery room drains,
g. Chemical system overflows and tank drains, and
h. Laboratory drains.

The quantities of these wastes are summarized in Table 11.2-10. These chemical wastes are of such high conductivity and organic content as to preclude normal treatment by ion exchange,

and the radioactivity concentrations are variable. These wastes are processed by routing to a backwash tank or phase separator and to the floor drain system for further processing.

If the concentrators were ever activated, the evaporator concentrates would be processed by the solid waste management system and the distillate would be routed to the distillate tank. After analysis, the distillate could be routed through a polishing demineralizer to further reduce impurities, recycled through the evaporator, or sent directly to condensate storage for plant reuse. As with the other subsystems, when high purity water storage capacity is exceeded or the processed liquid is of substandard purity, liquid within 10 CFR 20 release limits could be discharged via the blowdown line.

11.2.2.2.4 Shared Equipment

Other than serving as mutual backup, main process equipment normally is not shared between subsystems. Auxiliary equipment not in the direct process stream is shared between subsystems. Shared equipment includes the following:

a. The waste precoat tank and waste precoat pump are shared between the waste collector filter and the floor drain filter,

b. The waste filter aid tank is shared between the waste collector filter and floor drain filter,

c. The resin addition tank is shared between the waste demineralizer, floor drain demineralizer, and polishing demineralizer, and

d. The chemical addition tanks, caustic and acid, and associated pumps are shared between the waste collector tank, floor drain collector tank, detergent tanks, and chemical waste tanks.

11.2.2.2.5 Surge Capacities

The radwaste system process data is the basis for sizing of the equipment. Tables 11.2-8, 11.2-9, and 11.2-10 list startup flows, daily flows, and maximum flows for the equipment drain subsystem, floor drain subsystem and chemical waste subsystem, respectively. Anticipated operational occurrences such as startup operations, equipment malfunction, and shutdown operations are accounted for in these tabulations. The bases for these types of values are presented in NEDO-10951, “Releases from BWR Radwaste Management Systems,” July 1973.

The surge storage and process capacities can be envisioned by comparing the normal and maximum daily volumes, listed in Table 11.2-11, with the design flow rates of pumps and tank volumes listed in Table 11.2-13. Alternate processing rather than bypass operations are used

during equipment downtime. The equipment and floor drain subsystems are sized such that with either subsystem inoperative, the remaining subsystem is capable of processing the maximum expected volume of both subsystems. Additionally, the waste surge tank provides reserve storage capacity. The chemical waste subsystem incorporates two parallel processing paths. Cross connections allow individual components from either process path to serve as a substitute in the other process path. The parallel path processing and adequate storage capacity ensures that inoperability of any component in this subsystem will not limit plant operation.

11.2.2.2.6 Design Features

The design pressures and temperatures for individual components are listed in Table 11.2-13. Collection and storage tanks are designed for atmospheric pressure. The mixed-bed demineralizer units, precoat filter units, and concentrators are pressure vessels. The quality classification for the system is Quality Group D+ as defined in Section 3.2.

Chapter 12 discusses the design features incorporated in the system to maintain occupational exposure ALARA. As shown in the general arrangement drawings in Section 1.2, the radwaste processing equipment is located in shielded rooms and cells. Process lines that penetrate shield walls are routed to prevent a direct radiation path from the tanks and equipment to normally occupied areas.

Control of the liquid waste management system is from a shielded radwaste control room and from shielded local operating galleries. Flanges are provided where required for maintenance in an otherwise all-welded system. The cells and concrete rooms provide secondary enclosures which facilitate collection of spills or leaks from the system for processing. Vessels that may have their contents mixed with air or that may have batches transferred into them by means of pneumatic transfer are closed and vented to the radwaste building ventilation system. Piping and tubing 2 in. and under is field routed but required to be in specified space envelopes for shielding and in-plant exposure consideration.

The liquid waste management system is designed to minimize the effects of equipment malfunction and operator error. After initiation by the operator, valve positioning, equipment startup, and system operation can be automatically performed by a process controller. Failure of any valve to properly align stops the sequence and prevents a pump start. Once on line, processing continues until the feed volume is processed or the capacity of a piece of process equipment is reached. In either case, automatic shutdown occurs with valves returning to the shutdown position. During initial system startup or in the event of controller failure, the processes are manually controlled. System variables, such as tank levels, flow rates, pressures, and conductivity are indicated and alarmed in the radwaste control room.

The discharge from the liquid waste subsystems to the blowdown line is normally isolated by a closed manual valve and a key-locked closed manual valve separated by a telltale drain. This path is further protected by air-operated isolation valves, flow control valves, flow indication

and radiation monitoring instrumentation. These design features are complimented by administrative procedures which prevent inadvertent radioactive liquid releases and releases of liquids that exceed the Technical Specifications release limit.

11.2.3 RADIOACTIVE RELEASES

The decision to recycle or to release a particular batch of processed liquid is based on plant water inventory and the type and concentration of chemical impurities being processed. It is not expected that tritium buildup in the plant will determine the frequency of releases. Concentration of radionuclides released to the environment will be significantly less than those values specified in 10 CFR 20. The quantities of radioactive materials released during normal operation will not result in dose rates to persons in unrestricted areas in excess of 10 CFR 50, Appendix I, values as implemented by the Technical Specifications release limits.

11.2.3.1 Release Point and Dilution

Excess liquid effluent is discharged to the circulating water blowdown line at a variable rate up to about 190 gpm. Dilution is furnished by circulating water system blowdown when required or when normal blowdown is in progress. The circulating water blowdown line terminates in the Columbia River. Applicable concentration limits specified in 10 CFR 20 apply at the point of discharge to the river.

The storm water drainage ponds described in Section 9.3.3.2.3.1 are a point of release of detectable radionuclides. For the reasons discussed in Section 11.2.1, most of the activity is believed to be either condensed gaseous effluent or material of external origin. The ponds are within a plant restricted area (see Section 2.1.1.3), and public access is restricted by a fence which surrounds the ponds.

11.2.3.2 Calculation of Releases of Radioactive Materials

Quantities of radioactive materials released with liquids were calculated for initial plant licensing using the GALE code presented in draft Regulatory Guide 1.CC to show compliance with Appendix I to 10 CFR 50 for normal operation plus anticipated operational occurrences. The plant operational parameters, including source terms, were those presented in Appendix B of the guide with waste stream flow rates adjusted to the Columbia Generating Station plant design. The calculated quantities of radioactive materials released with liquids used for initial plant licensing are presented in Table 11.2-14 in terms of curies per year. The radionuclide concentrations in the effluent are presented in Table 11.2-6 and are compared with the values of 10 CFR 20, Appendix B, Table II, Column 2.

Since becoming operational, releases of radioactive materials in liquid effluents have been determined using actual flow volumes and quantitative and qualitative laboratory analyses. Doses due to radioactive materials released in liquid effluents are determined to be in

compliance with Technical Specifications at specified intervals. Compliance is reported in the Annual Radioactive Effluent Release Report using the NRC LADTAP II computer code along with parameters outlined in the Offsite Dose Calculation Manual.

11.2.3.3 Exposure of Persons at or Beyond the Site Boundary

Estimated annual exposure of persons in unrestricted areas resulting from liquid effluents (Table 11.2-6) is discussed in Section 5.2 of the Environmental Report, Operating License stage. The estimated total body dose of 2.3 mrem per year and largest calculated single organ dose of 1.6 mrem per year to the bone are well within the guidelines of 10 CFR 50, Appendix I.

11.2.3.4 Cost-Benefit Analysis

It has been determined that a cost-benefit analysis, as described in Appendix I to 10 CFR 50 Section II.D is not required for Columbia Generating Station.

The question of eligibility of Columbia Generating Station to dispense with the Appendix I cost-benefit analysis as to "per site" limitations was reviewed by the NRC in connection with their review of WPPSS Nuclear Project No. 1 (WNP-1) and WPPSS Nuclear Project No. 4 (WNP-4), Docket Nos. 50-460 and 50-513. The Staff concluded in its testimony at the Atomic Safety and Licensing Board (ASLB) hearing for WNP-1 and WNP-4 held on November 11, 1975, that

> The aggregate doses associated with WPPSS Nuclear Project No. 1, WPPSS Project No. 2, and WPPSS Nuclear Project No. 4 operation meet the RM-50-2 (i.e., Annex) design objectives [Tr. 724-727].

These conclusions were ratified by the ASLB on its decision pertaining to WNP-1 and WNP-4 of December 22, 1975, RAI-75/12 922, 934.

Table 11.2-1

Liquid Waste Management System Radioisotope Inventory
Equipment Drain Subsystem

Radioisotope	Radioisotope Inventory (μCi)			
	Waste Collector Tank[a]	Waste Filter[a]	Waste Demineralizer[a]	Waste Sample Tank[a]
^{83}Br	4.53E4		3.15E4	3.18E2
^{84}Br	4.67E4		9.19E3	9.28E1
^{85}Br	1.17E4		4.17E-4	4.21E-6
^{89}Sr	4.35E4		1.88E6	4.35E2
^{90}Sr	3.25E3		2.12E5	3.25E1
^{91}Sr	7.12E5		1.10E6	9.13E3
^{92}Sr	3.40E5		2.48E5	2.50E3
^{90}Y	3.53E2		2.00E5	3.92E0
^{91}Y	4.21E3		5.10E5	4.69E1
^{91M}Y	4.12E5		7.31E5	5.82E3
^{92}Y	2.15E5		2.37E5	2.33E3
^{95}Zr	5.63E2	2.81E2	1.31E4	5.62E0
^{97}Zr	6.94E1	1.73E1	1.70E1	3.46E-1
^{95}Nb	5.93E2	2.97E2	1.69E4	5.93E0
^{99}Mo	2.79E5	1.38E5	6.13E5	2.75E3
^{99M}Tc	1.70E6	8.18E3	2.02E6	1.51E4
^{101}Tc	2.05E5		5.15E3	5.20E1
^{103}Ru	2.66E2	1.33E2	5.17E3	2.66E0
^{106}Ru	3.67E1	1.84E1	1.12E3	3.67E-1
^{129M}Te	5.60E2		2.00E4	5.59E0
^{129}Te	3.03E2		1.27E4	3.31E0
^{132}Te	6.30E5		3.21E6	6.23E3
^{129}I	2.83E-8		9.76E-5	3.16E-10
^{131}I	1.77E5		2.12E6	1.76E3
^{132}I	8.53E5		3.45E6	7.83E3
^{133}I	9.03E5		1.57E6	8.67E3
^{134}I	4.66E5		1.75E5	1.77E3
^{135}I	7.73E5		7.35E5	6.80E3
^{134}Cs	2.26E3		1.44E5	2.26E1
^{135}Cs	4.19E-5		1.97E-2	5.95E-7
^{136}Cs	1.52E3		2.80E4	1.51E1
^{137}Cs	3.39E3		2.21E5	3.39E1
^{138}Cs	3.29E5		6.54E4	6.60E2
^{137M}Ba	2.99E3		2.07E5	3.17E1

Table 11.2-1

Liquid Waste Management System Radioisotope Inventory
Equipment Drain Subsystem (Continued)

Radioisotope	Radioisotope Inventory (μCi)			
	Waste Collector Tank[a]	Waste Filter[a]	Waste Demineralizer[a]	Waste Sample Tank[a]
^{139}Ba	3.67E5		1.98E5	2.00E3
^{140}Ba	1.24E5		2.29E6	1.24E3
^{141}Ba	2.64E5		1.50E4	1.52E2
^{142}Ba	2.32E5		2.14E3	2.16E1
^{141}Ce	1.18E3	5.99E2	3.04E4	1.22E1
^{143}Ce	3.98E2	1.94E2	4.80E2	3.88E0
^{144}Ce	4.94E2	2.47E2	1.48E4	4.94E0
^{140}La	2.07E4		2.28E6	2.29E2
^{141}La	3.66E4		4.61E4	4.60E2
^{142}La	2.34E4		3.08E4	3.11E2
^{143}Pr	5.37E2	2.69E2	1.71E4	5.37E0
^{144}Pr	4.40E2	2.46E2	1.48E4	4.91E0
^{147}Nd	1.92E2	9.59E1	1.52E3	1.92E0
^{239}NP	2.98E6		1.13E7	2.93E4
^{24}Na	1.80E4		2.52E4	1.70E2
^{32}P	2.77E2		5.54E3	2.76E0
^{51}Cr	6.99E3	3.49E3	1.10E5	6.98E1
^{54}Mn	5.65E2	2.82E2	1.70E4	5.65E0
^{56}Mn	1.58E5	5.67E4	5.57E4	1.13E3
^{58}Co	7.04E4	3.52E4	1.68E6	7.03E2
^{60}Co	7.07E3	3.53E4	2.26E5	7.07E1
^{59}Fe	1.12E3	5.61E2	2.30E4	1.12E1
^{65}Ni	9.40E2	3.36E2	3.30E2	6.73E0
^{65}Zn	2.82E1		1.67E3	2.82E-1
^{69M}Zn	2.63E2		3.52E2	2.47E0
^{110M}Ag	8.47E2	4.24E2	2.52E4	8.47E0
^{187}W	3.16E4	1.53E4	2.99E4	3.05E2

[a] The radioisotope inventory listed above for the waste collector tank, waste filter, waste demineralizer, and waste sample tank is assumed to be applicable to the floor drain collector tank, floor drain filter, floor drain demineralizer, and floor drain sample tank, respectively. This is done for the purpose of shielding analysis. In actuality, the radionuclide inventory in any component of the floor drain system is discussed in Sections 11.2.2.2.1 and 11.2.2.2.2. The values from the equipment drain subsystem are maximum due to cross-ties for alternate processing.

Table 11.2-2

Liquid Waste Management System Radioisotope Inventory
Chemical Waste Subsystem

Radioisotope	Radioisotope Inventory (μCi)				
	Chemical Waste Tank	DSC Waste Tank[a]	DSC Waste Measuring Tank[b]	Distillate Tank	Decon. Solution Concentrator
^{83}Br					
^{84}Br					
^{85}Br					
^{89}Sr	1.11E3	2.82E3	1.74E3	1.11E0	1.48E3
^{90}Sr	7.87E1	2.00E2	1.23E2	7.87E-2	1.05E2
^{91}Sr					
^{92}Sr					
^{90}Y					
^{91}Y					
^{91M}Y					
^{92}Y					
^{95}Zr	1.43E1	3.65E1	2.25E1	1.43E-2	1.92E1
^{97}Zr					
^{95}Bv					
^{99}Mo	7.87E3	2.00E4	1.23E4	7.87E0	1.05E4
^{99M}Tc					
^{101}Tc					
^{103}Ru					
^{106}Ru					
^{129M}Te	1.43E1	3.65E1	2.25E1	1.43E-2	1.92E1
^{129}Te					
^{132}Te	1.75E4	4.45E4	2.75E4	1.75E1	2.34E4
^{129}I					
^{131}I	4.65E3	1.13E4	6.95E3	4.65E0	5.92E3
^{132}I					
^{133}I					
^{134}I					
^{135}I					
^{134}Cs	5.75E1	1.46E2	9.02E1	5.75E-2	7.68E1
^{135}Cs					
^{136}Cs	3.94E1	1.00E2	6.19E1	3.94E-2	5.27E1
^{137}Cs	8.60E1	2.19E2	1.35E2	8.60E-2	1.15E2

Table 11.2-2

Liquid Waste Management System Radioisotope Inventory
Chemical Waste Subsystem (Continued)

Radioisotope	Radioisotope Inventory (μCi)				
	Chemical Waste Tank	DSC Waste Tank[a]	DSC Waste Measuring Tank[b]	Distillate Tank	Decon. Solution Concentrator
^{138}Cs					
^{137M}Ba					
^{139}Ba					
^{140}Ba	3.22E3	8.18E3	5.05E3	3.22E0	4.30E3
^{141}Ba					
^{142}Ba					
^{141}Ce	1.40E1	3.56E1	2.20E1	1.40E-2	1.87E1
^{143}Ce	1.25E1	3.17E1	1.96E1	1.25E-2	1.67E1
^{144}Ce	1.25E1	3.17E1	1.96E1	1.25E-2	1.67E1
^{140}La					
^{141}La					
^{142}La					
^{143}Pr	1.36E1	3.44E1	2.12E1	1.36E-2	1.81E1
^{144}Pr					
^{147}Nd	5.01E0	1.27E1	7.85E0	5.01E-3	6.69E0
^{239}Np	8.60E4	2.19E5	1.35E5	8.60E1	1.15E5
^{24}Na					
^{32}P	7.18E0	1.82E1	1.13E1	7.18E-3	9.58E0
^{51}Cr	1.79E2	4.54E2	2.80E2	1.79E-1	2.38E2
^{54}Mn	1.43E1	3.65E1	2.25E1	1.43E-2	1.92E1
^{56}Mn					
^{58}Co	1.79E3	4.54E3	2.80E3	1.79E0	2.38E3
^{60}Co	1.79E2	4.54E2	2.80E2	1.79E-1	2.38E2
^{59}Fe	2.87E1	7.26E1	4.48E1	2.87E-2	3.82E1
^{65}Ni					
^{65}Zn	7.18E-1	1.82E0	1.13E0	7.18E-4	9.58E-1
^{69M}Zn					
^{110M}Ag	2.15E1	5.46E1	3.37E1	2.15E-2	2.87E1
^{187}W					

[a] Decontamination solution concentrator waste tank.
[b] Decontamination solution concentrator waste measuring tank.

Table 11.2-3

Liquid Waste Management System Radioisotope Inventory
RWCU and Condensate Filter Demineralizers

Radioisotope	Radioisotope Inventory (μCi)	
	RWCU Filter Demineralizer	Condensate Filter Demineralizer
^{83}Br	2.50E6	2.30E5
^{84}Br	9.00E5	9.12E4
^{85}Br	3.00E4	5.42E3
^{89}Sr	1.40E7	3.00E5
^{90}Sr	1.10E6	2.56E4
^{91}Sr	3.20E7	2.13E5
^{92}Sr	1.60E7	9.12E4
^{90}Y	1.10E6	2.09E4
^{91}Y		4.54E4
^{91M}Y	2.20E7	1.28E5
^{92}Y		9.12E4
^{95}Zr	1.80E5	2.59E4
^{97}Zr	2.40E4	8.19E1
^{95}Nb	5.60E5	2.97E4
^{99}Mo	5.50E7	3.04E6
^{99M}Tc	2.40E7	3.18E6
^{101}Tc	1.90E6	1.04E4
^{103}Ru	9.00E4	1.16E4
^{106}Ru	1.20E4	1.85E3
^{129M}Te	1.50E6	3.60E3
^{129}Te		2.29E3
^{132}Te	3.70E7	1.20E6
^{129}I		4.89E-6
^{131}I	8.50E7	1.34E7
^{132}I	5.80E7	2.96E6
^{133}I	1.40E8	1.19E7
^{134}I	1.40E7	1.35E6
^{135}I	6.50E7	5.55E6
^{134}Cs	7.50E5	1.77E4
^{135}Cs		2.48E-2
^{136}Cs	4.20E5	7.37E3
^{137}Cs	1.20E6	2.67E4
^{138}Cs	5.50E6	3.23E4
^{127M}Ba	1.20E6	2.50E4

Table 11.2-3

Liquid Waste Management System Radioisotope Inventory
RWCU and Condensate Filter Demineralizers (Continued)

Radioisotope	Radioisotope Inventory (μCi)	
	RWCU Filter Demineralizer	Condensate Filter Demineralizer
^{139}Ba	1.20E7	7.23E4
^{140}Ba	3.60E7	6.03E5
^{141}Ba	3.00E6	1.63E4
^{142}Ba	1.60E6	9.94E3
^{141}Ce	1.80E5	2.86E4
^{143}Ce	5.00E4	2.40E3
^{144}Ce	1.60E5	2.47E4
^{140}La	4.10E7	5.60E5
^{141}La		1.63E4
^{142}La		9.94E3
^{143}Pr	1.60E5	2.74E4
^{144}Pr		2.47E4
^{147}Nd	5.50E4	5.65E3
^{239}Np	5.00E8	4.27E6
^{24}Na	1.50E6	9.56E3
^{32}P	9.50E4	1.40E3
^{51}Cr	2.60E6	2.80E5
^{54}Mn	2.20E5	2.83E4
^{56}Mn	6.00E6	2.69E5
^{58}Co	2.60E7	3.27E6
^{60}Co	2.80E6	3.61E5
^{59}Fe	4.30E5	4.96E4
^{65}Ni	3.80E4	1.59E3
^{65}Zn	1.10E4	2.16E2
^{69M}Zn	2.00E4	1.34E2
^{110M}Ag	3.40E5	4.24E4
^{187}W	3.60E6	1.49E5

Table 11.2-4

Liquid Waste Management System Radioisotope Inventory
Spent Resin and Condensate Backwash Receiving Tanks

Radioisotope	Radioisotope Inventory (μCi)	
	Spent Resin Tank	Condensate Backwash Receiving Tank
^{83}Br	3.15E4	2.30E5
^{84}Br	9.19E3	9.12E4
^{85}Br	4.17E-4	5.42E3
^{89}Sr	1.88E6	3.00E5
^{90}Sr	2.12E5	2.56E4
91Sr	1.10E6	2.13E5
^{92}Sr	2.48E5	9.12E4
^{90}Y	2.00E5	2.09E4
^{91}Y	5.10E5	4.54E4
^{91M}Y	7.31E5	1.28E5
^{92}Y	2.37E5	9.12E4
^{95}Zr	1.31E4	2.59E4
^{97}Zr	1.70E1	8.19E1
^{95}Nb	1.69E4	2.97E4
^{99}Mo	6.13E5	3.04E6
^{99M}Tc	2.02E6	3.18E6
^{101}Tc	5.15E3	1.04E4
^{103}Ru	5.17E3	1.16E4
^{106}Ru	1.12E3	1.85E3
^{129M}Te	2.00E4	3.60E3
^{129}Te	1.27E4	2.29E3
^{132}Te	3.21E6	1.20E6
^{129}I	9.76E-5	4.89E-6
^{131}I	2.12E6	1.34E7
^{132}I	3.45E6	2.96E6
^{133}I	1.57E6	1.19E7
^{134}I	1.75E4	1.35E6
^{135}I	7.35E4	5.55E6
^{134}Cs	1.44E4	1.77E4
^{135}Cs	1.97E-2	2.48E-2
^{136}Cs	2.80E4	7.37E3
^{137}Cs	2.21E5	2.67E4
^{138}Cs	6.54E4	3.23E4
^{137M}Ba	2.07E5	2.50E4

Table 11.2-4

Liquid Waste Management System Radioisotope Inventory
Spent Resin and Condensate Backwash Receiving Tanks (Continued)

	Radioisotope Inventory (μCi)	
Radioisotope	Spent Resin Tank	Condensate Backwash Receiving Tank
^{139}Ba	1.98E5	7.23E4
^{140}Ba	2.29E6	6.03E5
^{141}Ba	1.50E4	1.63E4
^{142}Ba	2.14E3	9.94E3
^{141}Ce	3.04E4	2.86E4
^{143}Ce	4.80E2	2.40E3
^{140}Ce	1.48E4	2.47E4
^{141}La	2.28E6	5.60E5
^{141}La	4.61E4	1.63E4
^{142}La	3.08E4	9.94E3
^{143}Pr	1.71E4	2.74E4
^{144}Pr	1.48E4	2.47E4
^{147}Nd	1.52E3	5.65E3
^{239}Np	1.13E7	4.27E6
^{24}Na	2.52E4	9.56E3
^{32}P	5.54E3	1.40E3
^{51}Cr	1.10E5	2.80E5
^{54}Mn	1.70E4	2.83E4
^{56}Mn	5.57E4	2.69E5
^{58}Co	1.68E6	3.27E6
^{60}Co	2.26E5	3.61E5
^{59}Fe	2.30E4	4.96E4
^{65}Ni	3.30E2	1.59E3
^{65}Zn	1.67E3	2.16E2
^{69M}Zn	3.52E2	1.34E2
^{110M}Ag	2.52E4	4.24E4
^{187}W	2.99E4	1.49E5

Table 11.2-5

Liquid Waste Management System Radioisotope Inventory
Phase Separators

Radioisotope	Radioisotope Inventory (μCi)		
	RWCU Phase Separator	Condensate Phase Separator	Waste Sludge Phase Separator
^{83}Br	4.95E6	2.26E5	
^{84}Br	1.71E6	7.04E4	
^{85}Br	3.73E4	8.21E2	
^{89}Sr	3.48E8	2.94E6	
^{90}Sr	3.95E7	3.23E5	
^{91}Sr	6.40E7	2.21E5	
^{92}Sr	3.17E7	8.98E4	
^{90}Y	3.95E7	3.15E5	
^{91}Y	5.66E6	4.76E5	
^{91M}Y	4.40E7	1.35E5	
^{92}Y	2.33E5	9.57E4	
^{95}Zr	4.83E6	2.55E5	3.70E1
^{97}Zr	4.70E4	7.14E1	4.53E-1
^{95}Nb	1.41E7	3.29E5	3.96E1
^{99}Mo	1.93E8	5.23E6	1.35E4
^{99M}Tc	1.27E8	2.17E6	8.76E3
^{101}Tc	3.41E6	7.67E-2	
^{103}Ru	2.06E6	1.36E5	1.74E1
^{106}Ru	4.09E5	2.14E4	2.45E0
^{129M}Te	3.15E7	3.48E-4	
^{129}Te	1.82E7	2.07E3	
^{132}Te	1.43E8	2.40E6	
^{129}I	1.29E-1	6.20E-5	
^{131}I	6.70E8	5.31E7	
^{132}I	1.87E8	4.16E6	
^{133}I	3.00E8	1.33E7	
^{134}I	2.72E7	1.18E6	
^{135}I	1.30E8	5.69E6	
^{134}Cs	2.63E7	2.19E5	
^{135}Cs	1.98E-1	3.48E-1	
^{136}Cs	4.00E6	4.07E4	
^{137}Cs	4.31E7	3.37E5	
^{138}Cs	1.05E7	2.50E4	

Table 11.2-5

Liquid Waste Management System Radioisotope Inventory
Phase Separators (Continued)

Radioisotope	Radioisotope Inventory (μCi)		
	RWCU Phase Separator	Condensate Phase Separator	Waste Sludge Phase Separator
^{137M}Ba	4.00E7	3.14E5	
^{139}Ba	2.36E7	6.74E4	
^{140}Ba	4.18E8	3.33E6	
^{141}Ba	5.51E6	1.01E4	
^{142}Ba	2.79E6	4.69E3	
^{141}Ce	3.80E6	2.37E5	7.78E1
^{143}Ce	1.22E5	2.88E3	1.46E1
^{144}Ce	5.37E6	2.83E5	3.29E1
^{140}La	4.78E8	3.64E6	
^{141}La	3.86E4	5.32E2	
^{142}La	4.89E4	6.33E2	
^{143}Pr	5.68E6	3.26E5	3.59E1
^{144}Pr	5.08E6	2.83E5	3.29E1
^{147}Nd	5.61E5	2.66E4	1.18E1
^{239}Np	1.57E9	6.92E6	
^{24}Na	3.07E6	1.02E4	
^{32}P	1.19E6	8.19E3	
^{51}Cr	4.93E7	2.15E6	4.50E2
^{54}Mn	7.43E6	3.26E5	3.76E1
^{56}Mn	1.19E7	2.52E5	1.52E3
^{58}Co	7.14E8	3.27E7	4.64E3
^{60}Co	9.98E7	4.30E6	4.72E2
^{59}Fe	1.03E7	4.50E5	7.35E1
^{65}Ni	7.52E4	1.49E3	9.01E0
^{65}Zn	3.63E5	2.57E3	
^{69M}Zn	4.06E4	1.41E2	
^{110M}Ag	1.14E7	4.84E5	5.64E1
^{187}W	7.95E6	1.63E5	9.81E2

Table 11.2-6

Annual Average Concentration of Radionuclides in Liquid Effluent

Nuclide	Effluent Concentration (μCi/ml)	10 CFR 20 Table II (μCi/ml)	Fraction of 10 CFR 20 Effluent/10 CFR 20
^{24}Na	1.3E-9	5E-5	0.000026
^{32}P	5.2E-11	9E-6	0.00000575
^{51}Cr	1.3E-9	5E-4	0.0000026
^{54}Mn	1.6E-11	3E-5	0.000000533
^{56}Mn	1.4E-9	7E-5	0.00002
^{55}Fe	2.8E-10	1E-4	0.0000028
^{59}Fe	8E-12	1E-5	0.00000008
^{58}Co	5.4E-11	2E-5	0.0000027
^{60}Co	1.1E-10	3E-6	0.0000367
^{65}Ni	8E-12	1E-4	0.00000008
^{64}Cu	4E-9	2E-4	0.00002
^{65}Zn	5.4E-11	5E-6	0.000011
^{69m}Zn	2.8E-10	6E-5	0.00000467
^{69}Zn	3E-10	8E-4	0.000000375
^{187}W	5.4E-11	3E-5	0.0000018
^{239}Np	1.6E-9	2E-5	0.00008
^{83}Br	7.2E-11	9E-4	0.00000008
^{84}Br	6E-12	4E-4	0.000000015
^{89}Rb	4.2E-11	9E-4	0.0000000467
^{89}Sr	2.8E-11	8E-6	0.0000035
^{91}Sr	4.4E-10	2E-5	0.000022
^{91m}Y	2.8E-10	2E-3	0.00000014
^{91}Y	1.4E-11	8E-6	0.00000175
^{92}Sr	3E-10	4E-5	0.0000075
^{92}Y	6.2E-10	4E-5	0.0000155
^{93}Y	4.6E-10	2E-5	0.000023
^{98}Nb	1.6E-11	2E-4	0.00000008
^{99}Mo	4.6E-10	2E-5	0.000023
^{99m}Tc	1.8E-9	1E-3	0.0000018
^{101}Tc	4E-12	2E-3	0.000000002
^{103}Ru	6E-12	3E-5	0.0000002

Table 11.2-6

Annual Average Concentration of Radionuclides in Liquid Effluent (Continued)

Nuclide	Effluent Concentration (μCi/ml)	10 CFR 20 Table II (μCi/ml)	Fraction of 10 CFR 20 Effluent/10 CFR 20
^{103m}Rh	6E-12	6E-3	0.000000001
^{104}Tc	1.2E-11	4E-4	0.00000003
^{105}Ru	1.1E-10	7E-5	0.00000165
^{105m}Rh	1.2E-10	--	--
^{105}Rh	3.8E-11	5E-5	0.00000076
^{129m}Te	1E-11	7E-6	0.00000150
^{129}Te	6E-12	4E-4	0.00000016
^{131m}Te	2E-11	8E-6	0.0000026
^{131}Te	4E-12	8E-5	0.00000005
^{131}I	1.4E-9	1E-6	0.0014
^{132}Te	2E-12	9E-6	0.000000233
^{132}I	7E-10	1E-4	0.000007
^{133}I	3.6E-9	7E-6	0.000510
^{134}I	2.9E-10	4E-4	0.0000007
^{134}Cs	1.6E-9	9E-7	0.00175
^{135}I	1.7E-9	3E-5	0.0000580
^{136}Cs	9.9E-10	6E-6	0.000165
^{137}Cs	3.6E-9	1E-6	0.0036
^{137m}Ba	3.4E-9	--	--
^{138}Cs	1.5E-9	4E-4	0.0000037
^{139}Ba	1E-10	2E-4	0.0000005
^{140}Ba	1E-10	8E-6	0.0000131
^{140}La	2.3E-11	9E-6	0.00000256
^{141}La	3.6E-11	5E-5	0.00000071
^{141}Ce	8E-12	3E-5	0.000000280
^{142}La	7.6E-11	1E-4	0.00000076
^{143}Ce	6E-12	2E-5	0.0000003
^{143}Pr	1E-11	2E-5	0.0000005
All others	1.5E-11	1E-8	0.0015
		Total	0.009020094

Table 11.2-7

Tank Design Features

Component	High Level Alarm	Overflows and Drains
Floor drain collector tank	Yes (RCR)	Floor drain sump, W-2
Waste sludge phase separator	Yes (RCR)	Floor drain sump, W-2
Floor drain sample tank	Yes (RCR)	Floor drain sump, W-2
Waste collector tank	Yes (RCR)	Floor drain sump, W-2
Spent resin tank	Yes (RCR)	Floor drain sump, W-2
Waste surge tank	Yes (RCR)	Equipment drain sump, W-3
Waste sample tanks	Yes (RCR)	Equipment drain sump, W-3
Detergent drain tanks[b]	No[a]	Chemical waste sump, W-4
Chemical waste tanks	Yes (RCR)	Chemical waste sump, W-4
Distillate tanks[b]	No[a]	Chemical waste sump, W-4
Nonoperational decontamination solution concentrator waste tanks[b]	No	Chemical waste sump, W-4
Condensate backwash receiving tank	Yes (RCR)	Equipment drain sump, W-3
Condensate phase separators	Yes (RCR)	Equipment drain sump, W-3
Decontamination solution concentrator waste measuring tank[c]	No[a]	Decontamination solution concentrator waste tanks
RWCU phase separators	Yes (RCR)	Equipment drain sump, W-3
Condensate storage tanks	Yes (MCR)	Floor drain sump, T4

[a] Alarms not installed.
[b] Nonoperational.
[c] Not used.

LEGEND
RCR - radwaste control room
MCR - main control room

Table 11.2-8

Equipment Drain Subsystem Sources

Source	Startup Flows (gpd)	Regular Daily Flows (gpd)	Irregular Flows (gpd)	Maximum Daily Flows (gpd)
Equipment drains				
Drywell	3,860	3,860		28,800
Reactor building	3,755	3,755		14,400
Turbine building	5,726	5,726		5,726
Radwaste building	1,000	1,000		1,000
Reactor hydrotest and water level reduction to operating state	56,720	0		0
RHR system flush water			4,000[a]	0
Condensate demineralizer backwash	27,000		13,500[b]	40,500[c]
Cleanup demineralizer backwash	2,430		1,215[d]	
Water inleakage to condenser	0	0		14,400
Total	100,491	14,341		104,826

[a] Occurs every shutdown prior to placing the RHR system in operation for shutdown cooling.

[b] Under normal operating conditions, one condensate filter demineralizer would be precoated every 4 days.

[c] The maximum daily flow is based on a main condenser inleakage of 10 gpm, which corresponds to 3 condensate demineralizer precoatings daily and maximum leak and drain inflows. Up to 36 gpm of condenser inleakage can be accommodated. This requires precoating of one condensate demineralizer every 3 hr. This inleakage rate would result in overloading the equipment drain subsystem but could be tolerated for short periods of time during location and repair of the leak.

[d] Under normal operating conditions, each cleanup demineralizer may be precoated every 6.8 days.

Table 11.2-9

Floor Drain Subsystem Sources

Source	Regular Daily Flows (gpd)	Irregular Flows (gpd)	Maximum Daily Flows (gpd)
Floor drains			
Drywell	700		28,800
Reactor building	2,000		15,000
Radwaste building	1,000		1,000
Turbine building	2,000		2,000
Waste sludge phase separator decant	0	8,489[a]	8,489
Total	5,700	8,489	55,289

[a] Under normal operating conditions, the waste sludge phase separator tank is decanted every 3-4 days.

Table 11.2-10

Chemical Waste Subsystem Sources

Source	Regular Daily Flow (gpd)	Irregular Flow (gpd)	Maximum Daily Flow (gpd)
Detergents drains/shop decontamination solutions	1,000		2,000
Laboratory drains	500		500
Decontamination drains reactor and turbine buildings		1,000	1,000
From floor drain or equipment drain subsystem		15,000	15,000
Filter demineralizer chemical cleaning solutions		Infrequent 2,000	2,000
Battery room drains		Infrequent 100	100
Total	1,500		20,600

Table 11.2-11

Radwaste System Process Flow Diagram Data

	Equipment Drain Subsystem						
Flow path	1	2	3	4	5	6	8
Batches/day (normal)	8.5	4.1	1.1	6.3	1.0/6.8	4.0/7.4	1.0/30.0
Batches/day (maximum)	63.4	15.8	1.1	6.3	1.0	4.0/2.0	-
Volume/batch (gal)	455	909	909	909	2430	13,500	4000
Normal daily volume (gal)	3860	3755	1000	5726	-	-	-
Normal activity ($\mu Ci/cm^3$)	5.16E-2	1.30E-2	7.18E-5	1.91E-5	1.59E-2	7.18E-6	9.22E-7
Maximum daily volume (gal)	28,800	14,400	1000	5726	2430	27,000	4000
Maximum activity ($\mu Ci/cm^3$)	1.72E0	4.32E-1	2.39E-3	1.26E-2	5.32E-1	3.59E-4	3.08E-5
Flow rate (gal/minute)	50	50	50	50	53	450	Batch
Daily activity (μCi/day)							
Normal	7.19E5	1.76E5	2.59E2	3.94E2	-	-	-
Maximum	2.51E7	6.14E6	9.05E3	2.73E3	4.89E6	3.67E4	4.66E2

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

	Equipment Drain Subsystem (Continued)						
Flow path	9	33	12	14	17	18	19
Batches/day (normal)	1.0	1.0	1.0	1.0	1.5	2.2	1.1
Batches/day (maximum)	7.0	7.0	7.0	7.0	63.4	16.5	1.1
Volume/batch (gal)	15,000	15,000	15,000	15,000	455	909	909
Normal daily volume (gal)	15,000	15,000	15,000	15,000	700	2000	1000
Normal activity ($\mu Ci/cm^3$)	1.16E-2	5.81E-3	1.16E-4	1.16E-4	7.18E-6	1.05E-6	1.44E-5
Maximum daily volume (gal)	104,826	104,826	104,826	104,826	28,800	15,000	1000
Maximum activity ($\mu Ci/cm^3$)	3.59E-1	3.53E-1	3.53E-3	3.53E-3	6.84E-2	1.00E-3	4.78E-4
Flow rate (gal/minute)	190	190	190	190	50	100	50
Daily activity (μCi/day)							
Normal	6.30E5	3.14E5	6.30E3	6.30E3	1.81E1	7.57E0	5.18E1
Maximum	2.04E7	2.00E7	2.03E5	2.04E5	1.81E5	7.57E3	1.81E3

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

	Equipment Drain Subsystem (Continued)					
Flow path	20	117	21	23	107	108
Batches/day (normal)	2.2	10./3.4	1.0/2.6	1.0/2.6	1.0/2.6	1.0/2.6
Batches/day (maximum)	2.2	1.0	3.7	3.74	3.7	3.7
Volume/batch (gal)	909	8489	15,000	15,000	15,000	15,000
Normal daily volume (gal)	2000		5700	5700	5700	5700
Normal activity ($\mu Ci/cm^3$)	7.18E-7	7.18E-6	1.00E-6	4.99E-5	1.00E-6	1.00E-6
Maximum daily volume (gal)	2000	8489	55,489	55,489	55,489	55,489
Maximum activity ($\mu Ci/cm^3$)	2.39E-5	3.59E-4	1.08E-2	1.08E-2	1.08E-4	1.08E-4
Flow rate (gal/minute)	50	50	190	190	190	190
Daily activity (μCi/day)						
Normal	5.18E0	-	2.56E3	1.03E3	2.06E1	2.06E1
Maximum	1.81E2	-	2.33E5	2.33E5	2.33E3	2.33E3

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

	Waste Surge Subsystem	
Flow path	104	37
Batches/day (normal)	1.0/yr	1.0/yr
Batches/day (maximum)	1.0	1.0
Volume/batch (gal)	56,720	56,720
Normal daily volume (gal)		
Normal activity ($\mu Ci/cm^3$)	7.18E-6	7.18E-6
Maximum daily volume (gal)	56,720	56,720
Maximum activity ($\mu Ci/cm^3$)	3.59E-4	3.59E-4
Flow rate (gal/minute)	Batch	190

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

	Chemical Waste Subsystem				
Flow path	27	109	120	121	122
Batches/day (normal)	1.0				
Batches/day (maximum)	2.0	2.0	2.0	1.0	3.7
Volume/batch solids(lb)			581	1162	314
liquids (gal)	1000	12,153	12,153	615	168
Normal daily volume (gal)	1000				
Normal activity solids (μCi/batch)					
liquids (μCi/cm^3)	1.05E-5	2.76E-3	2.71E-3	1.07E-1	1.07E-1
Maximum daily volume (gal)	2000	24,306	24,306		
Maximum activity solids (μCi/batch)					
liquids (μCi/cm^3)	1.05E-5	-	2.71E-3	1.07E-1	1.07E-1
Flow rate (gal/minute)	25	190	10	31	20

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

	Waste Sludge Subsystem (Condensate Backwash)			
Flow path	56	58	60	6
Batches/day (normal)	4.0/7.4	4.0/7.4	1/18.5	4/7.4
Batches/day (maximum)	4.0	4.0	1	4.0/2.0
Volume/batch solids (lb)	330	330	3300	-
liquids (gal)	13,500	13,500	7527	13,500
Normal daily volume (gal)	7300	7300	-	-
Normal activity solids (μCi/batch)	2.20E6	2.20E6	1.26E7	-
liquids (μCi/cm^3)	7.18E-6	7.18E-6	7.18E-6	7.18E-6
Maximum daily volume (gal)	54,000	54,000	7527	27,000
Maximum activity solids (μCi/batch)	5.26E7	5.26E7	2.67E7	-
liquids (μCi/cm^3)	3.59E-4	3.59E-4	3.59E-4	3.59E-4
Flow rate (gal/minute)	2500	450	20	450

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

	Waste Sludge Subsystem (Cleanup Backwash)		
Flow path	54	59	5
Batches/day (normal)	2.0/6.8	1.0/60	1.0/6.8
Batches/day (maximum)	2.0	1.0/60	1.0
Volume/batch solids (lb)	29.7	524	-
liquids (gal)	1215	1196	2430
Normal daily volume (gal)	360	-	-
Normal activity solids(μCi/batch)	2.55E7	1.19E8	-
liquids (μCi/cm^3)	1.59E-2	1.59E-2	1.59E-2
Maximum daily volume (gal)	2430	1196	2430
Maximum activity solids (μCi/batch)	7.68E8	1.80E8	-
liquids (μCi/cm^3)	5.32E-1	5.32E-1	5.32E-1
Flow rate (gal/minute)	270	20	53

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

	Waste Sludge Subsystem (Spent Resin)			
Flow path	64	119	69	71
Batches/day (normal)	1.0/66	1.0/67	1.0/165	-
Batches/day (maximum)	1.0/29	1.0/49	1.0/100	-
Volume/batch solids (lb)	1539	1539	1539	1539
liquids (gal)	746	746	746	3510
Normal activity solids (μCi/batch)	2.39E6	6.15E3	1.16E2	-
liquids (μCi/cm^3)	7.18E-6	7.18E-6	7.18E-6	7.18E-6
Maximum activity solids (μCi/batch)	2.74E7	2.34E5	1.26E2	-
liquids (μCi/cm^3)	3.59E-4	3.59E-4	3.59E-4	3.59E-4
Flow rate (gal/minute)	37	37	37	20

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

	Equipment Drain Subsystem				
Flow path	61	63	62	-	65
Batches/day (normal)	1.0	1.0/3.4	1.0/5.2	-	1.0/3.4
Batches/day (maximum)	2.9	1.1/1.0	1.0/5.2	-	1.0
Volume/batch solids (lb)	41.36	41.36	59.4	-	220
liquids (gal)	1692	1692	2430	-	501
Normal daily volume (gal)	1692	-	-	-	-
Normal activity solids (μCi/batch)	3.30E4	8.42E1	1.47E4	-	4.87E4
liquids (μCi/cm^3)	7.18E-6	7.18E-6	7.18E-6	-	7.18E-6
Maximum daily volume (gal)	4906	1861	-	-	501
Maximum activity solids (μCi/batch)	3.30E4	8.42E1	1.47E6	-	7.33E5
liquids (μCi/cm^3)	3.59E-4	3.59E-4	3.59E-4	-	9.88E-4
Flow rate (gal/minute)	376	376	540	-	20

Table 11.2-11

Radwaste System Process Flow Diagram Data (Continued)

NOTES:

Process Diagram, Figure 11.2-1, forms part of this data.

The following definitions are used for this data:

Normal volume - Expected flow during steady state normal operation

Maximum volume - Maximum expected flow during unsteady state operation such as startup, shutdown, etc.

Normal activity - Activity level expected during operation with no fuel leaks and corrosion product reactor water activity concentration of 0.1 $\mu Ci/cm^3$

Maximum activity - Activity level expected during operation with fuel leak rate equivalent to reactor water activity concentration of 2.3 $\mu Ci/cm^3$ and design basis noble radiogas release rate of 100,000 $\mu Ci/sec$ (corrosion and fission products present)

Maximum volume and maximum activity are not concurrent.

For activity values: E-1 = number x 10^{-1}; E1 = number x 10^{1}
E-4 = number x 10^{-4}; E4 = number x 10^{4}

Fractional values on tables denote the number of items per occurrence divided by the number of days between each occurrence (i.e., 1/30 batches/day means one batch processed every 30 days).

Waste system input activities are based on a reactor water-to-steam partition coefficient of 1.0E-3.

Values for design purposes only. Actual flows and activities may differ.

Table 11.2-12

Radwaste Process Equipment Design Basis

Equipment	Radioactivity Decontamination Factor	
Deep bed demineralizers	Soluble	100
	Insoluble	50
Precoat filters	Soluble	1
	Insoluble	2
Evaporators (influent/distillate)[a]	1000	

[a] Concentration factor (influent/bottoms) = 0.6/32.

Table 11.2-13

Liquid Radwaste Equipment

	Material of Construction	Quantity	Size or Capacity	Remarks
EQUIPMENT DRAIN SUBSYSTEM				
Waste collector tank	Carbon steel	1	20,000 gal	Note 11
Waste collector pump	Stainless steel	1	190 gal/minute @ 155 ft TDH	
Waste sample tanks	Carbon steel	2	20,000 gal	Note 11
Waste sample pumps	Stainless steel	2	190 gal/minute @ 100 ft TDH	
Waste collector filter	Carbon steel	1	188 ft^2 of filter area	Notes 3, 4
Waste filter hold pump	Stainless steel	1	75 gal/minute @ 50 ft TDH	Note 3
Waste demineralizer	Carbon steel	1	65 ft^3 resin bed	Note 2
Waste surge tank	Carbon steel	1	75,000 gal	Note 11
Waste surge pump	Stainless steel	1	190 gal/minute @ 155 ft TDH	
FLOOR DRAIN SUBSYSTEM				
Floor drain collector tank	Carbon steel	1	20,000 gal	Note 11
Floor drain collector pump	Stainless steel	1	190 gal/minute @ 155 ft TDH	
Floor drain sample tank	Carbon steel	1	20,000 gal	Note 11
Floor drain sample pump	Stainless steel	1	190 gal/minute @ 100 ft TDH	
Floor drain filter	Carbon steel	1	188 ft^2 of filter area	Notes 3, 4
Floor drain filter hold pump	Stainless steel	1	75 gal/minute @ 50 ft TDH	Note 3
Floor drain demineralizer	Carbon steel	1	65 ft^3 resin bed	Note 2

Table 11.2-13

Liquid Radwaste Equipment (Continued)

	Material of Construction	Quantity	Size or Capacity	Remarks
CHEMICAL WASTE SUBSYSTEM				
Detergent drain tank	Carbon steel	2	1600 gal	Note 12
Detergent drain pumps	Stainless steel	2	27 gal/minute @ 90 ft TDH	
Detergent drain filter	Carbon steel	1	50 gal/minute	
Chemical waste tanks	Stainless steel	2	15,000 gal	Note 12
Chemical waste pumps	Stainless steel	2	285 gal/minute @ 200 ft TDH	
Concentrator feed pumps	Stainless steel	2	30 gal/minute @ 95 ft TDH	
Concentrator (evaporator)	Stainless steel	2	10 gal/minute	Note 6
Heating element	Stainless steel/carbon steel	2		Note 10
Concentrator recycle pump	Stainless steel	2	2300 gal/minute @ 40 ft TDH	Note 5
Concentrator condenser	Stainless steel/carbon steel	2		Note 7
Distillate tanks	Stainless steel	2	15,000 gal	Note 12
Distillate pumps	Stainless steel	2	142 gal/minute @ 230 ft TDH	
Polishing demineralizer	Carbon steel	1	65 ft^3 resin bed	Note 2
AUXILIARY EQUIPMENT				
Waste precoat tank	Carbon steel	1	210 gal	Notes 4, 8
Waste precoat pump	Ductile iron	1	325 gal/minute @ 50 ft TDH	Note 3
Waste filter aid tank	Carbon steel	1	630 gal	Notes 4, 8
Waste filter aid metering pump	Stainless steel	2	0 to 154 gal/hr @ 145 psi	
Resin addition tank	Stainless steel	1	1200 gal	Note 12
Chemical addition tanks	Carbon steel	2	200 gal	Note 9
Chemical addition pumps	Carpenter 20 steel	2	79 gal/hr	

Table 11.2-13

Liquid Radwaste Equipment (Continued)

NOTES:

1.	Unless otherwise noted, the design pressure of the equipment is 150 psig, design temperature is 150°F.
2.	Vessel is rubber lined, internals are stainless steel.
3.	Design temperature is 220°F.
4.	Vessel is lined with phenolic/epoxy cross linked, alkaline polymerized coating, internals are stainless steel.
5.	Design temperature is 300°F.
6.	Design pressure is 30 psig, design temperature is 274°F.
7.	Design flow is: coolant inlet-750 gpm @ 105°F, outlet -120°F; design pressure: shell side - 30 psig, tube side - 150 psig; design temperature - 274°F.
8.	Design pressure is atmospheric.
9.	Vessel is lined with acid-resistant high-bake phenolic coating.
10.	Design flows: shell side - 6300 lb/hr @ 274°F, tube side - 2300 gpm @ 218°F; design pressures: shell side - 60 psig, tube side - 75 psig; design temperature - 307°F.
11.	Tank design pressure is atmospheric plus static head pressure, design temperature is 140°F.
12.	Tank design pressure is atmospheric plus static head pressure, design temperature is 150°F.

Table 11.2-14

Annual Releases of Radioactive Material as Liquid

Nuclide	Half-Life (days)	Concentration in Primary Coolant (μCi/ml)	High Purity (Ci)	Low Purity (Ci)	Total Laws (Ci)	Adjusted Total (Ci/yr)[a]	Total (Ci/yr)
CORROSION AND ACTIVATION PRODUCTS							
^{24}Na	6.25E-1	8.79E-3	0.00036	0.00043	0.00079	0.00688	0.00692
^{32}P	1.43E1	2.06E-4	0.00001	0.00002	0.00003	0.00027	0.00027
^{51}Cr	2.78E1	5.14E-3	0.00027	0.00052	0.00081	0.00704	0.00703
^{54}Mn	3.03E2	6.18E-5	0.00000	0.00001	0.00001	0.00008	0.00008
^{56}Mn	1.07E-1	4.28E-2	0.00052	0.00034	0.00085	0.00747	0.00745
^{55}Fe	9.50E2	1.03E-3	0.00005	0.00010	0.00017	0.00143	0.00147
^{59}Fe	4.50E1	3.08E-5	0.00000	0.00000	0.00000	0.00004	0.00004
^{58}Co	7.13E1	2.06E-4	0.00001	0.00002	0.00003	0.00028	0.00028
^{60}Co	1.92E2	4.12E-0	0.00002	0.00004	0.00006	0.00058	0.00058
^{65}Ni	1.07E-1	2.57E-4	0.00000	0.00000	0.00000	0.00004	0.00004
^{64}Cu	5.33E-1	2.91E-2	0.00111	0.00128	0.00239	0.02098	0.02098
^{65}Zn	2.45E2	2.06E-4	0.00001	0.00002	0.00003	0.00028	0.00028
^{69m}Zn	5.57E-1	1.94E-3	0.00007	0.00009	0.00017	0.00146	0.00147
^{69}Zn	3.96E-2	0.0	0.00008	0.00009	0.00018	0.00153	0.00157
^{187}W	9.96E-1	2.98E-4	0.00001	0.00002	0.00003	0.00028	0.00028
^{239}Np	2.35E0	7.10E-3	0.00036	0.00060	0.00094	0.00831	0.00790
FISSION PRODUCTS							
^{83}Br	1.00E-1	2.42E-3	0.00003	0.00002	0.00004	0.00039	0.00039
^{84}Br	2.21E-2	3.67E-3	0.00000	0.00000	0.00000	0.00003	0.00003
^{89}Rb	1.07E-2	3.60E-3	0.00001	0.00002	0.00002	0.00022	0.00022
^{89}Sr	5.20E1	1.03E-5	0.00001	0.00001	0.00002	0.00015	0.00015
^{91}Sr	4.03E-1	3.82E-3	0.00013	0.00014	0.00026	0.00233	0.00231

Table 11.2-14

Annual Releases of Radioactive Material as Liquid (Continued)

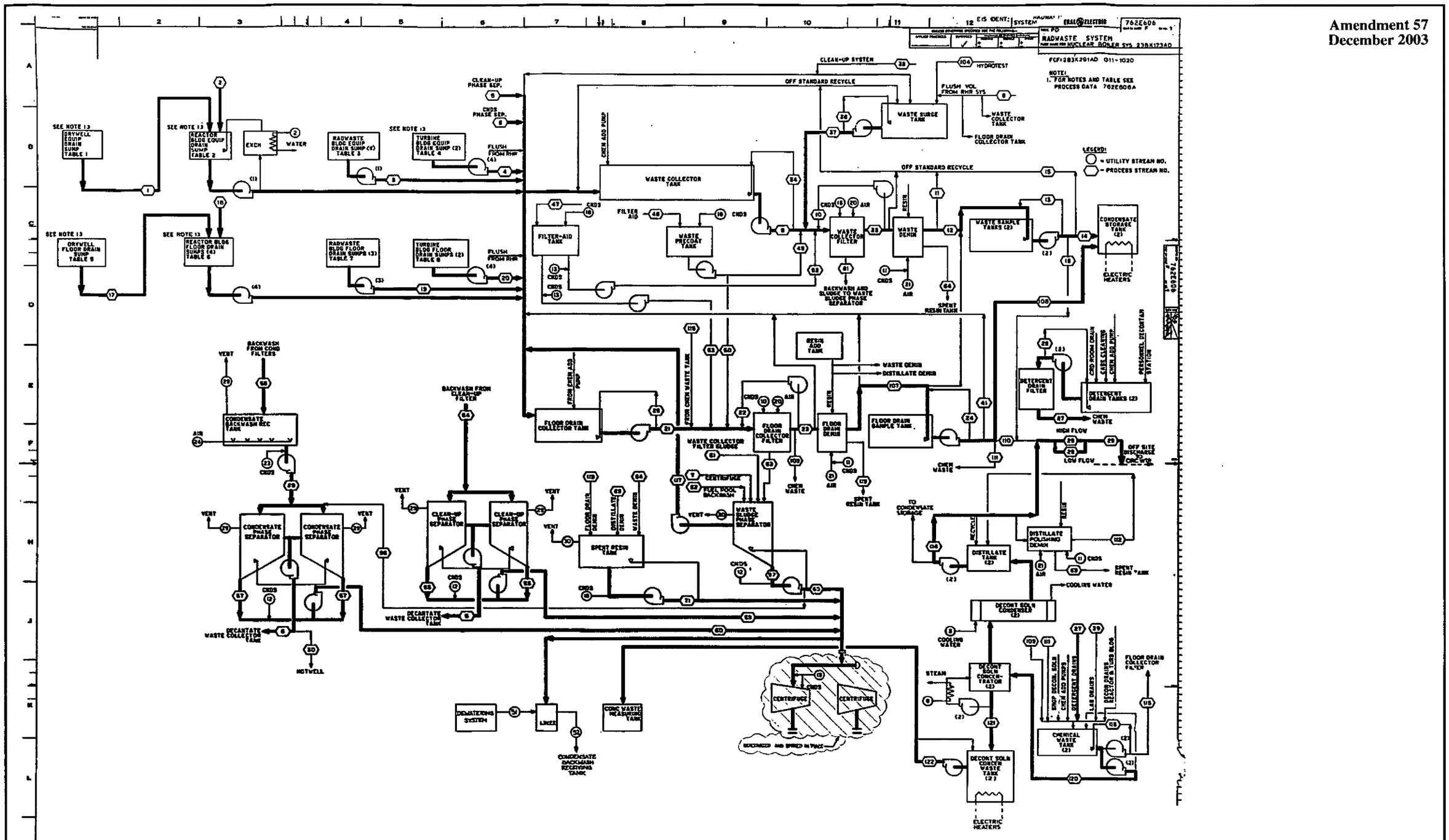
Nuclide	Half-Life (days)	Concentration in Primary Coolant (μCi/ml)	High Purity (Ci)	Low Purity (Ci)	Total Laws (Ci)	Adjusted Total (Ci/yr)[a]	Total (Ci/yr)
^{91m}Y	3.47E-2	0.0	0.00008	0.00008	0.00017	0.00146	0.00147
^{91}Y	5.88E1	4.12E-5	0.00000	0.00001	0.00001	0.00007	0.00007
^{92}Sr	1.13E-1	8.60E-3	0.00012	0.00007	0.00018	0.00159	0.00157
^{92}Y	1.47E-1	5.29E-3	0.00021	0.00016	0.00038	0.00328	0.00325
^{93}Y	4.25E-1	3.83E-3	0.00014	0.00014	0.00027	0.00241	0.00241
^{98}Nb	3.54E-2	3.11E-3	0.00001	0.00000	0.00001	0.00008	0.00008
^{99}Mo	2.798E0	2.04E-3	0.00010	0.00018	0.00028	0.00244	0.00241
^{99m}Tc	2.5E-1	1.85E-2	0.00054	0.00054	0.00107	0.00893	0.00937
^{101}Tc	9.72E-3	6.56E-2	0.00000	0.00000	0.00000	0.00002	0.00002
^{103}Ru	3.96E1	2.06E-5	0.00000	0.00000	0.00000	0.00003	0.00003
^{103m}Rh	3.96E-2	0.0	0.00000	0.00000	0.00000	0.00003	0.00003
^{104}Tc	1.25E-2	5.87E-2	0.00000	0.00000	0.00001	0.00006	0.00006
^{105}Ru	1.85E-1	1.80E-3	0.00004	0.00003	0.00006	0.00055	0.00058
^{105m}Rh	5.21E-4	0.0	0.00004	0.00003	0.00006	0.00056	0.00059
^{105}Rh	1.50E0	0.0	0.00001	0.00001	0.00002	0.00018	0.00019
^{129m}Te	3.40E1	4.11E-5	0.00000	0.00000	0.00001	0.00005	0.00005
^{129}Te	4.79E-2	0.0	0.00000	0.00000	0.00000	0.00003	0.00003
^{131m}Te	1.25E0	1.00E-4	0.00000	0.00001	0.00001	0.00010	0.00010
^{131}Te	1.74E-2	0.0	0.00000	0.00000	0.00000	0.00002	0.00002
^{131}I	8.05E0	5.11E-3	0.00027	0.00049	0.00077	0.00643	0.00671
^{132}Te	3.25E0	1.02E-5	0.00000	0.00000	0.00000	0.00001	0.00001
^{132}I	9.58E-2	2.41E-2	0.00026	0.00016	0.00041	0.00345	0.00367
^{133}I	8.75E-1	1.93E-2	0.00084	0.00115	0.00199	0.01667	0.01783
^{134}I	3.67E-2	5.26E-2	0.00010	0.00006	0.00017	0.00144	0.00147

Table 11.2-14

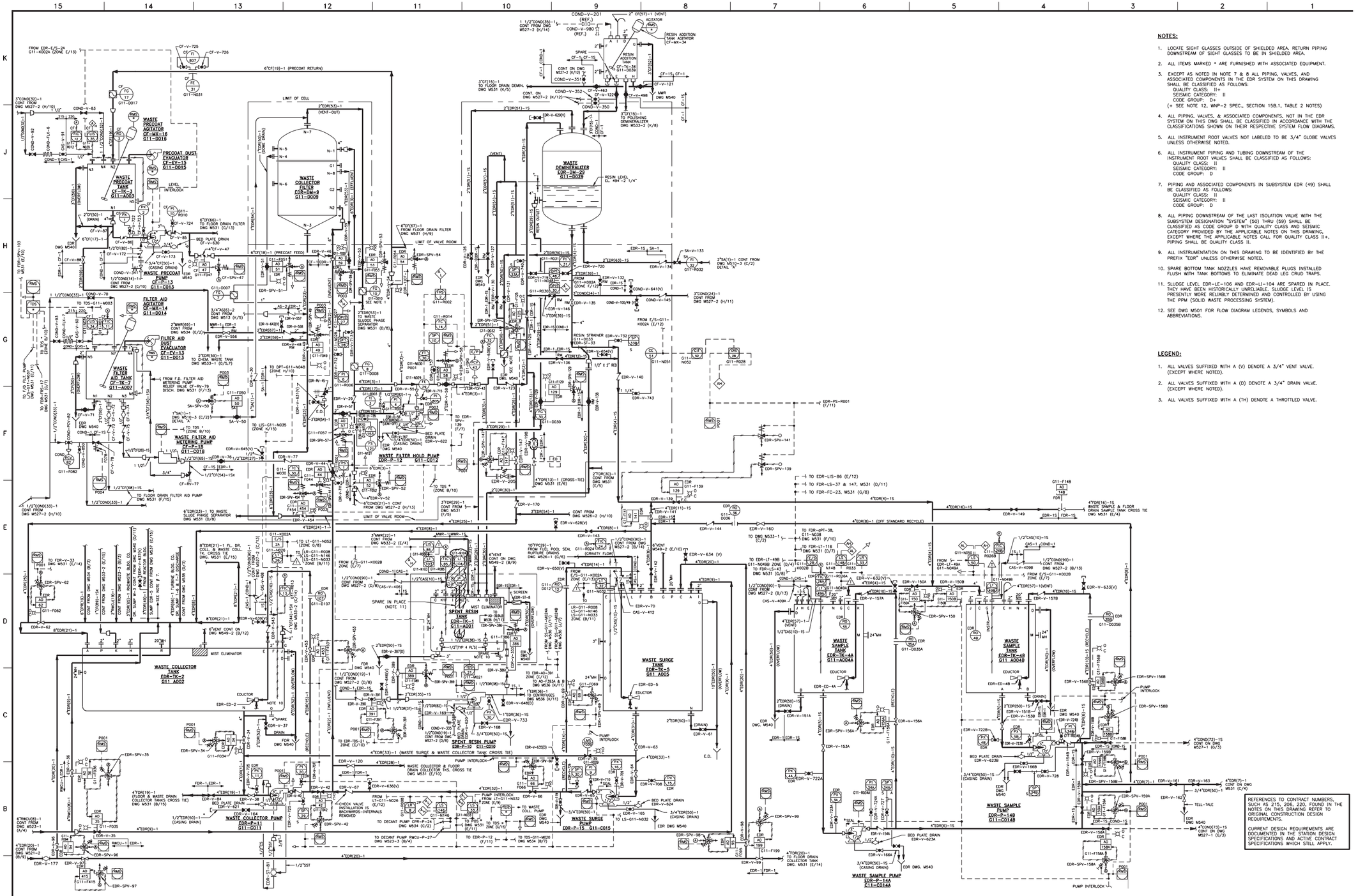
Annual Releases of Radioactive Material as Liquid (Continued)

Nuclide	Half-Life (days)	Concentration in Primary Coolant (μCi/ml)	High Purity (Ci)	Low Purity (Ci)	Total Laws (Ci)	Adjusted Total (Ci/yr)[a]	Total (Ci/yr)
^{134}Cs	7.49E2	3.09E-5	0.00008	0.00081	0.00089	0.00741	0.00776
^{135}I	2.79E-1	1.77E-2	0.00050	0.00044	0.00094	0.00788	0.00829
^{136}Cs	1.30E1	2.05E-5	0.00005	0.00051	0.00057	0.00473	0.00493
^{137}Cs	1.10E4	7.21E-5	0.00020	0.00188	0.00206	0.01730	0.01783
^{137M} Ba	1.77E-3	0.0	0.00018	0.00175	0.00193	0.01618	0.01678
^{138}Cs	2.24E-2	7.34E-3	0.00022	0.00065	0.00087	0.00724	0.00755
^{139}Ba	5.76E-2	8.09E-3	0.00004	0.00002	0.00006	0.00053	0.00056
^{140}Ba	1.28E1	4.11E-4	0.00002	0.00004	0.00006	0.00053	0.00056
^{140}La	1.67E0	0.0	0.00000	0.00001	0.00001	0.00011	0.00011
^{141}La	1.62E-1	0.0	0.00001	0.00001	0.00002	0.00017	0.00018
^{141}Ce	3.24E1	3.08E-5	0.00000	0.00000	0.00000	0.00004	0.00004
^{142}La	6.39E-2	4.08E-3	0.00003	0.00002	0.00004	0.00036	0.00038
^{143}Ce	1.38E0	3.01E-5	0.00000	0.00000	0.00000	0.00003	0.00003
^{143}Pr	1.37E1	4.11E-5	0.00000	0.00000	0.00001	0.00005	0.00005
All others	1.32E-2	1.38E-2	0.00000	0.00001	0.00007	0.00007	
Total (except tritium)		4.31E-01	0.00719	0.01307	0.02026	0.17962	0.17834
Total release						13.0	13.0

[a] Adjusted total includes an additional 0.15 ci/yr with the same isotopic distribution as the calculated source term to account for anticipated occurrences such as operator errors resulting in unplanned releases.



Columbia Generating Station Final Safety Analysis Report	Radwaste System Process Diagram		
	Draw. No. 2G11-04,6	Rev. 8	Figure 11.2-1


WASTE PRECOAT AGITATOR CF-MC-16 G11-D016
PRECOAT DUST EVACUATOR CF-EV-15 G11-D015
WASTE PRECOAT TANK CF-TK-3 G11-A003
LEVEL INTERLOCK
WASTE PRECOAT PUMP CF-P-13 G11-C013
FILTER AID AGITATOR CF-MC-14 G11-D014
FILTER AID DUST EVACUATOR CF-EV-13 G11-D013
WASTE FILTER AID TANK CF-TK-7 G11-A007
WASTE FILTER AID METERING PUMP CF-P-15 G11-C018
WASTE COLLECTOR FILTER EDR-DM-9 G11-D009
WASTE DEMINERALIZER EDR-DM-29 G11-D029
RESIN LEVEL EL. 484'-2 1/4"
SPENT RESIN TANK EDR-TK-1 G11-A001
WASTE COLLECTOR TANK EDR-TK-2 G11 A002
WASTE SURGE TANK EDR-TK-5 G11 A005
WASTE SAMPLE TANK EDR-TK-4A G11-A004A
WASTE SAMPLE TANK EDR-TK-4B G11 A004B
WASTE SAMPLE PUMP EDR-P-14B G11-C014B
WASTE SAMPLE PUMP EDR-P-14A G11-C014A
WASTE COLLECTOR PUMP EDR-P-11 G11-C011
SPENT RESIN PUMP EDR-P-10 G11-C010
WASTE FILTER HOLD PUMP EDR-P-12 G11-C012
WASTE SURGE PUMP EDR-P-15 G11-C015
MIST ELIMINATOR
BED PLATE DRAIN CF-V-630
PUMP INTERLOCK
LIMIT OF CELL
LIMIT OF VALVE ROOM
SPARE
SPARE IN PLACE (NOTE 11)
TO FDR-LIS-86 (E/12)
TO FDR-LS-37 & 147, M531 (D/11)
TO FDR-FC-23, M531 (G/8)
4"EDR(8)-1 (OFF STANDARD RECYCLE)
TO DWG M533-1 (C/2)
TO FDR-dPT-38, G11-N038 DWG M531 (F/10)
TO FDR-LT-118 DWG M531 (D/7)
TO FDR-LT-49B & G11-N049B ZONE (D/4)
TO FDR-LS-43 DWG M531 (D/8)
TO FDR-dPT-140 G11-N140 DWG M531 (D/6)
TO FDR-LT-49A & G11-N049A ZONE (D/4)
TO FDR-LT-49C & G11-N049C ZONE (D/4)
TO FDR-LS-148 G11-N148 DWG M531 (D/6)
TO FDR-LS-149 G11-N149 DWG M531 (D/6)
TO FDR-LS-150 G11-N150 DWG M531 (D/6)
TO FDR-LS-151 G11-N151 DWG M531 (D/6)
TO FDR-LS-152 G11-N152 DWG M531 (D/6)
TO FDR-LS-153 G11-N153 DWG M531 (D/6)
TO

NOTES:

1. LOCATE SIGHT GLASSES OUTSIDE OF SHIELDED AREA. RETURN PIPING DOWN STREAM OF SIGHT GLASSES TO BE IN SHIELDED AREA.
2. ALL ITEMS MARKED * ARE FURNISHED WITH ASSOCIATED EQUIPMENT.
3. ALL INSTRUMENT ROOT VALVES NOT LABELED WILL BE 3/4" GLOBE VALVES UNLESS SPECIFICALLY NOTED OTHERWISE.
4. EXCEPT AS NOTED IN NOTE 7 & 8 ALL PIPING, VALVES, AND ASSOCIATED COMPONENTS IN THE FDR SYSTEM ON THIS DRAWING SHALL BE CLASSIFIED AS FOLLOWS: QUALITY CLASS II+ SEISMIC CATEGORY II CODE GROUP D+
5. ALL PIPING, VALVES, AND ASSOCIATED COMPONENTS NOT IN THE FDR SYSTEM ON THIS DWG. SHALL BE CLASSIFIED IN ACCORDANCE WITH THE CLASSIFICATIONS SHOWN ON THEIR RESPECTIVE SYSTEM FLOW DIAGRAMS.
6. ALL INSTRUMENT PIPING AND TUBING DOWNSTREAM OF THE INSTRUMENT ROOT VALVES SHALL BE CLASSIFIED AS FOLLOWS: QUALITY CLASS II SEISMIC CATEGORY II CODE GROUP D
7. PIPING AND ASSOCIATED COMPONENTS IN SUBSYSTEMS FDR(40) & FDR(41) AND FDR(7) DOWNSTREAM OF FDR-V-187 AND FDR-V-188 SHALL BE CLASSIFIED AS FOLLOWS: QUALITY CLASS II SEISMIC CATEGORY II CODE GROUP D
8. ALL PIPING DOWNSTREAM OF THE LAST ISOLATION VALVE AND OPEN TO THE ATMOSPHERE WITH THE SUBSYSTEM DESIGNATION "SYSTEM"(50) THRU (59) SHALL BE CLASSIFIED AS CODE GROUP D WITH QUALITY CLASS AND SEISMIC CATEGORY PROVIDED BY THE APPLICABLE NOTES ON THIS DRAWING, EXCEPT WHERE THE APPLICABLE NOTES CALL FOR QUALITY CLASS II+, PIPING SHALL BE QUALITY CLASS II.
9. ALL INSTRUMENTATION ON THIS DRAWING TO BE IDENTIFIED BY THE PREFIX "FDR" UNLESS OTHERWISE NOTED. (+SEE NOTE 12, WNP-2 SPEC, SECTION 15B.1, TABLE 2 NOTES.)
10. SPARE BOTTOM TANK NOZZLES HAVE REMOVABLE PLUGS INSTALLED FLUSH WITH TANK BOTTOMS TO ELIMINATE DEAD LEG CRUD TRAPS.
11. FDR-PS-120 CONTACTS JUMPERED CLOSED TO INDICATE NORMAL SUCTION PRESSURE.
12. SLUDGE LEVEL FDR-LE-107 AND FDR-LS-109 ARE SPARED IN PLACE. THEY HAVE BEEN HISTORICALLY UNRELIABLE. SLUDGE LEVEL IS PRESENTLY MORE RELIABLY DETERMINED AND CONTROLLED BY USING THE PPM (SOLID WASTE PROCESSING SYSTEM).
13. SEE DWG M501 FOR FLOW DIAGRAM LEGENDS, SYMBOLS AND ABBREVIATIONS.

LEGEND

1. ALL VALVES SUFFIXED WITH A (V) DENOTE A 3/4" VENT VALVE.
2. ALL VALVES SUFFIXED WITH A (D) DENOTE A 3/4" DRAIN VALVE.
3. ALL VALVES SUFFIXED WITH A (TH) DENOTE A THROTTLED VALVE.

REFERENCES TO CONTRACT NUMBERS, SUCH AS 215, 206, 220, FOUND IN THE NOTES ON THIS DRAWING REFER TO ORIGINAL CONSTRUCTION DESIGN REQUIREMENTS.

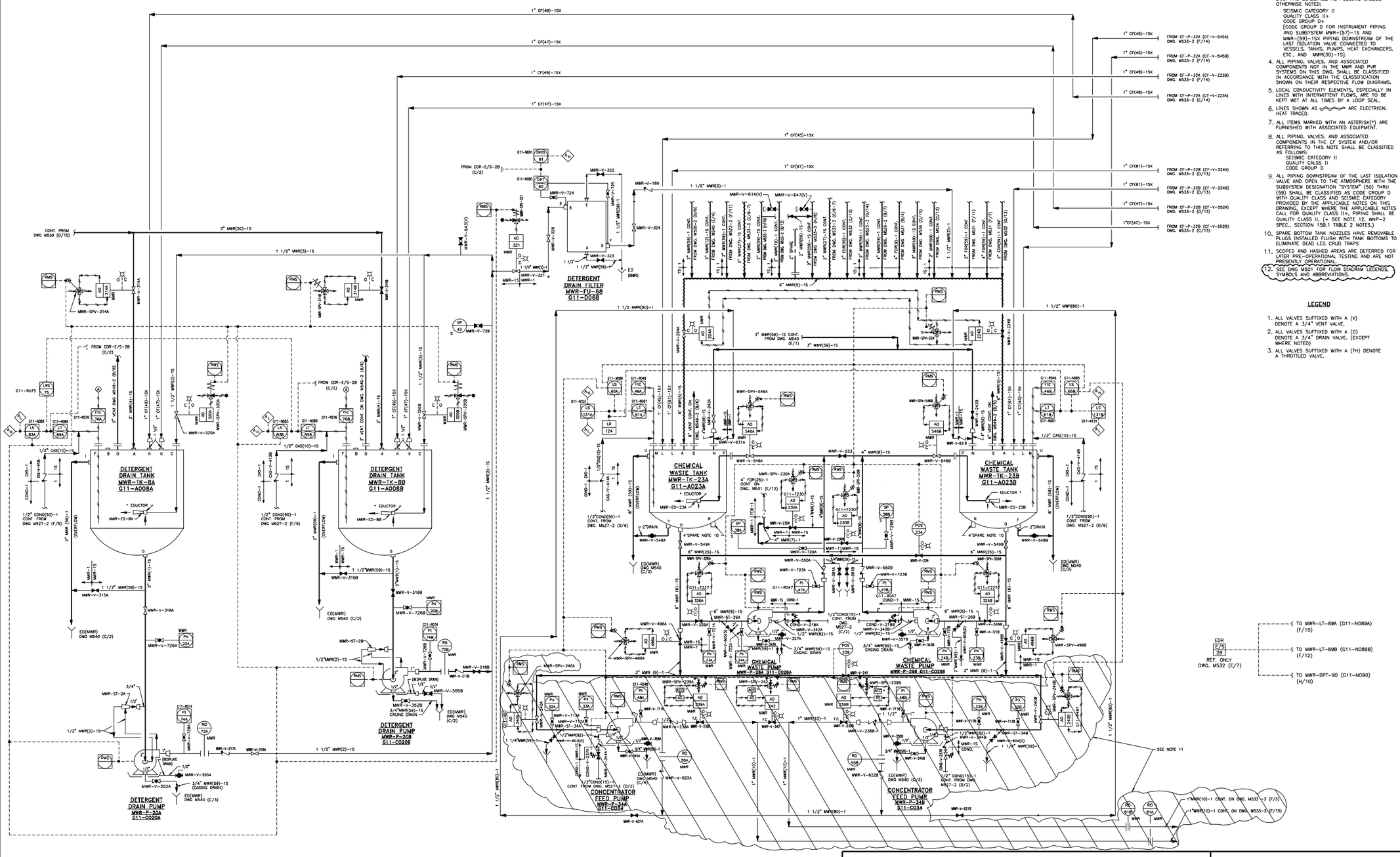
CURRENT DESIGN REQUIREMENTS ARE DOCUMENTED IN THE STATION DESIGN SPECIFICATIONS AND ACTIVE CONTRACT SPECIFICATIONS WHICH STILL APPLY.

FSAR FIG.

Columbia Generating Station
Final Safety Analysis Report

Flow Diagram Radioactive Waste System
Floor Drain Processing

Draw. No. M531	Rev. 80	Figure 11.2-3



NOTES:

1. LOCATE SIGHT GLASSES OUTSIDE OF SHIELDED AREA. RETURN PIPING DOWNSTREAM OF SIGHT GLASSES TO BE IN SHIELDED AREA.
2. ALL INSTRUMENT ROOT VALVES NOT LABELED WILL BE 3/4" GLOBE VALVES UNLESS SPECIFITALLY NOTED OTHERWISE.
3. ALL PIPING VALVES AND ASSOCIATED COMPONENTS IN THE MWR AND PVR SYSTEMS SHOWN ON THIS DWG. ARE CLASSIFIED AS FOLLOWS UNLESS OTHERWISE NOTED:
   SEISMIC CATEGORY II
   QUALITY CLASS II+
   CODE GROUP D+
   [CODE GROUP D FOR INSTRUMENT PIPING AND SUBSYSTEM MWR-(57)-1S AND MWR-(59)-1SX PIPING DOWNSTREAM OF THE LAST ISOLATION VALVE CONNECTED TO VESSELS, TANKS, PUMPS, HEAT EXCHANGERS, ETC., AND MWR(30)-1S].
4. ALL PIPING, VALVES, AND ASSOCIATED COMPONENTS NOT IN THE MWR AND PVR SYSTEMS ON THIS DWG. SHALL BE CLASSIFIED IN ACCORDANCE WITH THE CLASSIFICATION SHOWN ON THEIR RESPECTIVE FLOW DIAGRAMS.
5. LOCAL CONDUCTIVITY ELEMENTS, ESPECIALLY IN LINES WITH INTERMITTENT FLOWS, ARE TO BE KEPT WET AT ALL TIMES BY A LOOP SEAL.
6. LINES SHOWN AS ~~~~ ARE ELECTRICAL HEAT TRACED.
7. ALL ITEMS MARKED WITH AN ASTERISK(*) ARE FURNISHED WITH ASSOCIATED EQUIPMENT.
8. ALL PIPING, VALVES, AND ASSOCIATED COMPONENTS IN THE CF SYSTEM AND/OR REFERRING TO THIS NOTE SHALL BE CLASSIFIED AS FOLLOWS:
   SEISMIC CATEGORY II
   QUALITY CALSS II
   CODE GROUP D
9. ALL PIPING DOWNSTREAM OF THE LAST ISOLATION VALVE AND OPEN TO THE ATMOSPHERE WITH THE SUBSYSTEM DESIGNATION "SYSTEM" (50) THRU (59) SHALL BE CLASSIFIED AS CODE GROUP D WITH QUALITY CLASS AND SEISMIC CATEGORY PROVIDED BY THE APPLICABLE NOTES ON THIS DRAWING, EXCEPT WHERE THE APPLICABLE NOTES CALL FOR QUALITY CLASS II+, PIPING SHALL BE QUALITY CLASS II, (+ SEE NOTE 12, MWP-2 SPEC., SECTION 15B.1 TABLE 2 NOTES.)
10. SPARE BOTTOM TANK NOZZLES HAVE REMOVABLE PLUGS INSTALLED FLUSH WITH TANK BOTTOMS TO ELIMINATE DEAD LEG CRUD TRAPS.
11. SCOPED AND HASHED AREAS ARE DEFERRED FOR LATER PRE-OPERATIONAL TESTING AND ARE NOT PRESENTLY OPERATIONAL.
12. SEE DWG M501 FOR FLOW DIAGRAM LEGENDS, SYMBOLS AND ABBREVIATIONS.

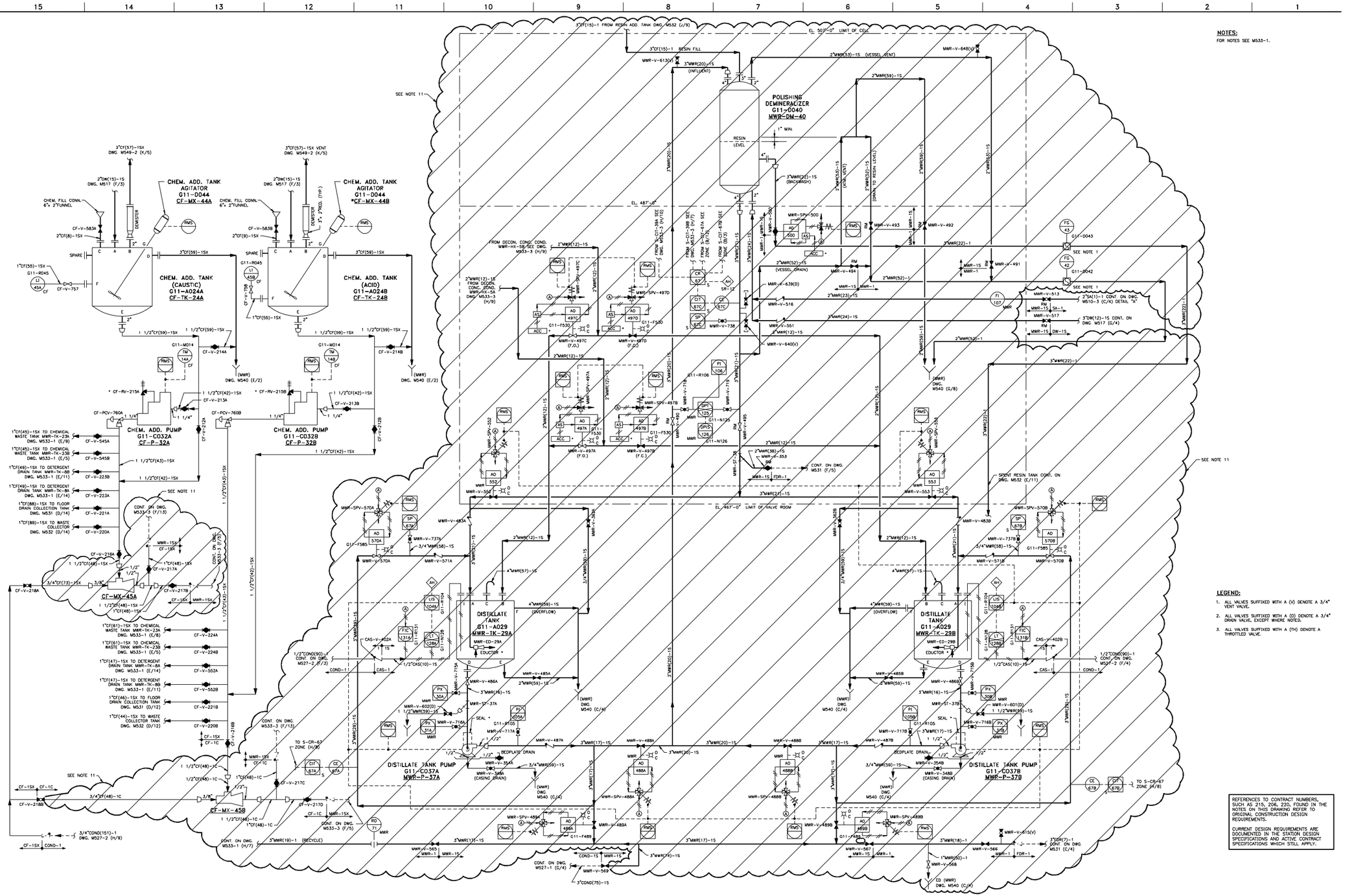
LEGEND

1. ALL VALVES SUFFIXED WITH A (V) DENOTE A 3/4" VENT VALVE.
2. ALL VALVES SUFFIXED WITH A (D) DENOTE A 3/4" DRAIN VALVE. (EXCEPT WHERE NOTED)
3. ALL VALVES SUFFIXED WITH A (Th) DENOTE A THROTTLED VALVE.

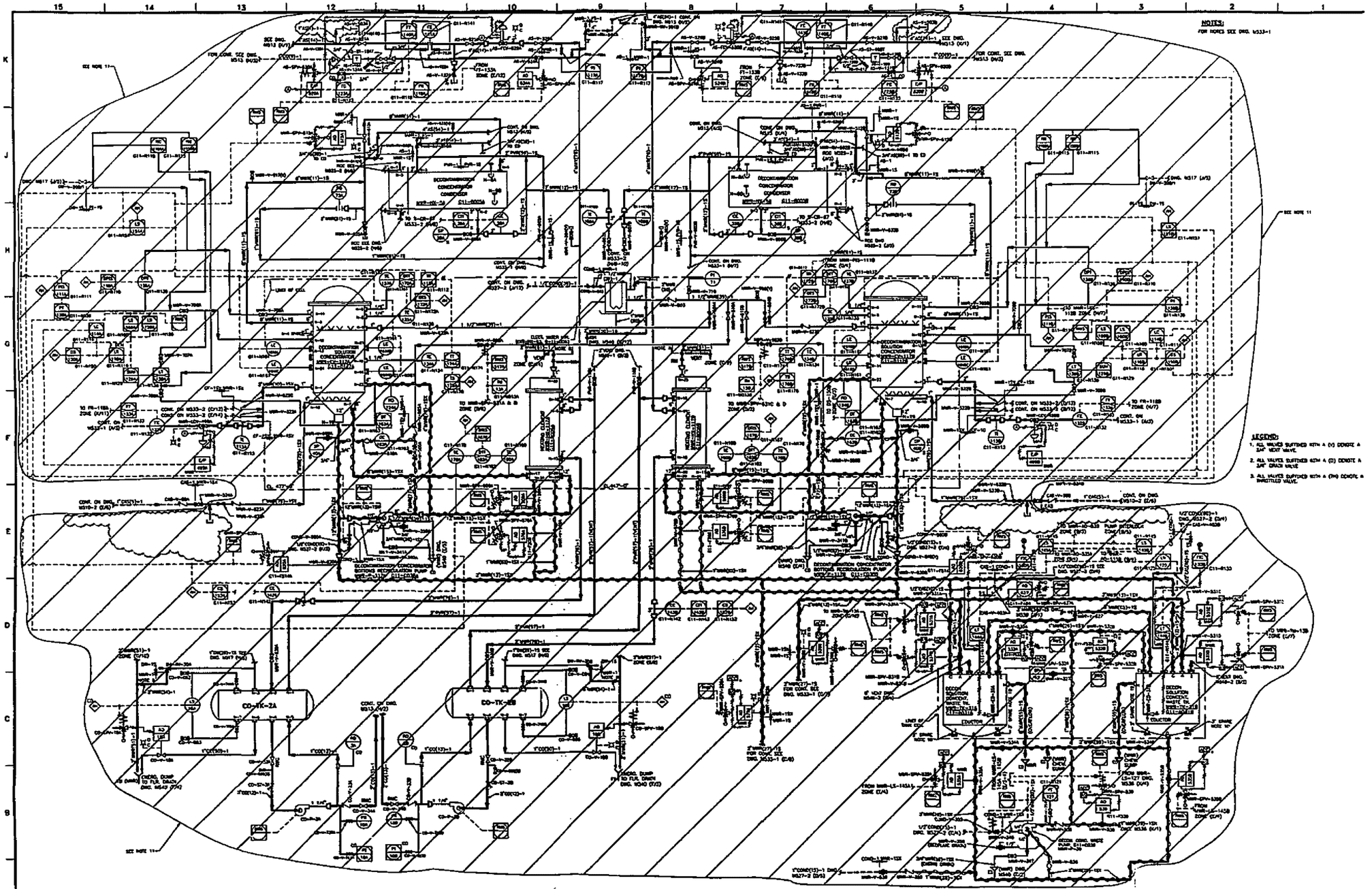
Columbia Generating Station
Final Safety Analysis Report

Flow Diagram Chemical Waste Processing

Draw. No. M533-1	Rev. 24	Figure 11.2-4.1



Columbia Generating Station **Final Safety Analysis Report**	**Flow Diagram Chemical Waste Processing**	
	Draw. No. M533-2	Rev. 22 / Figure 11.2-4.2



Columbia Generating Station
Final Safety Analysis Report

Flow Diagram Chemical Waste Processing

Draw. No.	M533-3	Rev.	29	Figure	11.2-4.3

11.3 GASEOUS WASTE MANAGEMENT SYSTEMS

11.3.1 DESIGN BASES

The objective of the gaseous waste management system is to process and control the release of gaseous radioactive effluents to the site environs so as to maintain as low as reasonably achievable the exposure of persons in unrestricted areas to radioactive gaseous effluents, during normal and anticipated operational occurrences. This is to be accomplished while maintaining occupational exposure as low as is reasonably achievable (ALARA) and without limiting plant operation or availability.

The gaseous waste management systems are designed to limit the dose to offsite persons from routine station releases to significantly less than the limits specified in 10 CFR 20 and to operate within the emission rate limits established in the Technical Specifications.

To evaluate the offgas system design compliance with the limits of 10 CFR 20, an annual average noble radiogas source term (based on 30-minute decay) of 100,000 µCi/sec of the "1971 Mixture" as discussed in Section 11.1 is used. The noble radiogas effluent release rate from the charcoal adsorbers is about 49-59 µCi/sec based upon 30 scfm air inleakage and injection. The isotopic composition is given in Table 11.3-1 in units of µCi/sec and Ci/yr.

To evaluate that the annual average exposure at the site boundary during normal operation and anticipated operational occurrences from gaseous effluents does not exceed the dose objectives of Appendix I to 10 CFR 50, an average source term release rate derived from field measurements (1993-2003) is used.

Equipment and components used to collect, process, or store gaseous radioactive waste are not designed as Seismic Category I. To evaluate that equipment failure will not result in offsite whole body doses exceeding 0.5 rem in two hours, the Technical Specification release rate limit was used, consistent with the guidance presented in Reference 11.3-7.

The gaseous radwaste equipment is selected, arranged, and shielded to maintain occupational exposure ALARA. The system was designed prior to the issuance of Regulatory Guide 8.8. However, the system incorporates substantially the guidance provided in this regulatory guide. The gaseous effluent treatment system conforms to the requirements of General Design Criteria 60 and 64 as specified in Section 3.1.

A list of the major offgas system components and design features is provided in Table 11.3-2. Equipment and piping is designed and constructed in accordance with the requirements of the applicable codes as given in Tables 3.2-1 and 3.2-2.

The quality group classifications of the various systems are shown in Table 3.2-1. Seismic category, safety class, quality assurance requirements, and principal construction codes

information are contained in Section 3.2. The system is designed to Quality Group Classification D+.

11.3.2 SYSTEM DESCRIPTION

11.3.2.1 Main Condenser Steam Jet Air Ejector RECHAR System

The offgas from the main condenser steam jet air ejector (SJAE) is treated by means of a system utilizing catalytic recombination and charcoal adsorption (RECHAR system) (see Figure 11.3-1). Descriptions of the major process components including design temperature and pressure are given in Table 11.3-2 and in the following.

Noncondensable radioactive offgas is continuously removed from the main condenser by the air ejector during plant operation. The air ejector offgas will normally contain activation gases, principally ^{16}N, ^{19}O, and ^{13}N. The ^{16}N and ^{19}O have short half-lives and are readily decayed. ^{13}N with a 10-minute half-life is present in small amounts that are further reduced by decay. Activation gas source terms are presented in Table 11.1-4.

The air ejector offgas will also contain radioactive noble gases including parents of biologically significant ^{89}Sr, ^{90}Sr, ^{140}Ba, and ^{137}Cs. The concentration of these noble gases depends on the amount of tramp uranium in the coolant and on the cladding surfaces (usually extremely small) and the number and size of fuel cladding defects.

A main condenser offgas treatment system has been incorporated in the plant design to reduce the radioactive gaseous effluents from the station. The offgas system uses a catalytic recombiner to recombine hydrogen and oxygen. After cooling (to approximately 130°F) to strip the condensables and reduce the volume, the remaining noncondensables (principally air with traces of krypton and xenon) are delayed in a holdup line. The gas is cooled to 45°F and processed through a HEPA filter. The gas is then passed through a desiccant dryer that reduces the dewpoint to approximately -90°F. Charcoal adsorption beds selectively adsorb and delay the xenon and krypton from the bulk carrier gas (principally dry air). With an air inleakage of 30 scfm, this treatment system results in a delay of 15 hr for krypton and 9.5 days for xenon. After the delay, the gas is again passed through a HEPA filter and discharged to the environment through the reactor building elevated release duct.

Figure 11.3-1 is the process flow diagram for the system. The process data for startup and normal operating conditions are submitted as proprietary data under separate cover as Table 11.3-3. The information supporting the process data is presented in Reference 11.3-2. The system is mechanically capable of processing three times the source terms of Table 11.3-1 without affecting delay time of the noble gases.

Table 11.3-1 also lists isotopic activities at the discharge of the system, from which the decontamination factor for each noble gas isotope can be determined.

The flow diagram is shown in Figure 11.3-2. The main process routing is indicated by a heavy line.

The basis for sizing the recombiner is to maintain the hydrogen concentration below 4% (including steam) at the inlet and below 1% at the outlet on a dry basis. The exit hydrogen concentration is normally well below the 1% maximum allowed. The hydrogen generation rate of the reactor is based on data from nine boiling water reactors (BWR). The hydrogen generation rate is given in Table 11.3-3.

The krypton and xenon holdup time is closely approximated by the following equation:

$$T = \frac{K_D M}{V}$$

where:

T = holdup time of a given gas, seconds.

K_D = dynamic adsorption coefficient for the given gas, $\frac{cm^3}{gram}$

M = weight of charcoal, grams

V = flow rate of the carrier gas, $\frac{cm^3}{sec}$

Dynamic adsorption coefficient values for xenon and krypton used to determine gaseous effluent releases are discussed in Reference 11.3-1. Moisture has a detrimental effect on adsorption coefficients. To prevent moisture from reaching the charcoal, fully redundant, adsorbent (-90°F dewpoint) air dryers are supplied. There are redundant moisture analyzers that will alarm on breakthrough of the dryer beds; however, breakthrough is not expected since the dryer beds will be regenerated on a time basis. The system is slightly pressurized, which together with very stringent leak rate requirements, prevents leakage of moist air into the charcoal.

After the hydrogen and oxygen are removed by the recombiner, the remaining carrier gas is primarily nitrogen with a small percentage of oxygen due to the air inleakage from the main condenser and air injection from the HWC system. Analyses have shown that the system can maintain condenser vacuum at combined air injection and air inleakage rates as high as 93 scfm. Reference 11.3-3, Par. S1 (c) (2) indicates that with certain conditions of stable operation and suitable construction, noncondensables (not including radiological decomposition products) should not exceed 6 scfm per condenser shell for large condensers. Dilution air is not added to the system unless the air inleakage is less than 6 scfm. In that event, 6 scfm is

added to provide for dilution of residual hydrogen from the recombiner. An initial bleed of oil-free air is added on startup until the recombiner reaches operating temperature.

The charcoal adsorbers design process flow sheet is for an ambient temperature. Operation at ambient temperature is sufficient to reduce gaseous radioactivity levels to a fraction of that allowed by 10 CFR 50 Appendix I. The decay heat is sufficiently small that, even in the no-flow condition, there is no significant loss of adsorbed noble gases due to temperature rise in the adsorbers. The adsorbers are located in a shielded room and maintained at ambient room temperature. A radiation monitor is provided to monitor the radiation level in the air handling room of the charcoal bed vault for high gas activity from a system leak. High radiation will cause an alarm in the control room.

Channeling in the charcoal adsorbers is prevented by supplying an effective flow distributor on the inlet, having long columns and having a high bed-to-particle diameter ratio of approximately 500. Underhill has stated that channeling or wall effects may reduce efficiency of the holdup bed if this ratio is not greater than 12 (Reference 11.3-4). During transfer of the charcoal into the charcoal adsorber vessels, radial sizing of the charcoal will be minimized by pouring the charcoal (by gravity or pneumatically) over a cone or other instrument to spread the granules over the surface.

A valve is provided to bypass the charcoal adsorbers. The main purpose of this bypass is to protect the charcoal during preoperational and startup testing when gas activity is zero or very low.

It may be desirable to use the bypass for short periods during startup or normal operations. This bypass mode would not be used for normal operation unless some unforeseen system malfunction would necessitate shutting down the power plant or operating in the bypass mode and remaining within the technical specification radioactivity release limits. The radioactivity released is controlled by a process monitor upstream of the vent isolation valve that will cause the bypass valve to close on a high radiation alarm. This interlock can be defeated only by a key lock switch. In addition, there is a high-high-high alarm setpoint on the same monitor that will cause isolation of the offgas system if established release rate limits are reached.

Leakage of radioactive gases from the system is limited by welding piping connections where possible and using bellows stem seals or equivalent leakage control valving. The system operates at a maximum of 7 psig during startup and less than 2 psig during normal operation so that the differential pressure to cause leakage is small.

Hydrogen concentration of gases from the air ejector is kept below the flammable limit by maintaining adequate process steam flow for dilution at all times. This steam flow rate is monitored and alarmed in the main control room.

Two parallel independent hydrogen analyzers are used to measure the hydrogen content of the offgas process flow downstream of the offgas condenser. The hydrogen concentration percentage output from the analyzers is indicated and recorded in the main control room along with independent alarm annunciation for a high hydrogen concentration. Each hydrogen analyzer continuously withdraws a sample of the process offgas, conditions the gas to a constant pressure, analyzes the hydrogen content, and returns the sample gas to the main condenser. The main condenser vacuum provides the pumping force to withdraw the sample gas from the offgas process line and through the hydrogen analyzer system. The analyzer element is a thermal conductivity cell type unit and does not serve as an ignition source to a detonable hydrogen-oxygen mixture.

Piping and tubing 2 in. and under is field routed, but required to be in specified space envelopes for shielding and in-plant exposure consideration. In the offgas system this includes drain lines, steam lines, and sample lines which are shown in Figure 11.3-2.

There are several liquid seals to prevent gas escape through drains shown in Figure 11.3-2. These seals are protected against permanent loss of liquid by an enlarged section downstream of the seal that can hold the seal volume and will drain by gravity back into the loop after a momentary pressure surge has passed. Each seal has a manual valve that is used to fill the loop with condensate after receiving a loop seal low level alarm.

Iodine input into the offgas system is small by virtue of its retention in reactor water and condensate. The iodine remaining is essentially removed by adsorption in the charcoal. This is supported by the fact that 2-in. charcoal filters remove more than 90% of the influent iodine, whereas this system has approximately 76 ft of charcoal in the flow path.

Particulates are removed with a 99.95% efficiency by a HEPA filter prior to the gas entering the charcoal adsorbers. The noble gas decays within the interstices of the activated charcoal and daughters are entrapped there. The charcoal serves as an excellent filter for particulates and essentially no particulates exit from the charcoal. The charcoal is followed with a HEPA filter which is a safeguard against escape of charcoal dust. Particulate activity discharged from this system is essentially zero.

With an airflow of about 30 scfm and the charcoal adsorbers bypassed, the delay time of the system decreases to approximately 10 minutes. This bypass line is only intended to be used during preoperational testing, and initial system startup operation until proper functioning of upstream equipment is established. This prevents possible degradation of the charcoal due to the introduction of excessive moisture or other contaminants.

In the unlikely event it is necessary to bypass the charcoal adsorbers to continue operation, the bypassing operation shall be allowed only if the radioactive effluents are within the release limits.

The isotopic inventory of each equipment piece, based on the source term discussed in Section 11.1, is given in Reference 11.3-5.

Performance of a similar system operating at ambient temperatures and the results of related experimental testing are discussed in Reference 11.3-2. The Tsuruga and Fukushima 1 plants in Japan have similar recombiners in service. Similar systems (ambient temperature charcoal) are in service at Dresden 2 and 3, Pilgrim, Quad Cities 1 and 2, Nuclenor, Hatch, Browns Ferry 1, 2, and 3, and Duane Arnold.

Design provisions are incorporated which preclude the uncontrolled release of radioactivity to the environment as a result of any single operator error or of any single failure. A comprehensive discussion of single failures is provided in Table 11.3-5.

Design precautions taken to prevent uncontrolled releases of radioactivity include the following:

a. The system design eliminates ignition sources so that a hydrogen detonation is highly unlikely even in the event of a recombiner failure;

b. The system pressure boundary is detonation resistant;

c. All discharge paths to the environment are monitored--the normal effluent path by the process radiation monitoring system and equipment areas by the area radiation monitoring system; and

d. Dilution steam flow to the SJAE is monitored and alarmed. Valve control logic causes the air ejector suction valves to close on low steam flow.

11.3.2.2 Other Radioactive Gas Sources

There are three buildings that contain radioactive gas sources. They are the reactor building, the turbine generator building, and the radwaste building. The ventilation systems for these three buildings are described in Section 9.4. Building volumes and ventilation flow rates are shown in Table 11.3-8. The sources of gaseous radioactivity in these buildings are discussed below. In-plant airborne radioactivity concentrations are discussed in Section 12.2.2.

The primary containment is divided into two sections which are designated as the drywell and suppression chamber. These are separated by the drywell floor which serves as a pressure barrier between the drywell and suppression chamber.

Radioactive halogens and noble gases can be introduced into the drywell atmosphere from two sources. One source is the leakage that occurs from the valves, especially the main steam isolation valves (MSIVs), the inner refueling bellows seal support drains and the main

recirculation pumps. This leakage is collected by means of leak-off lines which are directed to the drywell equipment drain sump.

The other source of activity results when a pressure transient causes the main steam relief valves to open with the resulting flow of steam into the suppression pool. When this occurs, there is a pressure buildup in the suppression chamber atmosphere. A tabulation of the expected relative frequency of pressure relief valve venting to the suppression pool is provided in Table 11.3-10. If the pressure differential across the drywell floor is greater than 0.5 psi, the vacuum breaker valves relieve this pressure into the drywell atmosphere. Thus, radioactivity in the suppression chamber atmosphere is introduced into the drywell. The drywell atmosphere is purged to the environment via the reactor building elevated release duct when access is required, either directly or through the standby gas treatment system, considering airborne radiation levels and release limits.

The reactor building ventilation system supplies fresh air to the secondary containment and exhausts air through the reactor building elevated release duct. There is a small amount of activity in the secondary containment atmosphere that emanates from the various reactor support systems. The reactor building ventilation exhaust is monitored and a radiation level, set by administrative control to ensure Technical Specifications compliance, will cause automatic heating, ventilating, and air-conditioning (HVAC) system isolation, startup of the standby gas treatment system (SGTS), and an alarm in the control room.

There are three sources of nontreated radioactive gas sources in the turbine generator building which are as follows:

a. Gland seal steam leakage and condenser exhaust,
b. Equipment leakage, and
c. Main condenser offgas during startup.

The activity from these sources is presented in Table 11.3-7.

Gland seal leakage and condenser exhaust contribute essentially no airborne radioactivity releases due to use of “clean” steam as discussed in Section 10.4. During startup, the mechanical vacuum pumps remove the gas present inside the main condenser. The exhaust from these pumps is discharged through the reactor building elevated release duct. Due to radioactive decay during shutdown, only a small amount of activity is exhausted by the vacuum pumps during startup.

Sources of gaseous radioactivity in the radwaste building include,

a. Offgas system leakage,
b. Liquid leakage to the radwaste building,

c. Liquid waste management system tank vents, and
d. Hydropneumatic transfer of resins.

Measures taken to minimize leakage from the offgas system are described in Section 11.3.2.1. Liquid leakage from equipment and floor drains in the radwaste building is collected in sumps. The sumps and tanks containing liquid radwaste are vented to the radwaste building exhaust system described in Section 9.4.3.2. The air blow during the backwash of the spent powdered resins to the phase separators generates airborne radioactivity which is vented to the radwaste building filtered exhaust system.

11.3.2.3 Cost-Benefit Analysis

The cost-benefit analysis is discussed in Section 11.2.3.4.

11.3.2.4 Design Features of the Offgas System

Design features of other gaseous waste management systems may be found in Section 9.4.

11.3.2.4.1 Maintainability

Design features which reduce or ease required maintenance include the following:

a. Redundant components for all active, in-process equipment pieces, and

b. No rotating equipment in the process stream and elsewhere in the system only where maintenance can be performed while the system is in operation.

11.3.2.4.2 Pressure Boundaries

Design features and requirements which reduce leakage and releases of radioactive material include the following:

a. Extremely stringent leak rate requirements placed on all equipment, piping, and instruments, and enforced by requiring as-installed leak tests of the entire process system,

b. Use of welded joints except where servicing access to equipment and instrumentation access dictates use of flanged joints,

c. Valve types with extremely low leak rate characteristics, i.e., bellows seals double stem seal, or diaphragm,

d. Use of loop seals with enlarged discharge section to avoid siphoning and to be self-refilling by gravity following a pressure surge, and

e. Stringent seat-leak characteristics for valves in lines discharging to the environment via other systems.

11.3.2.4.3 Building Seismic Design

The offgas system is located in the turbine generator and radwaste buildings. The portion of the turbine generator building housing the offgas system is a modified Seismic Category II structure designed to withstand the effects of a safe shutdown earthquake (SSE) and maintain its structural integrity. The portion of the radwaste building housing the offgas system is a Seismic Category I structure. The seismic classification of the turbine generator and radwaste buildings are discussed in Sections 3.8.4.1.3 and 3.8.4.1.2, respectively.

11.3.2.4.4 Construction of Process Systems

Pressure retaining components of process systems utilize welded construction to the maximum practicable extent. Process piping systems include the first root valve on sample and instrument lines. Process lines are not less than 0.75 in. nominal pipe size. Sample and instrument lines are not considered as portions of the process systems. Flanged joints or suitable rapid disconnect fittings are not used except where maintenance requirements clearly indicate that such construction is preferable. Screwed connections in which threads provide the only seal are not used. Screwed connections backed up by seal welding or mechanical joints are used only on lines of 0.75 in. nominal pipe size or less. In lines 0.75 in. or greater, but less than 2.5 in. nominal pipe size, socket type welds are used. In lines 2.5 in. nominal pipe size and larger, pipe welds are of the butt joint type.

11.3.2.4.5 Instrumentation and Control

The offgas system is monitored by flow, temperature, pressure, and humidity instrumentation, and by hydrogen analyzers to ensure correct operation and control. Table 11.3-4 lists the process parameters that are instrumented to alarm in the control room and indicates whether the parameters are recorded or just indicated.

The radioactivity of the gas entering and leaving the offgas system is continuously monitored. Thus, system performance is known to the operator at all times. A radiation monitor after the offgas condenser continuously monitors radioactivity release from the reactor and input to the charcoal adsorbers. This radiation monitor is used to provide an alarm on high radiation in the offgas. A sample rack with two radiation monitors is also provided at the outlet of the charcoal adsorbers to continuously monitor the radioactivity from the adsorber beds. These radiation monitors are used to isolate the offgas system on high radioactivity to prevent gas of

unacceptably high activity from entering the reactor building elevated release duct. Only one monitor is required to be operable.

The offgas system at the SJAE is sampled periodically in accordance with the Technical Specifications. Provision is made for sampling and periodic analysis of the influent and effluent gases for purposes of determining their compositions. This information is used in calibrating the monitors and in relating the release to calculated offsite doses. Process radiation instrumentation is described in detail in Sections 11.5 and 7.6.1.1.

11.3.2.4.6 Detonation Resistance

The pressure boundary of the system is designed to be detonation resistant. The pressure vessels are designed to withstand 350 psig static pressure, and piping and valves are designed to resist dynamic pressures encountered in long runs of piping at the design temperature. This analysis is covered in a proprietary report submitted to the NRC (Reference 11.3-6).

By this procedure a designer obtains the required wall thickness for specific equipment to sustain a hydrogen and oxygen detonation. The wall thickness is then translated by using the appropriate code calculation to the corresponding equipment that must contain the detonation static pressure. The method assumes the absence of simultaneous secondary events such as earthquakes.

11.3.2.4.7 Operator Exposure Criteria and Controls

This system is normally operated from the main control room. Equipment and process valves containing radioactive fluid are placed in shielded cells maintained at a pressure negative to normally occupied areas. Ventilation air flows from areas of low airborne contamination to areas of higher airborne contamination. Operating offgas process equipment does not require personnel access. Redundant equipment is located in separate cells, minimizing exposure of maintenance personnel. No process fluid is passing through instrumentation panels. Signals from the process streams are transmitted to the instrumentation panels by means of electrical signal converters. Design features minimizing occupational exposure are discussed in Sections 11.3.2.4.1 and 12.3.1.3.

11.3.2.4.8 Equipment Malfunction

Malfunction analyses indicating consequences and design precautions taken to accommodate failure of various components of the system are given in Table 11.3-5.

11.3.2.5 Offgas System Operating Procedure

11.3.2.5.1 Prestartup Preparations

Prior to starting the main condenser SJAEs, the glycol coolant is chilled to near 35°F and is circulated through the cooler condenser, a desiccant dryer is regenerated and valved in, the offgas condenser cooling water is valved in, and the recombiner heaters are turned on.

11.3.2.5.2 Startup

As the reactor is pressurized, preheater steam is supplied and air is bled through the preheater and recombiner. The recombiner is preheated to at least 300°F with this air bleed and/or by admitting steam to the final SJAE. With the recombiners preheated and the desiccant dryer and charcoal adsorbers valved in, the SJAE string is started. The bleed air is terminated. As the condenser is pumped down and the reactor power increases, the recombiner inlet stream is diluted to less than 4% hydrogen by volume by a fixed steam supply, and the offgas condenser outlet is maintained at less than 1% hydrogen by volume.

11.3.2.5.3 Normal Operation

After startup, the noncondensables pumped by the SJAE will stabilize. Recombiner performance is closely followed by the recorded temperature profile in the recombiner catalyst bed. The hydrogen effluent concentration is measured by a hydrogen analyzer.

Normal operation is terminated following a normal reactor shutdown or a scram by terminating steam to the SJAEs and the preheater.

11.3.2.6 Offgas System Performance Tests

This system is used on a routine basis and does not require specific testing to ensure operability. Monitoring equipment will be calibrated and maintained on a specific schedule and on indication of malfunction.

11.3.2.6.1 Recombiner

Recombiner performance is continuously monitored and recorded by catalyst bed thermocouples that monitor the bed temperature profile and by a hydrogen analyzer that measures the hydrogen concentration of the effluent.

11.3.2.6.2 Prefilter

These particulate filters were tested at the time of filter installation using dioctylphthalate (DOP) aerosol to determine whether an installed filter meets the minimum in-place efficiency 99.95% particle retention.

The DOP from filter testing is not allowed to enter into the desiccant or the activated charcoal. This equipment is isolated during filter DOP testing and is bypassed until the process lines have been purged clear of test material. Because the DOP may have a detrimental effect on the desiccant and charcoal, the prefilter will not be periodically tested. This is justified because the main function of this prefilter is to prevent the long-lived daughters of the radioactive xenons generated in the holdup pipe from depositing in the downstream equipment. Leakage through the filter would be unimportant to environmental release.

11.3.2.6.3 Desiccant Gas Dryer

Desiccant gas dryer performance is continuously monitored by an on-stream humidity analyzer.

11.3.2.6.4 Charcoal Performance

The ability of the charcoal to delay the noble gases can be continuously evaluated by comparing radioactivity measured and recorded by the process activity monitors at the exit of the offgas condenser and at the exit of the charcoal adsorbers.

Grab sample points are located upstream and downstream of the first charcoal bed and downstream of the last charcoal bed and can be used for periodic sampling if the monitoring equipment indicates degradation of system delay performance.

11.3.2.6.5 Post Filter

On installation these particulate filters were tested using a DOP aerosol test or equivalent, as described in Section 11.3.2.6.2.

11.3.3 RADIOACTIVE RELEASES

11.3.3.1 Release Points

The reactor building elevated release duct serves the following systems:

Offgas system,
Mechanical vacuum pump and gland seal condenser exhaust,
Reactor building ventilation exhaust,

Containment purge, and
Standby gas treatment system.

The reactor building elevated release duct location is shown in reactor building general arrangement drawings in Section 1.2.

The radwaste building ventilation exhausts through three louver houses 67 ft above plant grade. The location is shown on Figure 1.2-17.

The exhaust from the turbine generator building ventilation system is through four exhaust fans located on the radwaste building, 119 ft above plant grade. Their location is shown on Figure 1.2-16 (part plan of roof at el. 542 ft-0 in.) and in Figure 1.2-17.

The height, flow rate, heat content, and dimensions of the three release points are shown on Table 11.3-6.

11.3.3.2 Dilution Factors

The dispersion and dilution of gaseous radioactive effluents released from the plant depends on the meteorology of the site and its environs. To determine these parameters, onsite meteorological data has been obtained and analyzed as described in Section 2.3. Annual atmospheric dilution factors have been calculated to determine resultant annual doses and concentrations of radionuclides from normal operation.

The building ventilation exhaust ducts do not rise above the buildings; therefore, atmospheric releases for dose analysis purposes were considered ground level.

11.3.3.3 Estimated Releases

Releases of radioactive material in gaseous effluents for initial plant licensing were calculated using the GALE code presented in NUREG-0016 to show compliance with 10 CFR 20, Appendix B, and 10 CFR 50, Appendix I, for normal operation plus anticipated operational occurrences. The operational parameters including source terms were those presented in Appendix B. The values obtained are presented in Table 11.3-7. These values were used with the meteorology data from Section 2.3.5 to calculate maximum concentrations at the restricted area boundary and maximum individual dose offsite. These data provide maximum annual average (χ/Q) values and does not include a building wake factor.

Restricted area boundary concentrations used for this initial licensing analysis are tabulated by radionuclide and compared with 10 CFR 20 limits in Table 11.3-9. The estimated annual dose to persons offsite is presented in Section 5.2 of the Environmental Report and is well within the numerical guidelines of 10 CFR 50, Appendix I.

Since becoming operational, releases of radioactive materials in gaseous effluents have been determined using actual flow rates and quantitative and qualitative analyses. Doses due to radioactive materials in gaseous effluents are determined to show Technical Specifications compliance at specified intervals. Compliance is reported in the Annual Radioactive Effluent Release Report. Doses are ascertained using ODCM approved calculation methods, current meteorology (or historical meteorology if current data are unavailable), and parameters outlined in the Offsite Dose Calculation Manual.

Tritium contamination of the auxiliary boiler and associated components has resulted in release paths which were not intended (see Section 9.4.16.2). The released radioactivity attributable to the worst case tritium concentration (2E + 06 ρCi/liter) in the boiler water is an insignificant portion of the total activity released in liquid and gaseous effluents and has been analyzed to result in a correspondingly insignificant radiation dose. The tritium concentration levels and makeup water volume are monitored and evaluated to ensure that tritium effluent releases from the plant are adequately quantified.

11.3.4 REFERENCES

11.3-1 NUREG-0016 (BWR-GALE Code), January 1979

11.3-2 Miller, C. W., Experimental and Operational Confirmation of Off-Gas System Design Parameters, NEDO-10751, January 1973 (Proprietary).

11.3-3 Standards for Steam Surface Condensers, Sixth Edition, Heat Exchange Institute, New York, NY, 1970.

11.3-4 Underhill, Dwight, et al., “Design of Fission Gas Holdup Systems,” Processing of the Eleventh AEC Air Cleaning Conference, 1979, p. 217.

11.3-5 Miller, C. W., et al., A General Justification for Classification of Effluent Treatment System Equipment as Group D, NEDO-10734, February 1973.

11.3-6 Nesbitt, L. B., Design Basis for New Gas Systems, NEDE-11146, July 1971 (Proprietary).

11.3-7 NUREG-0800 USNRC Standard Review Plan, Revision 2, July 1981, Branch Technical Position ETSB 11-5.

Table 11.3-1

Design Air Ejector Offgas Release Rates (30 cfm inleakage) [a]

Isotope	Half-life	T=0 µCi/sec	T=30 Minutes µCi/sec	Normal Discharge from Charcoal Adsorbers µCi/sec	Normal Discharge from Charcoal Adsorbers Ci/yr[b]	Additional Discharge from Charcoal Adsorbers During Startup µCi/sec	Additional Discharge from Charcoal Adsorbers During Startup Ci/startup
^{83m}Kr	1.86 hr	3.4×10^{3}	2.9×10^{3}	-	-	-	-
^{85m}Kr	4.4 hr	6.1×10^{3}	5.6×10^{3}	4.3	1.2×10^{2}	1.1×10^{1}	1.4
^{85}Kr[c]	10.74 yr	10-20	10-20	10-20	280-560	0	0
^{87}Kr	76 minutes	2.0×10^{4}	1.5×10^{4}	-	-	-	-
^{88}Kr	2.79 hr	2.0×10^{4}	1.8×10^{4}	2.1×10^{-1}	6.0	1.4	1.7×10^{-1}
^{89}Kr	3.18 minutes	1.3×10^{5}	1.8×10^{2}	-	-	-	-
^{90}Kr	32.3 sec	2.8×10^{5}	-	-	-	-	-
^{91}Kr	8.6 sec	3.3×10^{5}	-	-	-	-	-
^{92}Kr	1.84 sec	3.3×10^{5}	-	-	-	-	-
^{93}Kr	1.29 sec	9.9×10^{4}	-	-	-	-	-
^{94}Kr	1.0 sec	2.3×10^{4}	-	-	-	-	-
^{95}Kr	0.5 sec	2.1×10^{3}	-	-	-	-	-
^{97}Kr	1 sec	1.4×10^{1}	-	-	-	-	-
^{131m}Xe	11.96 days	1.5×10^{1}	1.5×10^{1}	1.3	3.7×10^{1}	3.0×10^{-2}	1.07×10^{-1}
^{133m}Xe	2.26 days	2.9×10^{2}	2.8×10^{2}	-	-	-	-
^{133}Xe	5.27 days	8.2×10^{3}	8.2×10^{3}	3.3×10^{1}	9.4×10^{2}	1.9	6.8
^{135m}Xe	15.7 minutes	2.6×10^{4}	6.9×10^{3}	-	-	-	-
^{135}Xe	9.16 hr	2.2×10^{4}	2.2×10^{4}	-	-	-	-
^{137}Xe	3.82 minutes	1.5×10^{5}	6.7×10^{2}	-	-	-	-
^{138}Xe	14.2 minutes	8.9×10^{4}	2.1×10^{4}	-	-	-	-
^{139}Xe	40 sec	2.8×10^{5}	-	-	-	-	-
^{140}Xe	13.6 sec	3.0×10^{5}	-	-	-	-	-
^{141}Xe	1.72 sec	2.4×10^{5}	-	-	-	-	-
^{142}Xe	1.22 sec	7.3×10^{4}	-	-	-	-	-
^{143}Xe	0.96 sec	1.2×10^{4}	-	-	-	-	-
^{144}Xe	9 sec	5.6×10^{2}	-	-	-	-	-
	Totals	2.5×10^{6}	1.0×10^{5}	49-59	1383-1663	14.3	8.5

[a] Based on the 1971 mixture.
[b] This is based on curies present at time of release. No decay in environment is included.
[c] Estimated from experimental observations.

Table 11.3-2

Offgas System Major Equipment Items

Offgas Preheaters - two required

Construction: Stainless steel tubes and carbon steel shell. 350 psig shell design pressure, 1000 psig tube design pressure. 400°F shell design temperature, 575°F tube design temperature.

Catalytic Recombiners - two required

Construction: Stainless steel cartridge, carbon steel shell. Catalyst cartridge containing a precious metal catalyst on metal base. Catalyst cartridge to be replaceable without removing vessel. 350 psig design pressure, 900°F design temperature.

Offgas Condenser - one required

Construction: Low alloy steel shell. Stainless steel tubes. 350 psig shell design pressure. 250 psig tube design pressure, 900°F shell design temperature, 150°F tube design temperature.

Water Separator - one required

Construction: Carbon steel shell, stainless steel wire mesh. 350 psig design pressure, 250°F design temperature.

Cooler-Condenser - two required

Construction: Stainless steel shell. Stainless steel tubes. 100 psig tube design pressure, 350 psig shell design pressure. 32/150°F tube design temperature, 32/150°F shell design temperature.

Moisture Separators (downstream of cooler-condenser) - two required

Construction: Carbon steel shell, stainless steel wire mesh. 350 psig design pressure, 32/150°F design temperature.

Desiccant Dryer - four required

Construction: Carbon steel shell packed with Type 3A molecular sieve or equivalent. 350 psig design pressure, 32°F/ 500°F design temperature.

Desiccant Regeneration Skid - two required

Table 11.3-2

Offgas System Major Equipment Items (Continued)

Dryer Chiller - two required

Construction: Carbon steel shell, stainless steel tubes, design temperature 32°F/500°F, design pressure 50 psig.

Regenerator Blower - two required

Construction: Electrical, design pressure 50 psig design temperature 32°F/150°F.

Dryer Heater - two required

Construction: Electrical, design temperature 32°F/500°F, design pressure 50 psig.

Glycol Cooler Skid - one required

Glycol Storage Tank - one required

Construction: Carbon steel 3000 gal. Water-filled hydrostatic design pressure. 32°F design temperature. API-650.

Glycol Solution Refrigerators and Motor Drives - three required

Construction: Conventional refrigeration units. Glycol solution exit temperature 35°F.

Glycol Pumps and Motor Drives - three required

Construction: Cast iron, 0°F design temperature.

Prefilters and After Filters - two required of each type

Construction: Carbon steel shell. High-efficiency, moisture-resistant filter element. Flanged shell. 350 psig design pressure. -50/150°F design temperature.

Charcoal Adsorbers - eight beds

Construction: Carbon steel. Approximately 4-ft. O.D. x 21-ft vessels each containing approximately 3 tons of activated carbon. Design pressure 350 psig, design temperature -50/250°F.

Table 11.3-3

Process Data For The Offgas (RECHAR) System

The Information On This Page Is
Proprietary And Was Submitted
Under Separate Cover

(CVI DWG 02N64-04,11,1, Rev. 4 or 02N64-04,11,2, Rev. 2)

Table 11.3-4

Offgas System Alarmed Process Parameters

Parameters	Control Room	
	Indicated	Recorded
Air ejector discharge pressure - high	X	
Preheater discharge temperature - low	X	
Recombiner catalyst temperature - high/low		X
Offgas condenser water level (dual) - high/low	X	
Offgas condenser gas discharge temperature - high (local)	X	
H_2 analysis (offgas condenser discharge) - dual - high		X
Offgas condenser discharge radiation - high		X
Gas flow - high/low		X
Cooler - condenser discharge temperature - high/low		X
Glycol solution temperature - high/low		X
Glycol solution level - low		
Gas drier discharge humidity - high (local)	X	
Prefilter dP - high	X	
Charcoal adsorber temperature - high		X
Charcoal train flow - high/low		X
After filter dP - high	X	
Offgas (charcoal bed discharge) radiation - high		X
Steam flow - low	X	
Desiccant dryer outlet temperature - high/low		X
Dryer chiller outlet temperature - high (local)	X	
Dryer heater temperature - high		
Dryer heater outlet temperature - high (local)	X	
Loop seals water level - low		

Table 11.3-5

Equipment Malfunction Analysis

Equipment Item	Malfunction	Consequences	Design Precautions
Steam jet air ejectors	Low flow of motive high-pressure steam	When the hydrogen and oxygen concentrations exceed 4 and 5 vol %, respectively, the process gas becomes flammable.	Alarm provided on steam for low steam flow. Recombiner temperature alarm.
		Inadequate steam flow will cause overheating and deterioration of the catalyst.	Steam flow to be held at constant maximum flow regardless of plant level during operation.
	Wear of steam supply nozzle of ejector	Increased steam flow to recombiner. This could reduce degree of recombination at low power levels.	Low temperature alarms on preheater exit (recombiner inlet). Recombiner outlet H_2 analyzers.
Preheaters	Steam leak	Would further dilute process offgas. Steam consumption would increase.	Spare preheater.
	Low pressure steam supply	Recombiner performance would fall off at low power level and hydrogen content of recombiner gas discharge may increase, eventually to a combustible mixture.	Low-temperature alarms on preheater exit (recombiner inlet). Recombiner outlet H_2 analyzers.
Recombiners	Catalyst gradually deactivates	Temperature profile changes through catalyst. Eventually excess H_2 would be detected by H_2 analyzer or by gas flowmeter. Eventually the stripped gas could become combustible.	Temperature probes in recombiner and H_2 analyzer provided. Spare recombiner.
	Catalyst gets wet at start	H_2 conversion falls off and H_2 is detected by downstream analyzers. Eventually the gas could become combustible.	Condensate drains, temperature probes in recombiner. Air bleed system at startup. Recombiner thermal blanket, spare recombiner, and heater. Hydrogen analyzer.

Table 11.3-5

Equipment Malfunction Analysis (Continued)

Equipment Item	Malfunction	Consequences	Design Precautions
Offgas condenser	Cooling water leak	The coolant (reactor condensate) would leak to the process gas (shell) side. This would be detected if drain well liquid level increases. Moderate leakage would be of no concern from a process standpoint. (The process condensate drains to the hotwell).	None
	Liquid level instruments fail	If both drain valves fail to open, water will build up in the condenser and pressure drop will increase.	Two independent drain systems, each provided with high and low-level alarms.
		The high ΔP, if not detected by instrumentation, could cause pressure buildup in the main condenser and eventually initiate a reactor scram. If a drain valve fails to close, gas will recycle to the main condenser, increase the load on the SJAE, and increase operating pressure of the main condenser.	
Water separator	Corrosion of wire mesh element	Higher quantity of water collected in holdup line and routed to radwaste.	Stainless steel mesh specified.
Holdup line	Corrosion of line	Leakage to soil of gaseous and liquid fission products.	Outside of pipe dipped and wrapped. 0.25-in. corrosion allowance.
Cooler-condenser	Corrosion of tubes	Glycol-water solution would leak into process (shell) side and be discharged to clean radwaste. If not detected at radwaste, the glycol solution would discharge to the reactor condensate system.	Stainless-steel tubes specified. Low level alarm glycol tank level. Spare cooler condenser provided.

Table 11.3-5

Equipment Malfunction Analysis (Continued)

Equipment Item	Malfunction	Consequences	Design Precautions
Cooler-condenser (continued)	Icing up of the tubes	Shell side of cooler could plug up with ice, gradually building up pressure drop. If this happens, the spare unit could be activated. Complete blockage of both units would increase ΔP and lead to a reactor scram.	Design glycol-H_2O solution temperature well above freezing point. Spare unit provided. Temperature indication and low alarms on glycol temperature and process gas temperature.
Glycol refrigeration machines	Mechanical failure	If both spare units fail to operate, the glycol solution temperature will rise and the dehumidification system performance will deteriorate. This will require rapid regeneration cycles for the desiccant beds and may raise the gas dewpoint as it is discharged from the drier.	Two spare refrigerators during normal operation are provided. Glycol solution temperature alarms provided. Gas moisture detectors provided downstream of gas driers.
Moisture separators	Corrosion of wire mesh	Increased moisture would be retained in process gas routed to gas driers element. Over a long period, the desiccant drier cycle period would deteriorate as a result of moisture pickup. Pressure drop across prefilter may increase if filter media is wetted.	Stainless steel mesh specified. Spare unit provided. High ΔP alarm on prefilter.
Prefilters	Loss of integrity of filter media	More radioactivity would deposit on the drier desiccant. This would increase the radiation level in the drier vault and make maintenance more difficult, but would not affect releases to the environment.	Spare unit provided in separate vault. ΔP instrumentation provided.

Table 11.3-5

Equipment Malfunction Analysis (Continued)

Equipment Item	Malfunction	Consequences	Design Precautions
Desiccant drier	Moisture breakthrough	Increased moisture in air entering charcoal adsorbers would decrease adsorption effectiveness, thus reducing radioisotope retention time.	Drier cycles on timer. Redundant gas humidity analyzers and alarms supplied. Redundant gas drier system supplied. Gas drier and first charcoal bed can be bypassed through alternate drier to second charcoal bed.
Desiccant regeneration equipment	Mechanical failure	Inability to regenerate desiccant.	Redundant, shielded desiccant beds and drier equipment is supplied.
Charcoal adsorbers	Charcoal accumulates moisture	Charcoal performance will deteriorate gradually as moisture deposits. Holdup times for krypton and xenon would decrease, and plant emissions would increase. Provisions made for drying charcoal as required during annual outage.	Highly instrumented, mechanically simple gas dehumidification system with redundant equipment.
After filters	Loss of integrity of filter media	Probably of no real consequence. The charcoal media itself should be a good filter at the low air velocity.	ΔP instrumentation provided. Spare unit provided.
System	Internal detonation	Release of radioactivity if pressure boundary fails.	Main process equipment and piping are designed to contain a detonation.
	Earthquake damage	Release of radioactivity.	Dose consequences are within 10 CFR 20 limits. Analysis is included in Reference 11.3-5.

Table 11.3-6

Release Point Data

Note: Refer to ODCM Table 3-13.

Table 11.3-7

Gaseous Waste System Release

		Gaseous Release Rate[a] (Ci/yr)							
Nuclide	Coolant Conc. (μCi/g)	Containment Building	Turbine Building	Reactor Building	Radwaste Building	Gland Seal	Air Ejector	Mechanical Vac Pump	Total
^{83m}Kr	1.200E-03	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
^{85m}Kr	2.000E-03	3.1E 00	7.1E 01	3.1E 00	0.0	0.0	2.1E 00	0.0	8.0E 01
^{85}Kr	6.300E-06	0.0	0.0	0.0	0.0	0.0	2.8E 02	0.0	2.8E 02
^{87}Kr	6.900E-03	3.1E 00	2.0E 02	3.1E 00	0.0	0.0	0.0	0.0	2.1E 02
^{88}Kr	6.900E-03	3.1E 00	2.4E 02	3.1E 00	0.0	0.0	0.0	0.0	2.4E 02
^{89}Kr	4.300E-02	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
^{131m}Xe	4.900E-06	0.0	0.0	0.0	0.0	0.0	5.2E 00	0.0	5.2E 00
^{133m}Xe	9.400E-05	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
^{133}Xe	2.700E-03	6.9E 01	2.9E 02	6.9E 01	1.0E 01	0.0	2.3E 01	2.3E 03	2.8E 03
^{135m}Xe	8.800E-04	4.8E 01	6.8E 02	4.8E 01	0.0	0.0	0.0	0.0	7.8E 02
^{135}Xe	7.600E-03	3.6E 01	6.6E 02	3.6E 01	4.5E 01	0.0	0.0	3.5E 02	1.1E 03
^{137}Xe	4.900E-02	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
^{138}Xe	2.900E-02	7.3E 00	1.5E 03	7.3E 00	0.0	0.0	0.0	0.0	1.5E 03
Total noble gases									7.0E 03
^{131}I	3.618E-03	1.8E-02	2.0E-01	1.8E-01	5.0E-02	0.0	0.0	3.0E-02	4.8E-01
^{133}I	1.549E-02	7.1E-02	8.0E-01	7.1E-01	1.8E-01	0.0	0.0	0.0	1.8E-00
Tritium gaseous release									7.1E 01

NOTE: 0.0 appearing in the table indicates release is less than 1.0 Ci/yr for noble gas, 0.0001 Ci/yr for iodine.

Table 11.3-7

Gaseous Waste System Release (Continued)

	Airborne Particulate Release Rate (Ci/yr)					
Nuclide	Containment Building	Turbine Building	Reactor Building	Radwaste Building	Mechanical Vac Building	Total
^{51}Cr	3.1E-06	1.4E-02	3.1E-04	9.4E-05	0.0	1.4E-02
^{54}Mn	3.1E-05	5.2E-04	3.1E-03	4.7E-04	0.0	4.3E-03
^{59}Fe	4.2E-06	5.2E-04	4.2E-04	1.6E-04	0.0	1.1E-03
^{58}Co	6.3E-06	6.3E-04	6.3E-04	4.7E-05	0.0	1.4E-03
^{60}Co	1.0E-04	2.1E-03	1.0E-02	9.4E-04	0.0	1.5E-02
^{65}Zn	2.1E-05	2.1E-04	2.1E-03	1.6E-05	0.0	2.3E-03
^{89}Sr	9.4E-07	6.3E-03	9.4E-05	4.7E-06	0.0	6.4E-03
^{90}Sr	5.2E-08	2.1E-05	5.2E-06	3.1E-06	0.0	2.9E-05
^{95}Zr	4.2E-06	1.0E-04	4.2E-04	5.2E-07	0.0	5.2E-04
^{124}Sb	2.1E-06	3.1E-04	2.1E-04	5.2E-07	0.0	5.2E-04
^{134}Cs	4.2E-05	3.1E-04	4.2E-03	4.7E-05	3.1E-06	4.6E-03
^{136}Cs	3.1E-06	5.2E-05	3.1E-04	4.7E-06	2.1E-06	3.8E-04
^{137}Cs	5.8E-05	6.3E-04	5.8E-03	9.4E-05	1.0E-05	6.6E-03
^{140}Ba	4.2E-06	1.1E-02	4.2E-04	1.0E-06	1.1E-05	1.1E-02
^{141}Ce	1.0E-06	6.3E-04	1.0E-04	6.3E-05	0.0	8.0E-04

[a] Estimated release based on GALE code evaluation.

Table 11.3-8

Building Volume and Ventilation Rates

Building	Free Air Volume (ft^3)	Ventilation Rate (cfm)
Secondary containment (reactor building)	3.5×10^6	80,000
Radwaste building	2.0×10^6	83,000
Turbine building	5.7×10^6	360,000
Primary containment (drywell)	2.0×10^5	10,500[a]
Primary containment (wetwell)	1.4×10^5	7,500[a]

[a] During primary containment purge only.

Table 11.3-9

Maximum Sector Annual Average Concentrations of Gaseous Radioactive Materials at the Original Restricted Area Boundary

Nuclide	Annual Average Release Rate (Ci/sec)	Boundary Concentration[a] ($\mu Ci/cm^3$)	Derived Air Concentration (DAC)[b] ($\mu Ci/cm^3$)	Concentration/ DAC
^{3}H	2.3 E-6	1.8 E-11	1 E-7	0.00018
^{85m}Kr	2.5 E-6	1.9 E-11	1 E-7	0.0002
^{85}Kr	9.0 E-6	6.9 E-11	7 E-7	0.000099
^{87}Kr	6.6 E-6	5.1 E-11	2 E-8	0.002
^{88}Kr	8.0 E-6	6.2 E-11	9 E-9	0.00069
^{131m}Xe	1.7 E-7	1.3 E-12	2 E-6	0.0000006
^{133}Xe	9.0 E-5	6.9 E-10	6 E-7	0.0011
^{135m}Xe	2.4 E-5	1.9 E-10	4 E-8	0.0047
^{135}Xe	3.7 E-5	2.8 E-10	7 E-8	0.0041
^{138}Xe	4.6 E-5	3.6 E-10	2 E-8	0.018
^{131}I	1.6 E-8	1.3 E-13	2 E-10	0.0006
^{133}I	5.7 E-8	4.4 E-13	1 E-9	0.00044
^{51}Cr	4.3 E-10	3.4 E-15	3 E-8	0.00000011
^{54}Mn	1.3 E-10	9.7 E-16	1 E-9	0.00000097
^{59}Fe	3.7 E-11	2.8 E-16	5 E-10	0.00000057
^{58}Co	4.3 E-11	3.4 E-16	1 E-9	0.00000034
^{60}Co	4.3 E-10	3.4 E-15	5 E-11	0.000067
^{65}Zn	7.3 E-11	5.7 E-16	4 E-10	0.0000015
^{89}Sr	2.0 E-10	1.6 E-15	1 E-9	0.0000016
^{90}Sr	9.3 E-13	7.2 E-18	6 E-12	0.0000013
^{95}Zr	1.7 E-11	1.3 E-16	4 E-10	0.0000003
^{124}Sb	1.7 E-11	1.3 E-16	3 E-10	0.0000004
^{134}Ca	1.5 E-10	1.1 E-15	2 E-10	0.0000058
^{136}Cs	1.1 E-11	8.9 E-17	9 E-10	0.000000099
^{137}Cs	2.1 E-10	1.6 E-15	2 E-10	0.0000079
^{140}Ba	3.7 E-10	2.8 E-15	2 E-9	0.0000015
^{141}Ce	2.5 E-11	1.9 E-16	1 E-9	0.00000019
			MPC	0.03

[a] χ/Q factor of 7.7 x 10^{-6} at 0.5 miles distance in SE sector used. No building wake factor included; meteorological data from 4/74 through 3/75.

[b] 10 CFR 20, Appendix B, to 20.1001-20.2401, Table II Column I.

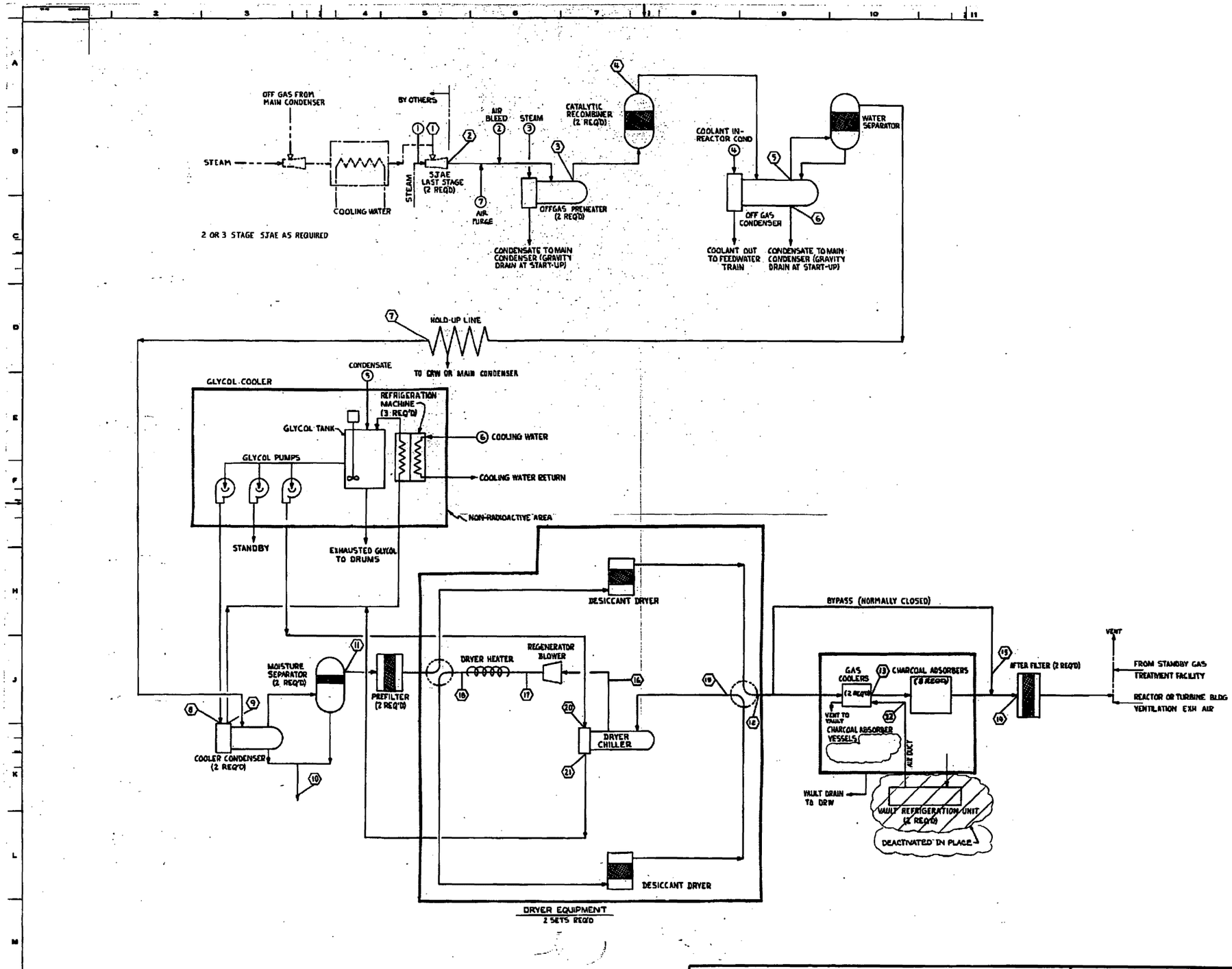
Table 11.3-10

Frequency and Quantity of Steam Discharged to Suppression Pool

Event	Frequency Category	Quantity of Steam (lb/event)
1. RCIC test (monthly)	Moderate	29,000
2. RCIC test (vessel injection at startup)	Moderate	116,000
3. Inadvertent RCIC injection	Moderate	5,000
4. SRV test	Moderate	118,000
5. Inadvertent SRV opening	Moderate	118,000
6. Trip of both recirculation pumps	Moderate	260,000[a]
7. Turbine trip	Moderate	18,000
8. Generator load rejection	Moderate	18,000
9. Pressure regulator failure - open	Moderate	256,000[a]
10. Recirculation flow control failure - decreasing	Moderate	260,000[a]
11. Loss of all feedwater flow	Moderate	267,000[a]
12. Inadvertent closure - all MSIV	Moderate	271,000[a]
13. Loss of condenser vacuum	Moderate	291,000[a]
14. Feedwater control failure - maximum demand	Moderate	270,000[a]
15. Loss of auxiliary transformer	Moderate	251,000[a]
16. Loss of all grid connections	Moderate	280,000[a]
17. Turbine trip w/o bypass	Moderate	126,000[a]
18. Generator load rejection w/o bypass	Moderate	127,000[a]
19. Stuck open SRV	Moderate	817,000

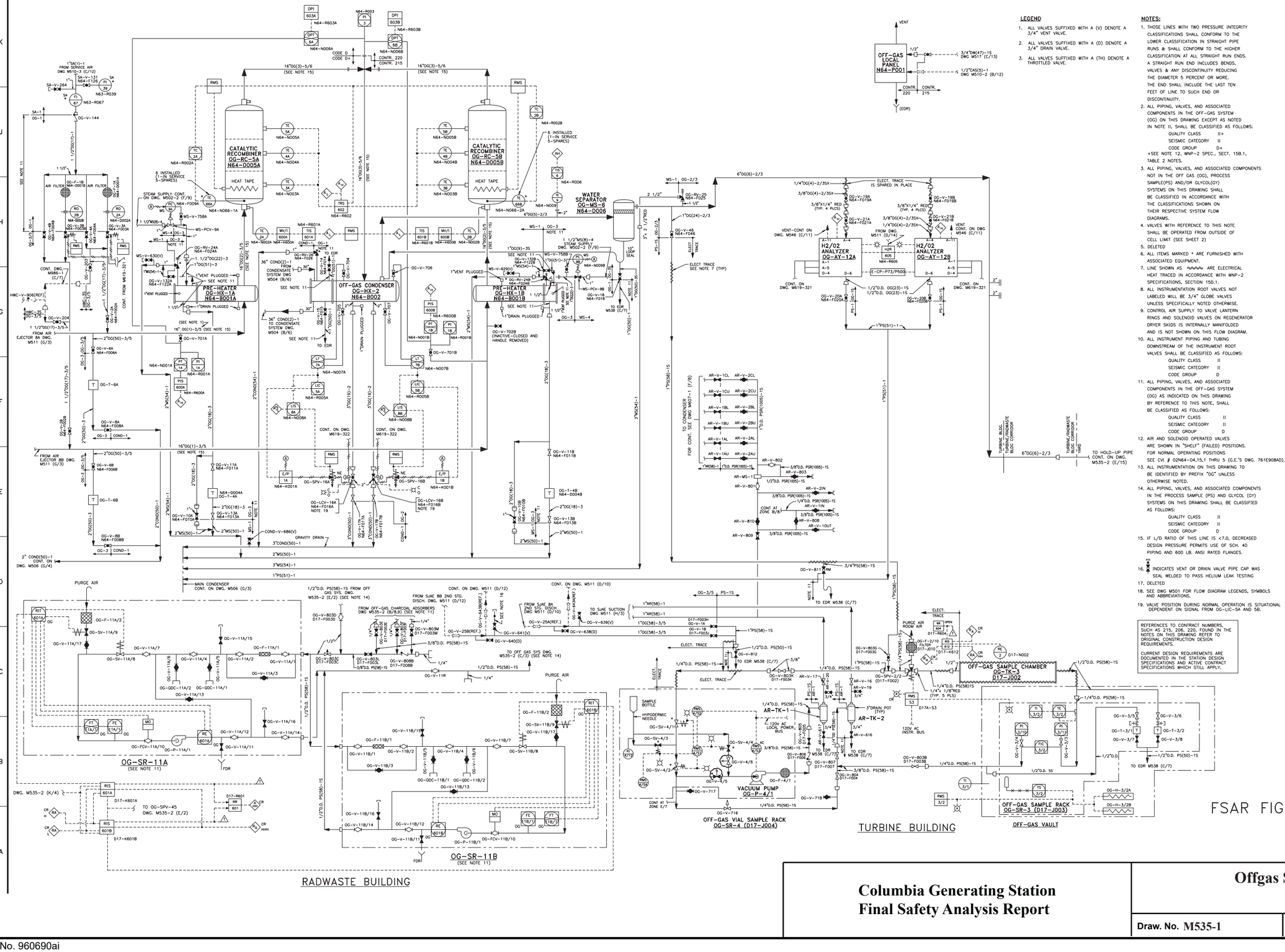
Notes: Events 1, 2, and 3 based on steam flow quantity during test mode per RCIC system process diagram.
Events 4 and 5 assuming test and inadvertent opening at 1000 psi reactor pressure for 10 minutes.
Events 6 through 18 based on event description from Chapter 15.
Event 19 based on results from 251 Standard Plant Suppression Pool Response Analyses.

[a] Isolation event. It is assumed that SRV cycling is terminated at T = 30 minutes and reactor depressurization.



Columbia Generating Station **Final Safety Analysis Report**	**Offgas System - Low Temperature**		
	Draw. No. 02N64-04,6,1	Rev. 3	Figure 11.3-1

Form No. 960690



LEGEND

1. ALL VALVES SUFFIXED WITH A (V) DENOTE A 3/4" VENT VALVE.
2. ALL VALVES SUFFIXED WITH A (D) DENOTE A 3/4" DRAIN VALVE.
3. ALL VALVES SUFFIXED WITH A (TH) DENOTE A THROTTLED VALVE.

NOTES:

1. THOSE LINES WITH TWO PRESSURE INTEGRITY CLASSIFICATIONS SHALL CONFORM TO THE LOWER CLASSIFICATION IN STRAIGHT PIPE RUNS & SHALL CONFORM TO THE HIGHER CLASSIFICATION AT ALL STRAIGHT RUN ENDS. A STRAIGHT RUN END INCLUDES BENDS, VALVES & ANY DISCONTINUITY REDUCING THE DIAMETER 5 PERCENT OR MORE. THE END SHALL INCLUDE THE LAST TEN FEET OF LINE TO SUCH END OR DISCONTINUITY.
2. ALL PIPING, VALVES, AND ASSOCIATED COMPONENTS IN THE OFF-GAS SYSTEM (OG) ON THIS DRAWING EXCEPT AS NOTED IN NOTE 11, SHALL BE CLASSIFIED AS FOLLOWS:
   QUALITY CLASS II+
   SEISMIC CATEGORY II
   CODE GROUP D+
   +SEE NOTE 12, WNP-2 SPEC., SECT. 15B.1, TABLE 2 NOTES.
3. ALL PIPING, VALVES, AND ASSOCIATED COMPONENTS NOT IN THE OFF GAS (OG), PROCESS SAMPLE(PS) AND/OR GLYCOL(GY) SYSTEMS ON THIS DRAWING SHALL BE CLASSIFIED IN ACCORDANCE WITH THE CLASSIFICATIONS SHOWN ON THEIR RESPECTIVE SYSTEM FLOW DIAGRAMS.
4. VALVES WITH REFERENCE TO THIS NOTE SHALL BE OPERATED FROM OUTSIDE OF CELL LIMIT (SEE SHEET 2)
5. DELETED
6. ALL ITEMS MARKED * ARE FURNISHED WITH ASSOCIATED EQUIPMENT.
7. LINE SHOWN AS ∿∿∿ ARE ELECTRICAL HEAT TRACED IN ACCORDANCE WITH WNP-2 SPECIFICATIONS, SECTION 150.1.
8. ALL INSTRUMENTATION ROOT VALVES NOT LABELED WILL BE 3/4" GLOBE VALVES UNLESS SPECIFICALLY NOTED OTHERWISE.
9. CONTROL AIR SUPPLY TO VALVE LANTERN RINGS AND SOLENOID VALVES ON REGENERATOR DRYER SKIDS IS INTERNALLY MANIFOLDED AND IS NOT SHOWN ON THIS FLOW DIAGRAM.
10. ALL INSTRUMENT PIPING AND TUBING DOWNSTREAM OF THE INSTRUMENT ROOT VALVES SHALL BE CLASSIFIED AS FOLLOWS:
    QUALITY CLASS II
    SEISMIC CATEGORY II
    CODE GROUP D
11. ALL PIPING, VALVES, AND ASSOCIATED COMPONENTS IN THE OFF-GAS SYSTEM (OG) AS INDICATED ON THIS DRAWING BY REFERENCE TO THIS NOTE, SHALL BE CLASSIFIED AS FOLLOWS:
    QUALITY CLASS II
    SEISMIC CATEGORY II
    CODE GROUP D
12. AIR AND SOLENOID OPERATED VALVES ARE SHOWN IN "SHELF" (FAILED) POSITIONS. FOR NORMAL OPERATING POSITIONS SEE CVI # 02N64-04,15,1 THRU 5 (G.E.'S DWG. 761E908AD).
13. ALL INSTRUMENTATION ON THIS DRAWING TO BE IDENTIFIED BY PREFIX "OG" UNLESS OTHERWISE NOTED.
14. ALL PIPING, VALVES, AND ASSOCIATED COMPONENTS IN THE PROCESS SAMPLE (PS) AND GLYCOL (GY) SYSTEMS ON THIS DRAWING SHALL BE CLASSIFIED AS FOLLOWS:
    QUALITY CLASS II
    SEISMIC CATEGORY II
    CODE GROUP D
15. IF L/D RATIO OF THIS LINE IS <7.0, DECREASED DESIGN PRESSURE PERMITS USE OF SCH. 40 PIPING AND 600 LB. ANSI RATED FLANGES.
16. ▼ INDICATES VENT OR DRAIN VALVE PIPE CAP WAS SEAL WELDED TO PASS HELIUM LEAK TESTING
17. DELETED
18. SEE DWG M501 FOR FLOW DIAGRAM LEGENDS, SYMBOLS AND ABBREVIATIONS.
19. VALVE POSITION DURING NORMAL OPERATION IS SITUATIONAL DEPENDENT ON SIGNAL FROM OG-LIC-5A AND 5B.

REFERENCES TO CONTRACT NUMBERS, SUCH AS 215, 206, 220, FOUND IN THE NOTES ON THIS DRAWING REFER TO ORIGINAL CONSTRUCTION DESIGN REQUIREMENTS.

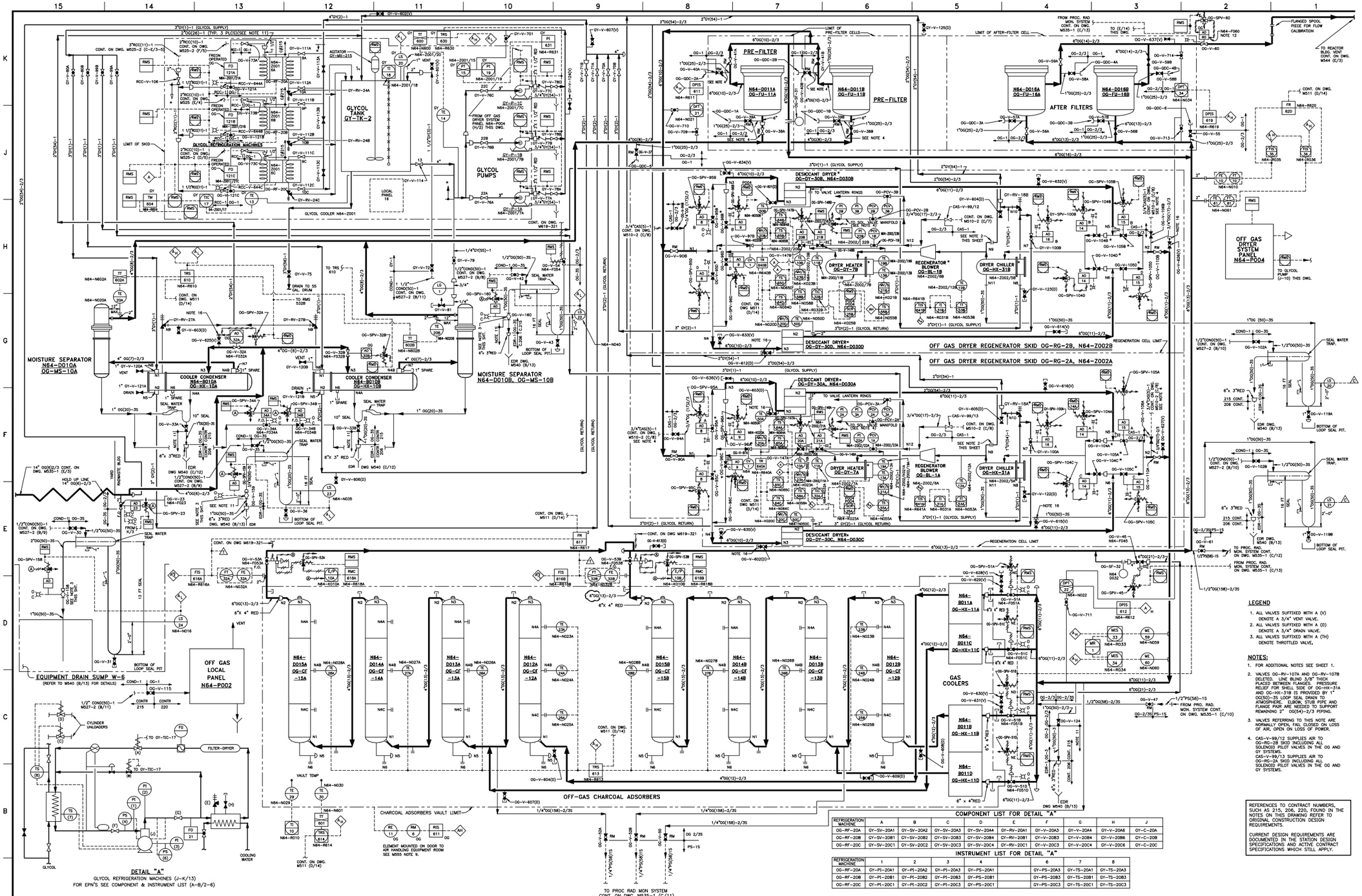
CURRENT DESIGN REQUIREMENTS ARE DOCUMENTED IN THE STATION DESIGN SPECIFICATIONS AND ACTIVE CONTRACT SPECIFICATIONS WHICH STILL APPLY.

FSAR FIG.

Columbia Generating Station
Final Safety Analysis Report

Offgas System P&ID

Draw. No. M535-1	Rev. 88	Figure 11.3-2.1


GLYCOL REFRIGERATION MACHINES
GLYCOL TANK GY-TK-2
GLYCOL PUMPS
GLYCOL COOLER N64-Z001
PRE-FILTER
N64-D011A OG-FU-11A
N64-D011B OG-FU-11B
AFTER FILTERS
N64-D016A OG-FU-16A
N64-D016B OG-FU-16B
DESICCANT DRYER* OG-DY-30B, N64-D030B
DESICCANT DRYER* OG-DY-30D, N64-D030D
DESICCANT DRYER* OG-DY-30A, N64-D030A
DESICCANT DRYER* OG-DY-30C, N64-D030C
DRYER HEATER OG-DY-7B
DRYER HEATER OG-DY-7A
REGENERATOR* BLOWER OG-BL-1B
REGENERATOR BLOWER OG-BL-1A
DRYER CHILLER OG-HX-31B
DRYER CHILLER OG-HX-31A
OFF GAS DRYER REGENERATOR SKID OG-RG-2B, N64-Z002B
OFF GAS DRYER REGENERATOR SKID OG-RG-2A, N64-Z002A
REGENERATION CELL LIMIT
MOISTURE SEPARATOR N64-D010A OG-MS-10A
MOISTURE SEPARATOR N64-D010B, OG-MS-10B
COOLER CONDENSER N64-B010A OG-HX-10A
COOLER CONDENSER N64-B010B OG-HX-10B
OFF GAS DRYER SYSTEM PANEL N64-P004
OFF GAS LOCAL PANEL N64-P002
OFF-GAS CHARCOAL ADSORBERS
N64-D015A OG-CF-15A
N64-D014A OG-CF-14A
N64-D013A OG-CF-13A
N64-D012A OG-CF-12A
N64-D012B OG-CF-12B
N64-D013B OG-CF-13B
N64-D014B OG-CF-14B
N64-D015B OG-CF-15B
GAS COOLERS
N64-B011A OG-HX-11A
N64-B011C OG-HX-11C
N64-B011B OG-HX-11B
N64-B011D OG-HX-11D
EQUIPMENT DRAIN SUMP W-6
(REFER TO M540 (B/13) FOR DETAILS)
DETAIL "A"
GLYCOL REFRIGERATION MACHINES (J-K/13)
FOR EPN'S SEE COMPONENT & INSTRUMENT LIST (A-B/2-6)
CHARCOAL ADSORBERS VAULT LIMIT
ELEMENT MOUNTED ON DOOR TO AIR HANDLING EQUIPMENT ROOM SEE M555 NOTE 9.
TO PROC RAD MON SYSTEM CONT. ON DWG. M535-1 (C/11)
TO REACTOR BLDG. VENT CONT. ON DWG. M544 (E/3)
FLANGED SPOOL PIECE FOR FLOW CALIBRATION
LIMIT OF SKID
LIMIT OF PRE-FILTER CELLS
LIMIT OF AFTER-FILTER CELL
LOCAL PANEL 16
EDR DWG M540 (B/13)
BOTTOM OF LOOP SEAL PIT.
VENT
DRAIN TO 55 GAL. DRUM
TO TRS 532B
SEAL WATER TRAP
SEE NOTE 2 THIS SHT.
SEE NOTE 3 THIS SHT.
SEE NOTE 4 THIS SHT.
SEE NOTE 11 THIS SHT.
SEE NOTE 16 THIS SHT.
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M519-321
CONT. ON DWG. M535-1 (C/5)
CONT. ON DWG. M525-2 (C/4-3-5)
CONT. ON DWG. M525-2 (D/5)
CONT. ON DWG. M525-2 (E/4)
CONT. ON DWG. M525-2 (G/5)
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M511 (E/14)
CONT. ON DWG. M511 (D/13)
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M511 (C/10)
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M511 (D/14)
CONT. ON DWG. M511 (D/14

11.4 SOLID WASTE MANAGEMENT SYSTEM

11.4.1 DESIGN BASIS

Power plant operation results in various solid radioactive wastes that require disposal. These wastes can be in the form of wet solids, such as powdered ion exchange resins from filter demineralizers, expended bead resins from deep bed demineralizers, small quantities of miscellaneous liquid, and miscellaneous dry materials such as paper, rags, plastic, and laboratory wastes.

The objective of the solid waste management system is to collect, monitor, process, and package these waste products in a suitable form for offsite shipment and burial. In designing the system to meet the stated objective, the following criteria were applied:

The system has the capacity to handle the volumes of waste from normal operations and anticipated operational occurrences. The expected annual volumes of wet solid wastes are shown in Table 11.4-1.

The system is designed to process the quantity of waste and concentration of radionuclides listed in Table 11.4-3 while maintaining occupational exposure as low as is reasonably achievable (ALARA). This is done by controlling the pipe run locations for shielding and exposure considerations, placing the process equipment in shielded areas, by providing remote operating stations, and by performing dewatering operations in radiation shields if required. The radwaste building shielding is designed for the highest radioactivity source, reactor water cleanup (RWCU) resins. The equipment layout is shown in general arrangement drawings in Section 1.2. Table 11.4-2 lists capacities, design pressure, and design temperature of the major equipment.

In keeping with the ALARA philosophy and Appendix I to 10 CFR 50, the solid waste management system's contribution to offsite doses is minimized by filtration and by directing the ventilation air flow from areas of low airborne contamination to areas of higher airborne contamination. The filtered ventilation radioactive releases are discussed in Section 11.3.

The solid waste management system operations and procedures are designed to limit the dose to offsite persons from station operations to significantly less than the limits specified in 10 CFR Part 20. Water separated in processing is returned to the liquid waste management system for treatment as described in Section 11.2.2.1 and shown in Figure 11.2-1.

The system can accommodate a variety of shipping container sizes and shapes with and without shields. Provisions are made for the detection and removal of loose surface contamination on the waste containers. The radiation levels of the waste containers are monitored so that

provisions can be made to ensure that shipping regulation radiation levels are not exceeded. Compliance with applicable regulations, e.g., 10 CFR Parts 61 and 71 and 49 CFR is discussed in Sections 11.4.2.9 and 11.4.3.

The safety class, quality group classification, quality class, and seismic category of radwaste systems are specified in Table 3.2-1. The solid waste management system is not designed to Seismic Category I. It is located in the Seismic Category I portion of the radwaste building. The seismic classification of the radwaste building is discussed in Section 3.8.4.1.2.

See Section 3.1.2.6.4 for a discussion of systems provided to meet General Design Criterion 63.

11.4.2 SYSTEM DESCRIPTION

A portable solid waste management system is used by Columbia Generating Station (CGS), as described in the following subsections.

It is required by 10 CFR 61 that, if certain activity criteria are exceeded, wet solid wastes be stabilized by solidification or processed to remove free standing liquids with containment in high integrity containers (HICs). The HICs may provide stability alone or in conjunction with an engineered barrier at the disposal site. The presently installed plant system is designed to interface with the portable dewatering/drying system which, when coupled with HICs, meet the requirements for stabilization in compliance with 10 CFR 61.

There are two resin dewatering systems used at CGS: the Self-Engaging Rapid Dewatering System (SERDS) and the Self Engaging Dewatering System (SEDS). The resin dewatering systems are vendor provided with NRC approved topical reports (RDS-25506-01-NP, Revision 1 for the SERDS and CNSI-DW-11118-01-P for the SEDS), and the systems are operated according to the technical requirements of this report. The types and quantities of waste to be processed are described below. The NRC licensed shipping casks and associated liners are used for processing, transporting, and disposing of wet solid wastes when required.

System operation is closely monitored by Energy Northwest personnel. If a vendor operates the dewatering system, the vendor will be required to submit their operating procedures for Energy Northwest review and approval. The vendor procedures will either be incorporated into a CGS procedure or will be approved “as is” by CGS prior to use.

Dewatering of resins to meet applicable dewatering criteria is conducted in accordance with approved procedures inside the radwaste building in the liner storage area or in shipping casks on trucks, where any spills are routed to existing floor drain sumps and the building ventilation filtration system ensures no unfiltered airborne releases occur from dewatering activities.

11.4.2.1 General

The sources of the various radioactive wet resin waste inputs to the system are shown in Figure 11.2-1. Table 11.2-11 shows the design basis expected frequency of input, the quantities of solids generated, the radioactivity level of the solids after accumulation, and the volume of liquid used in sluicing accumulated solids to the processing equipment. The excess liquid is subsequently returned to the liquid waste management system.

These values are based on experience from operational BWR nuclear power stations. Figure 11.4-1 shows the solid waste management system up to and including the portable portion of the system described in the vendor's topical reports. The phase separation and concentration portions of the system are shown on Figures 5.4-22, 10.4-5, 11.2-2, and 11.2-3. Tanks containing radioactive waste are provided with overflow connections which direct any overflow to drain sumps.

The solid waste processing areas are located in the radwaste building, where wet and dry solid wastes may be processed. Wet solid wastes include backwash resin from the RWCU system, the condensate filter demineralizer system, the fuel pool filter demineralizers, the floor drain and equipment drain filter demineralizers, and spent resin from the floor drain demineralizer and the waste demineralizer. Dry solid wastes include items such as rags, paper, plastics, small equipment parts, and laboratory wastes.

11.4.2.2 Radwaste Disposal System For Reactor Water Cleanup Resin

The backwash discharge from the cleanup filter demineralizers is collected and concentrated in two 4500-gal cleanup phase separators which are located below the cleanup demineralizers in the radwaste building. After several backwashes are accumulated, the waste is transferred to the portable dewatering system.

The cleanup phase separators are designed to concentrate the resin from 0.5% by weight solids to approximately 5% by weight solids by sedimentation and decantation of the slurry. While the working separator is filling, the other previously filled tank is held isolated to the extent practicable to allow for additional decay of resin activity.

After each backwash batch is received by the working separator, the batch is allowed to settle for a period of time and the decantate is then transferred by pumping to the waste collector tank. When sufficient resin has accumulated, the working separator is isolated and allowed to stand for a period to permit radioactive decay. At the end of this decay period the sludge is fluidized to approximately 5% weight solids and transferred by pumping to the portable dewatering system.

11.4.2.3 Radwaste Disposal System For Condensate Demineralizer Resin

The backwash discharge from the condensate filter demineralizers is collected in the condensate backwash receiving tank which is located below the condensate filter demineralizers in the radwaste building. After collection, the waste is transferred by pumping to one of the two condensate phase separators for processing.

Operation of the condensate phase separators is similar to that for the cleanup phase separators. Backwash resin is received at 0.5% by weight solids and concentrated to approximately 5% by weight solids, allowed to stand for a period of radioactive decay and then decanted and transferred by pumping to the portable dewatering system.

11.4.2.4 Radwaste Disposal System For Fuel Pool, Floor Drain, and Waste Collector Filter Resin

Backwash resin wastes from the fuel pool filter demineralizers, floor drain, and waste collector filter demineralizers are backwashed to the waste sludge phase separator tank. The waste sludge phase separator is designed to concentrate the resin from 0.5% by weight solids to approximately 5% by weight solids by sedimentation and decantation.

After each backwash batch is received by the separator, it is allowed to settle for a period of time and the decantate is then transferred by pumping to the floor drain collector tank.

When an appropriate quantity of resin is accumulated, the resin is fluidized to approximately 5% by weight solids and transferred by pumping to the portable dewatering system.

11.4.2.5 Radwaste Disposal System For Spent Resin

Spent bead resins from the floor and equipment drain polishing demineralizers are hydropneumatically transferred to the spent resin tank. The tank is designed to retain one batch of resins plus resin transfer water plus freeboard.

The decay time of any single batch is governed by the need to make the spent resin tank available to receive a subsequent batch from an alternate demineralizer. The frequency of spent resin discharge from the floor and equipment drain polishing demineralizers is estimated to be about once every 2 months. Each batch of the spent resin is transferred at approximately 40% by weight solids to the portable dewatering system.

11.4.2.6 Resin Container Handling and Storage

Filled containers from the dewatering operation in the container storage area are lifted by a crane and placed on a track-riding dolly. The dolly moves the containers from the storage area to the loading area where another crane lifts the container onto the truck for offsite shipment.

Alternatively, the dewatering operation for high activity resin waste can be performed in a cask on or off the truck bed. General locations and arrangements are shown in Figure 1.2-13.

11.4.2.7 Miscellaneous Dry Solid Waste System

Dry active waste may consist of air filtration media, miscellaneous paper, plastic and rags from contaminated areas, contaminated clothing, tools and equipment parts which cannot be effectively decontaminated, solid laboratory wastes, and other similar materials. The radioactivity of much of this waste is low enough to permit handling by contact. Compressible wastes may be compacted into metal containers to reduce their volumes. Alternately, container vans (C-vans) or other containers suitable for shipment may be used for dry radioactive waste shipped to a vendor for volume reduction services.

A relatively small quantity of high activity compactable and noncompactable dry active radioactive waste (DAW) may be loaded into the open top or encapsulation liners. Other containers whose geometry is compatible with shipment in shielded shipping casks may be used.

A locally controlled compactor system, if used, will include the following features: hydraulic pump with motor, hydraulic oil storage, high-efficiency filters, fan, and accessories. Ventilation air will be pulled across the top of the containers and then through high-efficiency filters by a fan during the compression process and exhausted to the radwaste building exhaust system described in Section 9.4.

Solid wastes and other nonliquid radioactive material and C-vans may be stored temporarily near the truck loading area outside the radwaste building, on the outdoor curbed pad adjacent to the radwaste building, or other suitable location. Noncompressible solid wastes are packaged in containers suitable for the waste. Low activity waste can be stored until enough is accumulated to permit economical transportation to an offsite disposal site or to a vendor for volume reduction of the waste prior to disposal.

Irradiated reactor components consisting of spent control rod blades, fuel channels, in-core ion chambers, and other equipment are stored in the spent fuel storage pool to allow for radioactive decay prior to shipment.

The packaging of dry solid wastes will be administratively controlled to ensure the 10 CFR 61 and/or burial facility free standing liquid criteria are met.

11.4.2.8 Expected Volumes

The design basis expected frequency of solids input, the quantities of solids generated, and the radioactivity of the solids after accumulation are shown in Table 11.2-11. Table 11.4-3 shows the expected solids production rate and significant nuclides associated with each batch of

dewatered waste. The distribution of these nuclides (dependent on the concentration in the reactor water) corresponds to the design basis offgas noble gas release rate as described in Section 11.3.1.

Table 11.4-1, excluding dry and compacted waste, shows the expected annual container production of solid wastes based on the process diagram data inputs. No decay after container filling operations has been considered.

11.4.2.9 Packaging

Radioactive wastes are packaged and shipped from CGS in containers that meet the requirements established in 49 CFR 171-180 for the Department of Transportation and 10 CFR 71 for the NRC.

Packaging of wastes being dewatered is typically performed remotely behind shielding as described below. Empty containers may be brought into the processing area using the dolly and filled with waste for processing. The quantity of wastes packaged in the container is controlled by operating procedures.

Dewatered liners are capped, surveyed for surface contamination, and decontaminated as necessary prior to shipment.

11.4.2.10 Storage Facilities

The general arrangement drawings in Section 1.2 show the layout of the radwaste handling areas in the radwaste building.

For the liners presently in use (of up to 210 ft^3 capacity), the storage area can accommodate about 15 filled liners. High activity containers can be stored to allow for 6 months decay prior to shipment if necessary. It is expected, however, that most radwaste containers will be shipped within 1 to 3 months of generation.

11.4.2.11 Shipment

The following describes a typical loading sequence for liners: A truck containing the cask is moved into the truck loading area. The dolly is moved to the storage area loading station and a capped container of waste is placed on the dolly by the storage area crane. The dolly is moved back to the truck loading area where the liner is lifted into the cask, and the cask lid is placed on the cask. The cask is decontaminated if necessary for shipment. Similar operations are performed when loading unshielded containers onto the truck. Dewatering of high activity wastes can also be performed in liners “in cask” in place on the shipping truck.

Any unshielded containers found to have external contamination exceeding 49 CFR limits are decontaminated. Further smear tests and cleaning are carried out as required until the activity on the container is within acceptable limits.

Radwaste tank failure and spent fuel cask drop incidents are discussed in Sections 15.7.3 and 15.7.5, respectively.

11.4.2.12 Process Monitoring

Process monitoring is performed by the dewatering system operator and the operator in the radwaste control room. They are in communication during waste transfer to the dewatering system. The dewatering system processing is monitored by remote closed-circuit TV cameras and other instrumentation as described in the topical reports for the process.

Each RWCU phase separator is equipped with one level indicating device for total liquid level. The total liquid level indicator utilizes an air bubbler and a pressure sensing level transmitter which drives a 0-100% level gauge and a high-level alarm in the radwaste control room. The level transmitter also drives a level indicator on the local control panel and provides control functions for the decant pump, the resin discharge pump, and the phase separator inlet selector valve.

The waste sludge phase separator has total liquid level indication. It uses an air bubbler and a pressure-sensing level transmitter. In addition to the level gauge and high-level alarm in the radwaste control room, the level transmitter provides control inputs to the decant pump, the stop and flush circuit on the sludge discharge pump, and the discharge valves from the waste collector and floor drain collector tanks to the waste sludge phase separator.

The condensate phase separators level instrumentation is the same as that described for the RWCU phase separators. Level indication for the spent resin tank is essentially the same as that described for the RWCU phase separators. The concentrated waste measuring and waste mixing tanks are not in service.

11.4.3 PROCESS CONTROL PROGRAM

11.4.3.1 Objective

The objectives of the process control program are to characterize and classify radwaste and ensure the complete solidification of all wet wastes being solidified and to ensure that dewatered wet wastes and disposal packages meet the free standing liquid and stability requirements of 10 CFR 61. To meet these objectives, the process control program has incorporated the recommendations set forth in NUREG-0800 and Branch Technical Position - ETSB 11-3.

11.4.3.2 Process Control Program

To ensure that acceptable waste forms are produced for disposal in compliance with the requirements in 10 CFR 61, the process control program provides for characterization of individual waste streams, classification of final waste products, proper disposal packaging, and verification that waste dewatering and/or solidification has been successful. In addition, nonstable waste form processing, waste storage, handling, and transportation activities take place under the process control program to ensure compliance with all applicable regulations in 10 CFR Parts 20, 61, and 71.

Stability of waste form at Columbia Generating Station is normally achieved through the combined properties of dewatered media and HICs or HICs plus engineered barriers. If solidification for stability is performed at Columbia Generating Station, it will be done in accordance with approved procedures to meet applicable disposal site license conditions and other applicable requirements.

Other processing methods such as portable deionization, solidification for nonstable waste form, evaporation, filtration, etc., may be used to process nonhazardous radioactively contaminated liquids. If these activities will result in the disposal of waste at a licensed burial site, they will be conducted in accordance with approved procedures to ensure that applicable disposal site license conditions and other requirements are met. Solidification agents listed in the burial facility license may be used to ensure free-standing liquid criteria are achieved.

The process control program contains the current formulas, sampling, analyses, test, and determinations to be made to ensure that processing and packaging of solid radioactive wastes based on demonstrated processing of actual or simulated wet solid wastes will be accomplished in such a way as to ensure compliance with 10 CFR Parts 20, 61, and 71, state regulations, burial ground requirements, and other requirements governing the disposal of solid radioactive wastes.

Changes to the process control program are documented, and records of reviews performed are retained for the duration of the operating license. This documentation contains sufficient information to support the change together with the appropriate analyses or evaluations justifying the change and a determination that the change will maintain the overall conformance of the solidified waste product to the existing requirements of federal, state, and other applicable regulations. Changes become effective after review and acceptance in accordance with the Operational Quality Assurance Program Description (OQAPD).

The radioactive waste process control program incorporates the following elements:

a. Waste stream descriptions,
b. Process controls,
c. Characterization,

d. Processing and stabilization,
e. Sampling,
f. Scaling factors,
g. Classification,
h. Computer code usage,
i. Analytical methods,
j. Packaging,
k. ALARA,
l. Shipping,
m. Documentation,
n. Equipment maintenance and calibration,
o. Minimization and segregation,
p. Storage,
q. Trending, and
r. Reporting.

Plant procedures implement the process control program.

11.4.3.3 Process Control Systems

The transfer of wet solid wastes to the processing system is monitored and controlled from the process control panel in the radwaste control room. The resin transfer system interfaces with the portable dewatering system at the point of connection with the dewatering equipment.

At the interface point, the applicable dewatering system is operated per the technical requirements of the NRC-approved topical report and is designed to ensure that processed waste in conjunction with the dewatering HIC or liner is prepared for burial and meets the 10 CFR Part 61 criteria. The process control program is described and controlled by procedure. The dewatering systems control panels are in a remote location shielded from the waste processing resin containing components. It is designed for automatic operation and provides indication and alarm for liner level, temperature, pressure, and other operating parameters.

If a waste processing vendor is used, they are monitored by Energy Northwest to ensure that applicable procedures, Energy Northwest or vendor, are being followed and that an acceptable end product is formed as described in the procedure used.

11.4.3.4 Waste Characterization

The wet wastes at CGS to be processed are characterized in individual streams for RWCU resins, equipment drain radioactive (EDR) and floor drain radioactive (FDR) powdered resins, EDR and FDR bead resins, and condensate resins. The four major systems producing wastes are described below.

a. Floor drain system - Wastes from the turbine building, reactor building, and radwaste building floor drain sumps are routinely monitored and collected for processing in the floor drain collector tank. The floor drain filter and demineralizer sludges are combined with equipment drain filters and sludges to form a mixture which is sampled prior to processing. Similarly, the EDR/FDR bead (polishing) resins are also sampled prior to processing.

b. Equipment drain system - Wastes from the turbine building, reactor building, and radwaste building equipment drain sumps are routinely monitored and collected in the waste collector tank. Sludges and resins from the high purity filter demineralizers are sampled prior to processing as described above in the FDR system description.

c. Condensate filter demineralizer system - The condensate polishing filter demineralizers use pressure precoated ion exchange media filters. Resins are sampled prior to processing.

d. Reactor water cleanup filter demineralizer system - The RWCU filter demineralizers use pressure precoated ion exchange media filters. Resins are sampled prior to processing.

Accumulations in collection tanks and phase separators will be tracked to aid in determining processing schedules and potential problems in system operation by tracking volumes. Selected coolant isotopic concentrations are trended to provide early indication of changes important to waste classification.

The waste streams in the foregoing systems are characterized by a periodic sampling and analysis program that establishes plant-specific isotopic correlation factors and relationships for inferring concentrations (i.e., scaling factors) of all 10 CFR 61 nuclides from easy to measure gamma-emitting species.

Individual waste stream activities and concentrations are determined for each batch prior to shipment for disposal. The evaluation of historical evidence is used as a screen to determine the appropriate disposal container followed by a formal evaluation based on actual samples or dose to curie determinations as appropriate.

11.4.3.5 Processing Methods (Wet Wastes)

The dewatering units are portable systems containing all necessary equipment and controls for removing the free water from ion-exchange resins and filter media.

Containers used for dewatering are furnished to CGS with factory installed "internals" functionally identical to those used during qualification testing. The determination of "functionally identical" may be performed by the vendor in accordance with their NRC approved Quality Assurance program or by CGS in accordance with the Quality Assurance program. The internals are free-standing and self-supporting, without protuberances which might damage the container. A fill head interfaces with the container and liner dewatering internals. The dewatering equipment is operated using the labeled controls on the control panel.

The resins that are below the Class A limits of 10 CFR 61 are normally dewatered in carbon steel containers. Resins exceeding Class A limit are dewatered in HICs to provide waste form stability.

The RDS process includes use of moisture indicators and the SEDS process uses dewatering verification to verify that free liquid criteria are met.

11.4.3.6 Control Instrumentation and Sampling Program

Processing of radwaste at CGS is conducted using instrumentation and controls for each batch to ensure that (a) suitable, well characterized waste is delivered to the various waste processing subsystems, (b) adequate process control information is provided to system operators to ensure adherence to proper operating parameter limits, such as tank levels, flow rates, release concentrations, etc., and (c) sufficient information is available to limit personnel exposures in conformance with the ALARA program.

The control instrumentation used at CGS includes in-process instruments, as well as portable radiation monitoring instruments.

The sampling program at CGS is a twofold program. Individual waste streams are characterized and classified by CGS personnel with the responsibility for shipping radioactive waste. Additionally, samples are sent to an offsite laboratory on a periodic basis to validate scaling factors.

11.4.3.7 Maintenance and Calibration

The control of the waste processing system is ensured by a thorough preventative and corrective maintenance program described in plant procedures for scheduled maintenance systems and maintenance work requests.

The control instrumentation is an integral part of the process system. The instruments providing the controlling functions are calibrated on a predetermined schedule and after each maintenance activity as applicable.

The periodic verification of calibration or recalibration assures that the process control program associated instruments are maintained and that conditions within the system are known.

The maintenance and calibration activities are performed in accordance with written procedures.

11.4.3.8 Waste Processing System Capacity

Wastes can be dewatered in up to 210 ft^3 liners in approximately 8 to 12 hr. This gives two shift operation a processing capacity of at least 100,000 ft^3 of waste per year. Similar processing capacity is available from portable solidification equipment which could be moved onsite if solidification to provide stabilization became necessary.

Dry wastes are segregated and monitored to reduce volumes where practicable. The expected design basis volume of radwaste is approximately 20,000 ft^3/year, however, actual volumes have been reduced to less than 10,000 ft^3/year. Forced outages and refueling outages increase volume but can be limited by preplanning material usage.

The radwaste processing capacity at CGS is sized to provide the needed capacity for anticipated occurrences and normal operation. This includes wet wastes, liquids, and solid wastes. Table 11.4-1 lists some of the major flowrates and capacities for several of the waste processes. Table 11.4-2 tabulates the major equipment items in the permanently installed waste processes.

11.4.3.9 Waste Storage Capacity

The storage capacity for DAW is adequate due to containerization allowing outside storage prior to shipping.

For the liners currently in use (up to 210 ft^3 capacity), approximately 15 liners can be readily accommodated. This storage space is available for the portable waste processing system and to store processed liners. Sufficient casks and transport packages have been factored into the planning to allow shipment of processed radwaste at a rate to preclude storage problems.

11.4.3.10 Compliance With ALARA Principles

a. Facility layout for the portable dewatering equipment - The portable dewatering systems are designed to meet Regulatory Guide 8.8. The waste container filling, capping, and decontamination operations are either performed remotely from a control area or as quickly as possible to minimize exposures. The system operator can view container filling and processing operations using a television monitoring system to prevent overfilling the waste container.

The portable system is located to limit the exposure of personnel to high dose rate piping and is designed to minimize the accumulation of crud deposits in the system. Piping and pumps are designed to allow complete flushing and, when possible, piping is flushed prior to maintenance. The control area is located in a relatively low radiation area away from the dewatering operation. Equipment with clean components is segregated from the areas containing waste except in the processing area. This allows maintenance on non-contaminated equipment to be performed in low radiation areas.

The placement of the portable processing system was determined based on criteria from Regulatory Guide 8.8 on ALARA and quality assurance provisions from Regulatory Guide 1.143. Processing radiation shields, in-cask processing and/or careful planning of processing activities contribute to efforts to minimize exposures from radwaste system operations.

b. Plant layout for ALARA - The installed radwaste processing system (excluding the portable dewatering system) was designed for remote operation from the radwaste control room in the 467-ft elevation of the radwaste building. The control room is designed with visual aids mimicking the processes being controlled. This allows for visual indication of processing status.

c. Exposure control - The radwaste operators control the processing of radioactive waste from the radwaste control room that is located in a low radiation area.

Most radwaste processing system components and systems can be remotely aligned from the radwaste control room without going into the radiation areas. This allows flexibility of operation and ensures processing capacity is maintained with reduced exposure to personnel. The systems and components can be flushed remotely from the control room reducing radiation levels for maintenance activities, also helping meet the ALARA concept.

The installed systems at CGS are sized with the capacity to process peak volumes of waste to reduce the effluents offsite, meeting criteria of 10 CFR 50, Appendix I.

The processing control from a remote control room and the ability to process all waste streams meets the intent of ALARA for both onsite and offsite.

11.4.3.11 Unanticipated Wastes

The waste streams at CGS have been characterized for normal expected waste components. Periodically, there may be waste produced from operations such as decontamination or cleaning that have not been characterized.

These wastes are classified and processed for disposal based on the practical experience of the CGS staff. It is unlikely that any wastes will be produced at CGS that cannot be prepared for disposal in accordance with 10 CFR 61. When changes in the system or process do occur that are out of the ordinary, processing requirements will be determined on a case-by-case basis.

11.4.3.12 Waste Classification

Waste is classified at CGS to determine if wastes meet 10 CFR 61 Class A, B, or C. The individual waste stream will be sampled and analyzed for nuclides in Tables 1 and 2 of 10 CFR 61 to establish the waste classification. This analysis may in many instances use scaling factors to tie difficult to measure nuclides, to more easily identified gamma emitting nuclides. The requirements for the determination of these scaling factors is administratively controlled by plant procedure.

11.4.3.13 Waste Packaging and Shipping

The radioactive waste shipped from CGS for disposal or further processing is containerized, prepared, and shipped in accordance with applicable state, DOT, and NRC regulations.

Table 11.4-1

Waste Processing Systems Capacities

Waste Processing Systems	Flow Rates and Capacities
Liquid Waste	
EDR subsystem	14,341 to 104,826 gpd
EDR storage	135,000 gal
FDR subsystem	5700 to 55,289 gpd
FDR storage	40,000 gal
Chemical waste subsystem	1500 to 20,600 gpd
Chemical waste storage	65,000 gal
Filter sludge and chemical waste concentrates	
Normal	8000 to 16,000 ft^3/year
Processed volume for disposal	8000 to 16,000 ft^3/year
Portable solidification system	
Normal	200 ft^3/day
Maximum	100,000 ft^3/year[a]
Storage	72 50-ft^3 liners
Dry compactible waste	
Compactible radwaste	20,000 ft^3/year
Dewatering system	
Normal	200 ft^3/12-hr shift
Maximum	100,000 ft^3/year[a]

[a] Figured on two shifts with increase in personnel.

Table 11.4-2

Solid Waste Management System Major Equipment Items

Equipment	Number Required	Construction	Design Pressure	Design Temperature	Capacity
Cleanup phase separators	2	Stainless steel shell and internals	Atmospheric	250°F	4500 gal each
Cleanup sludge discharge mixing pump	1	Stainless steel	150 psig	150°F	210 gpm at 170 ft TDH
Cleanup decant pump	1	Stainless steel	150 psig	150°F	53 gpm at 50 ft TDH
Condensate backwash receiving tank	1	Stainless steel shell and internals	Atmospheric	150°F	19,000 gal
Condensate backwash transfer pump	1	Stainless steel		150°F	450 gpm at 50 ft TDH
Condensate phase separator	2	Epoxy-coated carbon steel shell, stainless steel internals	Atmospheric	250°F	23,500 gal each
Condensate sludge discharge mixing pump	1	Stainless steel	150 psig	150°F	420 gpm at 160 ft TDH
Condensate decant pump	1	Stainless steel	150 psig	150°F	450 gpm at 50 ft TDH
Waste sludge phase separator tank	1	Epoxy-coated carbon steel, stainless steel internals	Atmospheric	150°F	13,000 gal
Waste decant pump	1	Stainless steel	150 psig	150°F	53 gpm at 50 ft TDH
Waste sludge discharge mixing pump	1	Stainless steel	150 psig	150°F	210 gpm at 105 ft TDH

Table 11.4-2

Solid Waste Management System Major Equipment Items (Continued)

Equipment	Number Required	Construction	Design Pressure	Design Temperature	Capacity
Spent resin tank	1	Stainless steel shell and internals	Atmospheric	150°F	1200 gal
Spent resin pump	1	Stainless steel	150 psig	150°F	21 gpm at 105 ft TDH
Decontamination solution concentrated waste tank[a]	2	Stainless steel shell and internals	Atmospheric	150°F	700 gal each
Decontamination solution concentrated waste pump[a]	1	Stainless steel	150 psig	150 °F	30 gpm at 70 ft TDH
Concentrated waste measuring tank[a]	1	Stainless steel	Atmospheric	150°F	400 gal
Transfer dolly[b]	1				

[a] Not in service.

[b] Track riding dolly for transfer of waste containers between the storage area and truck shipping area.

Table 11.4-3

Significant Isotope Activity in Dewatered Waste

Stream	Clean up Sludge	Waste Sludge	Equipment Drain Resin	Floor Drain Resin	Condensate Sludge
Batch Solid Production	524 lb/60 days	220 lb/3.4 days	1539 lb/66 days	1539 lb/67 days	3300 lb/18.5 days
Isotopes			Ci/ft^3		
^{89}Sr	0.63	--	9.0×10^{-3}	6.5×10^{-5}	4.8×10^{-3}
^{90}Sr	0.16	--	1.0×10^{-3}	7.2×10^{-6}	8.4×10^{-4}
^{91}Sr	--	--	5.2×10^{-3}	3.7×10^{-5}	--
^{92}Sr	0.46	--	1.0×10^{-3}	3.4×10^{-5}	7.1×10^{-6}
^{90}Y	0.16	--	1.0×10^{-3}	3.4×10^{-5}	8.4×10^{-4}
^{91}Y	0.023	--	2.3×10^{-3}	--	1.7×10^{-3}
^{91m}Y	--	--	3.6×10^{-3}	--	--
^{92}Y	--	--	1.0×10^{-3}	--	--
^{95}Zr	0.29	8.1×10^{-4}	--	--	4.2×10^{-4}
^{95}Nb	0.44	1.1×10^{-3}	8.4×10^{-4}	--	6.3×10^{-4}
^{99}Mo	--	2.0×10^{-3}	2.9×10^{-3}	2.4×10^{-5}	1.3×10^{-4}
^{99m}Tc	0.0015	1.3×10^{-3}	9.7×10^{-3}	1.1×10^{-4}	1.3×10^{-4}
^{103}Ru	--	--	--	--	2.1×10^{-4}
^{122}Sb	0.020	2.0×10^{-4}	1.1×10^{-5}	--	--
^{129m}Te	0.038	--	1.3×10^{-4}	6.9×10^{-7}	--
^{129}Te	0.024	--	--	--	--
^{132}Te	--	--	1.6×10^{-2}	1.3×10^{-4}	1.3×10^{-4}
^{83}Br	--	--	1.3×10^{-4}	4.5×10^{-6}	--

Table 11.4-3

Significant Isotope Activity in Dewatered Waste (Continued)

Stream	Clean up Sludge	Waste Sludge	Equipment Drain Resin	Floor Drain Resin	Condensate Sludge
Batch Solid Production	524 lb/60 days	220 lb/3.4 days	1539 lb/66 days	1539 lb/67 days	3300 lb/18.5 days
Isotopes			Ci/ft^3		
^{131}I	0.077	7.8×10^{-4}	1.01×10^{-2}	7.4×10^{-5}	2.3×10^{-2}
^{132}I	--	--	1.7×10^{-3}	3.5×10^{-5}	1.3×10^{-4}
^{133}I	--	--	7.3×10^{-3}	7.7×10^{-5}	--
^{134}I	--	--	8.4×10^{-4}	4.7×10^{-5}	--
^{135}I	--	--	3.6×10^{-3}	5.7×10^{-5}	--
^{134}Cs	0.17	3.0×10^{-3}	6.0×10^{-4}	4.9×10^{-6}	4.2×10^{-4}
^{136}Cs	0.0078	--	1.3×10^{-4}	--	--
^{137}Cs	0.18	4.4×10^{-3}	1.0×10^{-3}	7.6×10^{-6}	8.4×10^{-4}
^{138}Cs	--	--	2.1×10^{-4}	--	--
^{137m}Ba	0.17	--	1.0×10^{-3}	--	6.3×10^{-4}
^{139}Ba	--	--	8.4×10^{-4}	3.9×10^{-5}	--
^{140}Ba	0.076	--	1.0×10^{-2}	2.6×10^{-5}	2.7×10^{-3}
^{141}Ce	--	--	1.3×10^{-4}	--	3.8×10^{-3}
^{144}Ce	0.019	--	1.3×10^{-4}	4.7×10^{-6}	6.3×10^{-4}
^{140}La	0.087	3.9×10^{-5}	1.0×10^{-2}	--	2.9×10^{-3}
^{141}La	--	--	2.1×10^{-4}	--	--
^{142}La	--	--	1.3×10^{-4}	--	--
^{143}Pr	0.022	--	1.3×10^{-4}	--	8.4×10^{-4}

Table 11.4-3

Significant Isotope Activity in Dewatered Waste (Continued)

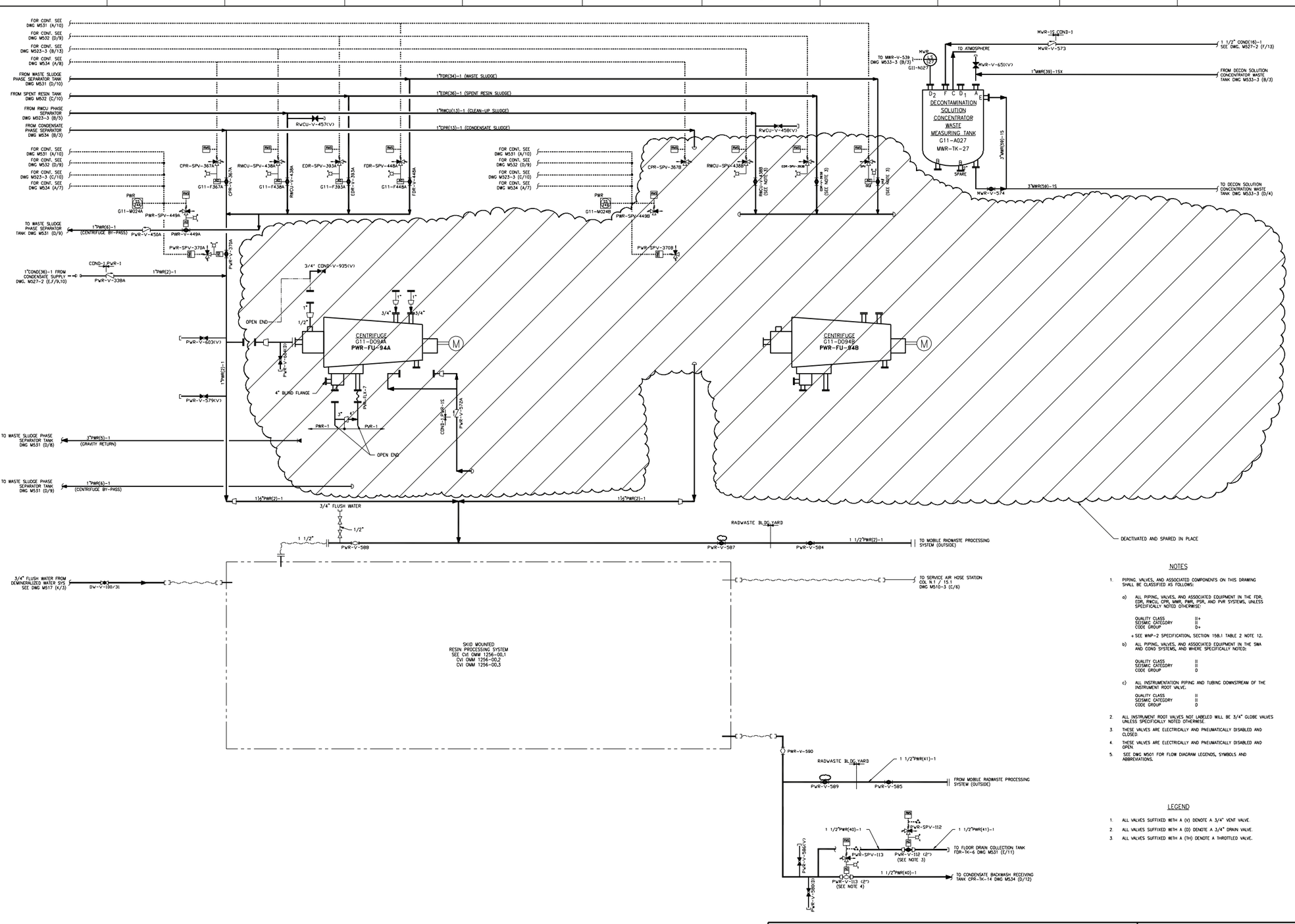
Stream	Clean up Sludge	Waste Sludge	Equipment Drain Resin	Floor Drain Resin	Condensate Sludge
Batch Solid Production	524 lb/60 days	220 lb/3.4 days	1539 lb/66 days	1539 lb/67 days	3300 lb/18.5 days
Isotopes			Ci/ft^3		
^{144}Pr	0.019	--	1.3×10^{-4}	--	6.3×10^{-4}
^{239}Np	--	--	5.4×10^{-2}	4.5×10^{-4}	--
^{51}Cr	1.8	1.5×10^{-2}	4.4×10^{-4}	4.1×10^{-6}	3.1×10^{-3}
^{54}Mn	0.089	2.1×10^{-3}	1.3×10^{-4}	7.0×10^{-5}	6.6×10^{-4}
^{56}Mn	--	2.3×10^{-4}	2.2×10^{-4}	7.8×10^{-6}	--
^{58}Co	1.8	3.3×10^{-3}	8.4×10^{-3}	6.1×10^{-5}	6.2×10^{-2}
^{60}Co	0.65	2.4×10^{-2}	1.1×10^{-3}	9.2×10^{-6}	9.7×10^{-3}
^{59}Fe	0.018	2.6×10^{-4}	1.3×10^{-4}	8.3×10^{-7}	8.8×10^{-4}
^{65}Zn	2.1	7.4×10^{-2}	3.5×10^{-4}	2.3×10^{-4}	4.4×10^{-4}
^{110m}Ag	0.14	8.8×10^{-6}	1.3×10^{-4}	9.0×10^{-6}	1.1×10^{-3}
^{187}W	--	1.5×10^{-4}	1.3×10^{-4}	1.4×10^{-6}	--
Total	9.67	1.4×10^{-1}	1.7×10^{-1}	1.7×10^{-3}	1.2×10^{-1}

Table 11.4-4

Expected Annual Production of Solids

	ft^3/year	Normal Activity Ci/Container[a]	Maximum Activity Ci/Container[a]
Cleanup filter demineralizer sludge	720	1196	1820
Condensate filter demineralizer sludge	7460	19	40
Waste floor drain and fuel pool filter demineralizer sludge	1000	0.2	3.4
Waste demineralizer resin	540	29	330
Total volume (ft^3)	9720		

[a] Based on 164 ft^3 of dewatered resin per container



NOTES

1. PIPING, VALVES, AND ASSOCIATED COMPONENTS ON THIS DRAWING SHALL BE CLASSIFIED AS FOLLOWS:
   a) ALL PIPING, VALVES, AND ASSOCIATED EQUIPMENT IN THE FDR, EDR, RWCU, CPR, MWR, PWR, PSR, AND PVR SYSTEMS, UNLESS SPECIFICALLY NOTED OTHERWISE:

QUALITY CLASS	II+
SEISMIC CATEGORY	II
CODE GROUP	D+

      + SEE WNP-2 SPECIFICATION, SECTION 15B.1 TABLE 2 NOTE 12.
   b) ALL PIPING, VALVES, AND ASSOCIATED EQUIPMENT IN THE SWA AND COND SYSTEMS, AND WHERE SPECIFICALLY NOTED:

QUALITY CLASS	II
SEISMIC CATEGORY	II
CODE GROUP	D

   c) ALL INSTRUMENTATION PIPING AND TUBING DOWNSTREAM OF THE INSTRUMENT ROOT VALVE.

QUALITY CLASS	II
SEISMIC CATEGORY	II
CODE GROUP	D

2. ALL INSTRUMENT ROOT VALVES NOT LABELED WILL BE 3/4" GLOBE VALVES UNLESS SPECIFICALLY NOTED OTHERWISE.
3. THESE VALVES ARE ELECTRICALLY AND PNEUMATICALLY DISABLED AND CLOSED.
4. THESE VALVES ARE ELECTRICALLY AND PNEUMATICALLY DISABLED AND OPEN.
5. SEE DWG M501 FOR FLOW DIAGRAM LEGENDS, SYMBOLS AND ABBREVIATIONS.

LEGEND

1. ALL VALVES SUFFIXED WITH A (V) DENOTE A 3/4" VENT VALVE.
2. ALL VALVES SUFFIXED WITH A (D) DENOTE A 3/4" DRAIN VALVE.
3. ALL VALVES SUFFIXED WITH A (TH) DENOTE A THROTTLED VALVE.

Columbia Generating Station
Final Safety Analysis Report

Flow Diagram Radioactive Waste Disposal Solids Handling System

Draw. No. M536	Rev. 33	Figure 11.4-1

11.5 PROCESS AND EFFLUENT RADIOLOGICAL MONITORING AND SAMPLING SYSTEMS

The process and effluent radiological monitoring and sampling systems are provided to allow determination of the content of radioactive material in various gaseous and liquid process and effluent streams. The design objective and criteria are primarily determined by the system designation of either

a. Instrumentation systems required for safety, or
b. Instrumentation systems required for plant operation.

11.5.1 DESIGN BASIS

11.5.1.1 Design Objectives

The process and effluent radiological monitoring and sampling system is designed to provide for compliance with the requirements of 10 CFR Part 50 including the General Design Criteria (GDC) of Appendix A and provides the monitoring and sampling required to make measurements, evaluations, and reports recommended by Regulatory Guide 1.21, Revision l.

11.5.1.1.1 Systems Required for Safety

The main objective of radiation monitoring systems (RMS) required for safety is to initiate appropriate protective action to limit the release of radioactive materials from the reactor vessel and reactor building if predetermined radiation levels are exceeded in major process/effluent streams and to limit inflow of airborne radioactivity to the control room following an accidental release. Additional objectives are to have these systems available under all operating conditions, including accidents and postaccidents, and to provide control room personnel with an indication of the radiation levels in the major process/effluent streams plus alarm annunciation if high radiation levels are detected.

The RMS provided to meet these objectives are

a. Main steam line RMS,
b. Reactor building ventilation exhaust plenum RMS, and
c. Standby service water RMS.

11.5.1.1.2 Systems Required for Plant Operation

The main objective of the RMS is to provide operating personnel with measurement of the content of radioactive materials in all effluent and important process streams. This allows demonstration of compliance with Technical Specifications by providing gross radiation level

monitoring and collection of halogens and particulates on cartridges and filters (gaseous effluents). Additional objectives are to initiate discharge valve isolation on the offgas, liquid radwaste, or drain systems if predetermined release rates are exceeded and to provide for sampling at certain radiation monitor locations to allow determination of specific radionuclide content.

The RMS provided to meet these objectives are

a. For gaseous process streams

   1. Offgas pretreatment RMS,
   2. Offgas post-treatment RMS,
   3. Charcoal bed vault RMS, and
   4. Mechanical vacuum pump exhaust RMS.

b. For gaseous effluent streams

   1. Reactor building elevated release duct RMS,
   2. Turbine generator building ventilation release duct RMS, and
   3. Radwaste building ventilation release duct RMS.

c. For liquid process streams

   1. Standby service water RMS and
   2. Reactor building closed cooling water RMS.

d. For liquid effluent streams

   1. Radwaste effluent RMS,
   2. Circulating water (blowdown) RMS,
   3. Standby service water RMS, and
   4. Plant service water (TSW) RMS.

e. For primary containment monitoring

   1. Leak detection RMS and
   2. Loss-of-coolant accident (LOCA) tracking RMS.

11.5.1.2 Design Criteria

11.5.1.2.1 Systems Required for Safety

The design criteria for the RMS required for safety are that the systems shall:

a. Withstand the effect of natural phenomena (e.g., earthquakes) without loss of capability to perform their functions,

b. Perform its intended safety function in the environment resulting from normal and postulated accident conditions,

c. Meet the reliability, testability, independence, and failure mode requirements of engineered safety features,

d. Provide continuous output on control room panels,

e. Permit checking of the operational availability of each channel during reactor operation with provision for calibration function and instrument checks,

f. Ensure an extremely high probability of accomplishing its safety function in the event of anticipated operational occurrences,

g. Initiate prompt protective action prior to exceeding Technical Specifications limits,

h. Provide warning of increasing radiation levels indicative of abnormal conditions by alarm annunciation,

i. Insofar as practical, provide self-monitoring of components to the extent that power failure or component malfunction causes annunciation and channel trip,

j. Register full scale output if radiation detection exceeds full scale, and

k. Have sensitivities and ranges compatible with anticipated radiation levels.

The applicable GDC are 1, 2, 3, 4, 13, 19, 20, 21, 22, 23, 24, 29, 60, and 64 (see Section 3.1). The systems shall meet the design requirements for Safety Class 2, Seismic Category I systems, along with the quality assurance requirements of 10 CFR Part 50, Appendix B.

11.5.1.2.2 Systems Required for Plant Operation

The design criteria for operational RMS are that the systems shall

a. Provide continuous indication of radiation levels in the main control room,

b. Provide warning of increasing radiation levels indicative of abnormal conditions by alarm annunciation,

c. Insofar as practical, provide self-monitoring of components to the extent that power failure or component malfunction causes annunciation and, for systems initiating protective action, channel trip,

d. Monitor a sample representative of the bulk stream or volume,

e. Have provisions for calibration, function and instrumentation checks,

f. Have sensitivities and ranges compatible with anticipated radiation levels and Technical Specifications limits, and

g. Register full scale output if radiation detection exceeds full scale.

The RMS monitors discharges from the gaseous and liquid radwaste treatment systems and nonradioactive sumps have provisions to alarm and to initiate automatic closure of the effluent discharge valves on the affected treatment systems prior to exceeding the normal operation limits specified in the Technical Specifications. Additionally, the primary containment monitoring system meets the criteria, except for item g.

The applicable GDC are 13, 60, and 64 (see Section 3.1).

11.5.2 SYSTEM DESCRIPTION

11.5.2.1 Systems Required for Safety

Information on these systems is presented in Tables 11.5-1 and 11.5-2 and the arrangements shown in Figures 11.5-1 through 11.5-4. The equipment is designed to Quality Class I and Seismic Category I requirements. High reliability is further achieved by the use of redundancy as noted below.

11.5.2.1.1 Main Steam Line Radiation Monitoring System

This system monitors the gamma radiation level exterior to the main steam lines. The normal radiation level is produced primarily by coolant activation gases plus smaller quantities of

fission gases being transported with the steam. In the event of a gross release of fission products from the core, this monitoring system provides signals to the following:

a. Reactor water sample valves,
b. Mechanical vacuum pumps,
c. Mechanical vacuum pumps isolation valves,
d. Gland seal exhausters, and
e. Control room annunciators.

The system consists of four redundant instrument channels. Each channel consists of a local detector (gamma-sensitive ion chamber) and a control room radiation recorder and a readout module with an auxiliary trip unit. Power for two channels (A and C) is supplied from the reactor protection system (RPS) bus A and for the other two channels (B and D) from RPS bus B. Channels A and C are physically and electrically independent of channels B and D.

The detectors are located near the main steam lines in the steam tunnel as it enters the turbine building. The detectors are geometrically arranged so that this system is capable of detecting significant increases in radiation level with any number of main steam lines in operation.

Each radiation monitor has four trip circuits: two upscale (high-high and high), one downscale (low), and one inoperative. Each trip is visually displayed on the affected radiation readout module. A high-high or inoperative trip results in a channel trip in the auxiliary unit which is an input to the reactor water sample valves, mechanical vacuum pump shutdown, and discharge valve closure. A high trip actuates a main steam line (MSL) high control room annunciator common to all channels. A downscale and inoperative trip actuates a MSL downscale/inoperative control room annunciator common to all channels. High and low trips do not result in a channel trip. Each radiation monitor displays the measured radiation level.

Arrangement details are shown in Figure 11.5-1.

11.5.2.1.2 Reactor Building Exhaust Plenum Radiation Monitoring System

This system monitors the radiation level of the reactor building ventilation system exhaust plenum prior to its discharge from the building into the elevated release duct. A high radioactivity level in the exhaust system could be due to fission gases from a leak or an accident.

The system consists of four redundant instrument channels. Each channel consists of a local detection assembly (a sensor and converter unit containing a Geiger-Mueller (GM) tube and electronics) and a control room radiation readout module. The 120-V ac power for channels (A and B) is provided from Division 1, and for channels (C and D) from Division 2 power panels; the multipoint strip chart recorder supplied from the Division 2 uninterruptible power supply (UPS) power panel records the output of all four channels. The detection assemblies

are located outside the exhaust air plenum upstream of the secondary containment discharge isolation valves. The distance upstream from the inboard discharge isolation valve is such that, at the maximum design flow, the transport time from the detector location to the inboard discharge valve is greater than the total time required to respond to trip level radiation and close the inboard discharge valve before exceeding 10 CFR 50.67 dose limits (see Section 9.4.2.3).

Each radiation monitor has two trip circuits: one upscale (high-high) and one downscale/inoperative (fail safe design). Two-out-of-two upscale/downscale trips (channels A and B) initiates closure of the reactor building ventilation outboard isolation valves and the primary containment outboard purge and vent valves, and initiates startup of standby gas treatment (SGT) system train B. The same condition for channels C and D initiates closure of the corresponding inboard valves and initiates startup of SGT train A.

An upscale trip is displayed on the affected radiation readout module and actuates a reactor building vent high-high radiation control room annunciator common to all channels.
A downscale trip is also displayed on the radiation readout module and actuates a reactor building vent downscale control room annunciator common to all channels.

An additional trip signal for high radiation alarm is provided by the recorder and actuates a reactor building vent high radiation control room annunciator. Each radiation monitor displays the measured radiation level.

Arrangement details are shown in Figure 11.5-2.

11.5.2.1.3 Deleted

11.5.2.1.4 Standby Service Water Radiation Monitoring System

This system monitors gamma radiation levels of the service water liquid process and effluent streams.

Each monitor system consists of a gamma scintillation detector inserted into an offline chamber to which a process stream sample is piped. The detector locations are selected to obtain a reasonable geometry and are positioned away from crud trap and associated high background regions. Lead shielding is provided to further reduce background levels.

At each liquid offline detector location, a continuous sample from the liquid process pipe passes through a shielded detection assembly for gross radiation monitoring and then is returned to the process pipe. The detection assembly consists of a detector mounted in a shielded sample chamber. The local radiation monitor and the meter in the control room display the measured gross radiation level. The sample chamber and lines can be drained to allow assessment of background buildup. The flow meter at the sample rack provides local sample line flow indication.

Sample flow for each detector is from the standby service piping downstream of each of the two residual heat removal (RHR) heat exchanger (loops A and B). These monitors are designed to detect any primary coolant leakage into the standby service water through the RHR heat exchanger, during operation of the RHR heat exchangers in the shutdown heat removal mode.

Additional details are shown in Figure 11.5-4.

11.5.2.2 Systems Required for Plant Operation

All systems associated with the plant process cycle provide for indication and recording of radiation levels in the main control room in conjunction with alarm annunciation features.

Information on these systems is presented in Tables 11.5-1 and 11.5-2 and the arrangements are shown in Figures 11.5-2 and 11.5-4 through 11.5-10.

11.5.2.2.1 Gaseous Process and Effluent Radiation Monitoring System

11.5.2.2.1.1 Offgas Pretreatment Radiation Monitoring System. This system monitors radioactivity at the outlet of the water separator downstream of the catalytic recombiners. The monitor detects the radiation level which is attributable to the fission gases produced in the reactor and transported with steam through the turbine to the condenser.

A continuous sample is extracted from the offgas pipe via a stainless steel sample line that is then passed through a sample chamber and a sample panel before being returned to the suction side of the steam jet air ejector (SJAE). The sample chamber is a steel pipe which is internally polished to minimize plateout. It can be purged with room air to check detector response by using a three-way solenoid-operated valve. The valve is controlled by a switch located in the main control room. The sample panel measures and indicates sample line flow.

The detector is a gamma-sensitive ionization chamber mounted external to the sample chamber. The channel has a logarithmic radiation readout module which provides a system alarm output and is provided with a recorder.

The 120-V ac UPS power for count ratemeter and trip auxiliary circuit is supplied from critical Division 2 panel; strip chart recorder receives reliable (Division B) instrument power, and local 120-V ac for the offgas sample and vial sampler control panel.

The radiation readout module has four trip circuits: two upscale (high-high and high), one downscale (low), and one inoperative. The trip outputs are used for alarm function only. Each trip is displayed on the radiation monitor and actuates a control room annunciator: offgas high-high, offgas high, and offgas downscale/inoperative. Sample line flow is displayed at the sample panel.

The radiation level output by the monitor can be directly correlated to the concentration of the noble gases by using the semiautomatic vial sampler panel to obtain a grab sample. To draw a sample, a serum bottle is inserted into a sample chamber, the sample lines are evacuated, and a solenoid-operated sample valve is opened to allow offgas to enter the bottle. After the sampling bottle is removed, the sample is analyzed in the counting room with a multichannel gamma analyzer to determine the concentration of the various noble gas radionuclides. A correlation between the observed activity and the monitor reading permits calibration of the monitor.

For arrangement details see Figure 11.5-5.

11.5.2.2.1.2 Offgas Posttreatment Radiation Monitoring System. This system monitors radioactivity in the offgas piping downstream of the offgas system charcoal vessels and upstream of the offgas system discharge valve. A continuous sample is extracted from the

offgas system piping, passed through the offgas posttreatment sample panels for monitoring and sampling, and returned to the offgas system piping. Each sample panel has a cartridge and filter (one for particulate collection and one for halogen collection) in series with filter bypass capabilities (with respect to flow) with two identical continuous gross radiation detection assemblies. Each gross radiation detection assembly consists of a shielded chamber, a radiation detector, and a check source. Two radiation monitors in the main control room display the measured gross radiation level.

The sample panels and shielded chambers can be purged with room air to check detector response by using solenoid valves that can be operated from the control room or at the sample panels. The sample panels measure and indicate sample line flow. A solenoid operated check source for each detection assembly operated from the control room or locally can be used to check operability of the gross radiation channel.

Channel A monitor receives reliable (Division A) 120-V ac power and channel B is supplied with Division 2 power; the ±24-V dc auxiliary trip circuits are supplied internally from the monitor power supply. Control room chart recorder receives reliable (Division B) instrument power. Offgas posttreatment sample panel 11A is provided 120-V ac (Division A) power and offgas posttreatment sample panel 11B is provided 120-V ac (Division B) power.

Each radiation readout module has four trip circuits: two upscale (high-high-high, and high), one downscale (low), and one inoperative. Each trip is visually displayed on the radiation monitor. The first three trips actuate corresponding control room annunciators: offgas posttreatment high-high-high radiation, offgas posttreatment high radiation, and offgas posttreatment downscale. A high-high trip from the recorder actuates an offgas posttreatment high radiation annunciator in the main control room. High or low sample flow measured at the sample panel actuates a control room offgas posttreatment trouble annunciator.

A trip auxiliary unit in the control room takes the high-high-high and downscale trip outputs and, if its logic is satisfied, initiates closure of the offgas system discharge and drain valves. The logic is satisfied if two high-high-high, one high-high-high and one downscale, or two downscale trips occur. The high-high-high trip setpoints are determined such that valve closure is initiated prior to exceeding Technical Specifications limits. Any one high upscale trip initiates closure of the offgas system bypass line valve and initiates opening of the charcoal adsorber treatment line valve.

Grab sampling functions required for isotopic analysis and gross monitoring calibrations can be performed at each redundant sample panel.

For arrangement details see Figure 11.5-5.

11.5.2.2.1.3 Offgas Charcoal Bed Vault Radiation Monitoring System. The charcoal bed vault air handling room is monitored for an increase in the gross gamma radiation level from

leakage of radioactive noble gases out of the treatment system. The channel includes a sensor indicator, trip unit, and a locally mounted auxiliary unit. The detector is mounted outside of the air handling room door. The insulated door only attenuates 80 keV gamma (^{133}Xe) by approximately 40%. An indicator and trip unit is located in the main control room. The channel provides for sensing and readout, both local and remote, of gamma radiation over a range of six logarithmic decades (10^{+0} to 10^{+6} mR/hr).

The indicator and trip unit has one adjustable upscale trip circuit for alarm and one downscale trip circuit for instrument inoperative which annunciates in the main control room. The trip circuits are capable of convenient operational verification by means of test signals or through the use of portable gamma sources. Power is supplied from the channel A power supply of the reactor handling ventilation exhaust plenum RMS.

For arrangement details see Figure 11.5-2.

11.5.2.2.1.4 Mechanical Vacuum Pump Exhaust Radiation Monitoring System. The radiation monitor on the mechanical vacuum pump exhaust is designed to alarm, stop, and isolate the mechanical vacuum pumps in the case of high level of radioactive gases in air being exhausted to the reactor building elevated release duct. The mechanical vacuum pump is operated during plant startups to remove bulk air from the condenser and is secured at the point where the steam jet air ejection suction is available and condenser offgases are routed through the recombiner charcoal process treatment system. In addition to monitoring discharges via the mechanical vacuum pumps, the turbine gland seal air exhauster system is continuously monitored via this process radiation monitoring system. Clean sealing steam is used on the turbine gland seals to maintain the releases of radionuclides to as low as is reasonably achievable (ALARA) limits. The monitor complies with GDC 64 and is Quality Class II and Seismic Category II.

The channel includes an energy-compensated GM detector, a readout module, and recorder in the main control room. The channel provides for sensing and readout, both local and remote, of gamma radiation over a range of four logarithmic decades (10^{-2} to 10^{2} mR/hr).

The indicator and trip unit has two adjustable trip circuits. The upscale trip circuit generates a high radiation alarm and stops the mechanical vacuum pumps. The adjustable downscale trip circuit generates an instrument alarm.

For arrangement details see Figure 11.5-6.

11.5.2.2.1.5 Reactor Building Elevated Release Duct Radiation Monitoring and Sampling System. This system is one of various plant gaseous effluent monitoring systems used to comply with regulatory requirement GDC 64 and 10 CFR 20. GDC 64 requires utilization facilities to provide a means to monitor effluent discharge paths for radioactivity that may be released into the outside atmosphere during normal operations and postulated accidents. Under

normal operating conditions, this criterion is met by applying applicable provisions found in Regulatory Guide 1.21, Rev. 1, Position C and Appendix A. Under postulated accident conditions, this criterion is met by applying applicable provisions found in NUREG-0737, Item 11.F.1, Attachments 1 and 2, and Regulatory Guide 1.97, Rev. 2, Position C and Table 1. 10 CFR 20 limits the release of radioactive materials to within specified quantities. These limits are specified in the ODCM. A necessary requirement of limiting the release of radioactive material is to continually measure radioactivity release points. The same provisions of Regulatory Guide 1.21, Rev. 1, provide an acceptable methodology to measure radioactivity. Additionally, this system and the LOCA detectors of Section 11.5.2.2.3.2 comply with GDC 13. This system is evaluated to ANSI N42.18-1980, Section 5.4. For this system, ANSI N13.1-1999 supersedes reference to ANSI N13.1-1969 in the above documents. ANSI N13.1-1999 methodology assures continuous representative sampling of radioactive gaseous effluents released from the duct.

The system provides detection capability for Regulatory Guide 1.97 variables/categories C.2, C.3, E.2 and E.3. These variables are given in Regulatory Guide 1.97, Rev. 2, Table 1. They are: Type C – Containment Effluent Radioactivity – Noble Gases (from identified release points including Standby Gas Treatment System Vent), Type C –Effluent Radioactivity – Noble Gases (from buildings as indicated above), Type E –Noble Gas and Vent Flow Rate Standby Gas Treatment System Purge (for Mark II plants), Type E – Noble Gas and Vent Flow Rate Secondary Containment Purge (for Mark II plants), Type E – Noble Gas and Vent Flow Rate Secondary Containment (reactor shield building annulus), and Type E – Particulates and Halogens All Identified Plant Release Points. Portions of this system which are necessary in order to monitor the Regulatory Guide 1.97 Category 2 variables stated above are Quality Class 2+. All other portions of the system are Quality Class 2.

Per the above regulations, four categories of effluent are monitored or sampled: noble gases, halogens, particulates, and tritium. Monitoring and sampling methods provide gross radioactivity concentration values and individual radionuclide activity concentration values. These values are direct inputs into plant analyses. Under normal conditions noble gas, halogen, particulate, and tritium values input into analyses that demonstrate radionuclide releases into the environment do not exceed limits found in the ODCM. Furthermore, the values demonstrate the adequacy and performance of containment and effluent controls. Under postulated accidents, monitoring the release of noble gases can show significant releases from the Reactor Building, allow for the assessment of a release, indicate containment breach and accomplishment of mitigation actions, and provide verification. Sampling of halogens and particulates can show significant releases from the Reactor Building, allow for the assessment of a release, and provide the ability to conduct long-term surveillance.

The system is not required to remain functional after a seismic event. SSCs inside the duct, cable chase C416, and the Main Control Room are mounted Seismic Category 1M to prevent interactions with Seismic Category 1 SSCs. The Remote Sampling System post-accident samplers, the transformer, and the noble gas monitoring skid (NGMS) are mounted Seismic

Category 1M because these SSCs are heavy and are mounted to floors that are Seismic Category 1. All system components are Seismic Category 2.

Per the regulatory guidance above, the system must have a reliable power source. To meet this requirement, the system is powered off of a power source which is backed by the emergency diesel generators and, therefore, is considered reliable under postulated accident conditions. The power source is not redundant, as Regulatory Guide 1.97 and NUREG-0737 do not require a redundant or a battery backed power source for this particular monitoring and sampling function.

Reactor Building Elevated Release Duct is located adjacent to the north wall of the Reactor Building. The duct takes supply from the gland seal and mechanical vacuum pumps, the Off Gas System, the Standby Gas Treatment System exhaust, and the Reactor Building Ventilation System exhaust. These effluents exhaust to the outside atmosphere at the top of the duct.

Both monitoring and sampling are performed continuously using offline detectors and sampling media. The sample effluent is taken at a nominal supply flow of 2 SCFM using a shrouded nozzle located near the top of the elevated release duct. The shrouded nozzle is located so as to obtain a representative sample of the process effluent. Design and location of the shrouded nozzle are per guidance given in ANSI N13.1-1999. The nozzle is connected to a stainless steel tube, which runs down the inside of the elevated release duct, exits the duct, runs down the northwest portion of the outside wall of the Reactor Building, and penetrates through the Reactor Building roof into corridor C402. The tubing connects to the remote sampling system, located in the southeast corner of the corridor. After passing through the remote sampling station, the sample effluent is routed, via a second run of tubing, along the south wall of the corridor, into the Radwaste and Control Building - first through the cable chase and then into room C405. In C405 the tubing connects to the NGMS. After passing through the NGMS, the sample effluent is routed through a small section of tubing and vented directly to the outside atmosphere. The sampling station is located sufficiently close to sampling point so as to meet the sampling efficiency guidelines given in ANSI N13.1-1999. The NGMS is located in C405 to avoid space restrictions and environmental concerns. The system is heat traced to maximize sample transport and mitigate the potential for condensation inside the sample line.

The noble gas monitoring skid is a GEMS-700 monitoring system. It includes three noble gas detectors: low, intermediate, and high. The low range detector is located on the normal sample line path of the skid. The intermediate and high range detectors are located on the accident sample line path of the skid. Under normal conditions, the normal sample line is open, the bypass line is closed, and the accident sample line is closed. On a signal of elevated radiation levels, the skid accident sample line opens, the bypass line opens, and the normal line closes. The low range detector is a plastic beta scintillation detector. The intermediate range and high range detectors are cadmium telluride semiconductor gamma detectors. Table 11.5-1

lists the design ranges and sensitivities of each detector. The GEMS-700 is equipped with a local processing and display unit and a remote display unit. Signal from the local processing and display unit is routed to the remote display unit and from the remote display unit to a digital recorder. The local processing and display unit is located on the GEMS-700. The remote display unit and digital recorder are located in the main control room. From the recorder, signal is sent to TDAS and four annunciator alarms. The alarms are: RB STACK GENERAL TROUBLE, RB STACK CRITICAL FAILURE, RB STACK RAD ALERT, and RB STACK RAD HIGH.

The remote sampling station consists of a normal sample line, a bypass sample line, and an accident sample line. The normal line divides into two redundant branches, each of which contains a filter paper medium and a charcoal cartridge. The filter paper collects particulates and the charcoal cartridge adsorbs halogens. Sampling frequency is shown on Table 11.5-3 and the ODCM. The accident line also divides into two redundant branches, each of which connects to a shielded particulate and iodine sampler. The sampler contains a filter paper medium to collect particulates and a silver zeolite cartridge to trap halogens. The redundant sampling media on the normal and accident lines allow for continuous sampling to occur. The samplers on the accident sample line are shielded to limit personnel radiation exposure, per provisions in NUREG-0737. A flow totalizer on the remote sampling station is used to calculate the volume of effluent passing through the sampling media per a given sample. The sampling media are transported to a laboratory for analyses. The activity concentrations of particulates and halogens are determined using the laboratory measured activity values and the flow totalizer measured effluent sample volume. Under normal conditions, the normal sample line is open and the bypass line is closed. On an elevated level of radiation signal from the NGMS, the normal sample line automatically closes and the bypass line automatically opens. The accident line branches off from the bypass line. Downstream of the sampling media all three branches recombine before connecting to a system pump, which drives the sample effluent through the system to the NGMS. A redundant pump, on the remote filter station ensures system reliability. See Figure 11.5-11 for a simplified drawing of the system.

The remote sampling station and the noble gas monitoring skid also provide for the connection of a tritium sampler. Sampling of tritium is performed during normal conditions.

Major electrical components are located in corridor C402 and room C405. Heat trace and conduit are located in the elevated release duct, outside, in corridor C402, the cable chase C416, and room C405. Both C402 and C405 are equipped with environmental monitors, which provide signal to the Main Control Room if the environments in either of these two areas are outside the allowable environmental operating limits of the electrical devices or if either environmental monitor becomes inoperable. This is done to meet the requirements of Regulatory Guide 1.89, Rev. 1, for augmented quality systems. A signal from either environmental monitor will cause a RB STACK GENERAL TROUBLE alarm or a RB STACK CRITICAL FAILURE alarm on the annunciation panel in the Main Control Room. Loss of either environmental monitor will cause a RB STACK CRITICAL FAILURE alarm.

Under normal conditions, the system is only used to monitor and sample the release of radioactive gaseous effluents and no setpoints or actions, manual or automatic, are associated with this system. However, under abnormal conditions the monitoring portion of the system provides input to Emergency Action Level (EAL) scheme for classification of an emergency. The EAL thresholds and their associated classifications are given in the Emergency Plan procedures.

The monitoring instrumentation is calibrated and inspected at regular intervals, per the recommendations of the vendor, operating experience with similar components, and ANSI N13.1-1999. Both calibrations and inspections are performed per plant surveillance test procedures. Preventive maintenance is also performed on components susceptible to failure. This subset of components was chosen per vendor recommendation and operating experience. Frequently replaced parts are maintained available on site. While less frequently used replacement parts are available from the vendor or the vendor's subcontractor as stated on the GEMS-700 procurement specification.

Note 1: The previous Reactor Building Elevated Release Duct monitoring system consisted of three HPGe detectors, associated processing equipment, and a recorder. During normal conditions, this system provided monitoring of noble gases. During postulated accident conditions, this system provided monitoring of noble gases, particulates, and halogens. The previous system has not been physically removed from the plant. However, the previous system has been de-terminated and therefore no longer functions. For a diagram of the previous monitoring system see Figure 11.5-11A.

Note 2: The previous Reactor Building Elevated Release Duct sampling system consisted of two holders for filter media and charcoal cartridges. During normal conditions, the previous system provided sampling of particulates and halogens. The previous system has not been physically removed from the plant. However, the previous system has been de-terminated and therefore no longer functions. For a diagram of the previous sampling system see Figure 11.5-6.

11.5.2.2.1.6 Turbine Generator Building Ventilation Release Duct Radiation Monitoring System. This monitoring system measures the radioactivity in the turbine building exhaust prior to its discharge to the environment and in doing so complies with Regulatory Guide 1.21, Revision 1, GDC 64, and NUREG-0737. This monitor detects the fission and activation products from the steam which may leak from the turbine or the other primary components in the building. The gaseous activity in the exhaust is expected to normally be below detectable levels. The particulate and iodine activity is typically accumulated on a filter and cartridge respectively for a week to obtain sufficient activity to be detectable. These filters are analyzed to determine the quantities of specific radionuclides present and the results, together with the gaseous activity strip chart recorder, provide a permanent record of radioactivity released to the environment.

A continuous representative sample is extracted from the exhaust vent through a multi-ported isokinetic probe, down a 30-ft tube to pass through a filter paper to collect particulates, and through an impregnated charcoal cartridge to collect iodine. The sample travels through two gas detectors, a mass flow transducer, a local flow meter with switch, and then a sample pump prior to being exhausted to the radwaste building roof area by the turbine building exhaust fans. The sample flow rate is automatically adjusted to compensate for effluent flow changes.

The gas monitoring system includes two local detectors mounted in one lead-shielded chamber, a local ratemeter with display, a remote control/display unit, and a recorder in the main control room. The normal sample flows through the high range detector followed by the low range detector. Arrangement details are shown on Figure 11.5-7.

The monitoring system includes four channels: channel 1, low-range (10^{-7} to 10^{-1} μCi/cc), channel 2, high-range (10^{-3} to 10^{+3} μCi/cc), channel 3, calculated composite range (10^{-7} to 10^{+3} μCi/cc), and channel 4, gas effluent release rate range (10^{+1} to 10^{+11} μCi/sec). A two-decade overlap is provided between the low and high range channels. The low and high range channels are equipped with a radioactive test source.

The gas monitoring system has no control functions. The monitor provides an alert radiation alarm and an instrument inoperative alarm, both of which annunciate in the main control room. The reliable power for low/high range radiation monitor(s) and recorder(s) is provided from Division A power panel and backed by standby power.

11.5.2.2.1.7 Radwaste Building Ventilation Release Ducts Radiation Monitoring System. This monitor system measures the radioactivity in the radwaste building ventilation air exhaust as it is being discharged to the environment and in doing so complies with Regulatory Guide 1.21, Revision 1, GDC 64, and NUREG-0737. Radioactivity originates from radwaste tank vents, from primary water processing equipment, and from laboratory sampling hoods, as well as various cubicles having liquid process treatment systems within the building. A continuous sample is drawn from each of the two out of three exhaust fans that are operating. The sampling uses isokinetic probes with a fixed flow rate and samples are isokinetic over the normal ventilation operating range. The representative sample is withdrawn through a multi-ported duct probe, through a tube passing through a particulate filter, and through a charcoal cartridge to collect particulate and iodine samples, which are removed at least weekly for laboratory radiochemical analyses. The filtered air sample streams from the operable exhaust fans are combined to pass through two gas detectors, a mass flow transducer, a local flow meter with switch and a sample pump prior to being exhausted to the room.

The gas monitoring system includes two local detectors mounted in one leaded-shielded chamber, a local ratemeter with display, a remote control/display unit, and a recorder in the main control room. The normal sample flows through the high range detector followed by the low range detector. Arrangement details are shown in Figure 11.5-7.

The monitoring system includes four channels: channel 1, low-range (10^{-6} to 10^{-1} μCi/cc), channel 2 high-range (10^{-3} to 10^{+3} μCi/cc), channel 3, calculated composite range (10^{-6} to 10^{+3} μCi/cc), and channel 4, gas effluent release rate range (10^{+1} to 10^{+10} μCi/sec). A two-decade overlap is provided between the low and high range channels.

The gas monitoring system has no control functions. The monitor provides an alert radiation alarm and an instrument inoperative alarm, both of which annunciate in the main control room. The reliable power for low/high range radiation monitor(s) and recorder(s) is provided from Division A, power panel and backed by standby power.

11.5.2.2.1.8 NRC Safety Evaluation Report, NUREG-0892 Acceptance. System comparisons to ANSI 13.10 (redesignated as ANSI N42.18) have been performed.

An inline gamma spectroscopy system provides an isotopic analysis of all effluents exiting the reactor building elevated release duct. See Section 11.5.2.2.1.5, Figure 11.5-11, and Table 11.5-1.

Particulate/iodine sampling of the other buildings (radwaste and turbine) exhausts will be handled by the normal effluent samplers where the postaccident release concentration is quite low.

If there were a reactor accident with a core fission product release, the reactor building (secondary containment) immediately isolates. The atmosphere is maintained at a 0.25 in. H_2O vacuum by the SGT system. The only potential airborne contamination that could reach the other buildings is from the SGT system bypass leakage pathways as listed in Table 6.2-16. Shielding needed for the radwaste and turbine building normal effluent sampling systems was evaluated using assumptions consistent with the discussion in Appendix J (see Sections J.2 and J.3). *For the purpose of defining the shielding requirements for the radwaste and turbine building, an iodine concentration of 3.7 x 10^{+4} $\mu Ci/cm^3$ is assumed in the inleakage air. This iodine concentration is based on a 50% core inventory release to the drywell atmosphere and a 50% plate-out factor. This plate-out factor is conservative for determining sample shielding requirements. With 0.35 scfh bypass leakage into the radwaste building, the building exhaust (83,000 cfm) concentration will be 2.6 x 10^{-3} $\mu Ci/cm^3$, ignoring building volume dilution. The normal effluent sampler operates at 3 cfm; therefore, the charcoal cartridge 30 minutes accumulation would be 6.7 mCi. This would result in a dose rate of 21 mR/hr at 1 ft from the cartridge. Doubling the dose rate to account for particulates yields 42 mR/hr at 1 ft from the sample assembly. Because of the high exhaust flow rate (260,000 cfm) and less inleakage (0.24 scfh), the turbine building exhaust is less concentrated.* The radwaste and turbine building normal effluent sampling systems are adequate for postaccident sampling and no shielding is necessary (see Section 11.5.2.2.1.5). The evaluation for shielding was not repeated for the implementation of the alternative source term (AST).

11.5.2.2.2 Liquid Process and Effluent Radiation Monitoring System

These systems monitor gamma radiation levels of liquid process and effluent streams.

Each monitor system consists of a gamma scintillation detector inserted into either a well in the process piping or a sump or an offline chamber to which a process stream sample is piped. The detector locations are selected to obtain a reasonable geometry and are positioned away from crud traps and associated high background regions. Lead shielding is provided to further reduce background levels.

At each liquid offline detector location a continuous sample is extracted from the liquid process pipe, passed through a shielded detection assembly for gross radiation monitoring, and then returned to the process pipe. The detection assembly consists of a detector mounted in a shielded sample chamber equipped with a check source. The meter and recorder in the control room displays the measured gross radiation level. The sample chamber and lines can be drained to allow assessment of background buildup. The flow meter at the sample rack provides local sample line flow indication. A solenoid-operated check source operated from the control room is used to check operability of the channel response.

The critical $\pm$24-V dc for monitor(s) is supplied from Division 1 (and 2) power panels, except for the plant service water monitor which receives 120-V ac from Division A panel, reliable

power is provided for control room strip chart recorders, sample control panel receives local 120-V ac.

The detector's local preamplifier unit is designed to remain fully operational in the expected environment. If exposed to radiation transients which exceed the channel range, the channel maintains full scale deflection and returns to normal functioning when the transient has subsided.

Each radiation monitor, except for the circulating water, has four trip circuits: two upscale (high-high and high), one downscale (low), and one inoperative. Each trip is visually displayed on the affected radiation monitor. Two of these trips actuate corresponding control room annunciators: one upscale (high radiation) and the downscale for the affected liquid monitoring channel. High or low sample flow measured at the sample panel actuates a control room high-low flow annunciator for the affected liquid channel.

All alarms are annunciated in a control room. Liquid monitoring system details are given in Table 11.5-2 and the monitor arrangements are shown in Figure 11.5-4.

11.5.2.2.2.1 Standby Service Water Radiation Monitoring System. See Section 11.5.2.1.4.

11.5.2.2.2.2 Reactor Building Closed Cooling Water Radiation Monitoring System. The radiation detector is located offline and samples the closed cooling water piping side of the reactor closed cooling water system (RCC) heat exchangers.

The monitor system is a diagnostic tool to verify that no inleakage of primary plant water has occurred from the reactor water cleanup system nonregenerative heat exchanger system which uses RCC as coolant. Since the RCC is a closed system, inleakage would be detected by the monitor system.

11.5.2.2.2.3 Radwaste Effluent Radiation Monitoring System. This monitor system measures the radioactivity in the radwaste effluent discharge prior to its entering the cooling tower blowdown line.

Liquid waste can be discharged from several radwaste processed water tanks such as the floor drain sample tank, waste sample tanks, or distillate drain tanks. These tanks contain water that has been processed through one or more treatment systems such as evaporation, filtration, and ion exchange. Prior to the discharge from any tank, the liquid in the appropriate tank is sampled and analyzed in the laboratory for radioactivity. Based on this analysis, discharge is permitted as specified in the Offsite Dose Calculation Manual (ODCM).

The radiation detector is located offline and samples the common discharge line from the liquid radwaste system through which all liquid radwaste is discharged to the blowdown line. The

piping arrangement is designed so that the sample well can be flushed to lower background levels.

The high-high upscale trip on the radwaste effluent radiation monitor is used to initiate closure of the radwaste system discharge valve. The trip point is set such that closure is initiated prior to exceeding ODCM limits for liquid effluents. The high upscale trip actuates an annunciator in the radwaste control room as well as the main control room.

11.5.2.2.2.4 Circulating Water and Plant Service Water Radiation Monitoring Systems. The circulating water monitor is located on the discharge side of the circulating water pumps for the main condenser in the coolant blowdown line to the Columbia River. The location of this monitor permits detection of radioactive material leaking to the circulating water from any source, including the TSW system. If an alarm condition exists, circulating water blowdown to the Columbia River is terminated by automatic closure of the circulating water blowdown valve in the circulating water pump house (see Section 9.2.1.2), and it annunciates in the control room.

During plant outages, water is discharged to the river via temporary pumps installed in the circulating water pump house and connected to the blowdown line. Automatic termination of blowdown is not provided for this temporary arrangement. The only possible source of radiation would be through the TSW system which is monitored for radiation. There is about a 2-hr holdup time in the circulating water pipe giving ample time for blowdown to be manually secured if an alarm is received.

The TSW radiation monitor is located in the TSW return header to the circulating water system. The monitor is an offline type that continuously measures the radioactivity level of the TSW returning flow to the circulating water system. The radiation detector is lead shielded to minimize background radiation effects. The signal from the detector is displayed on a readout module and recorded in the main control room.

11.5.2.2.3 Primary Containment Radiation Monitoring System

This monitoring system is composed of two parts, a sensitive two-channel leak detection system and a four-channel high activity LOCA tracking system.

11.5.2.2.3.1 Leak Detection Monitors. This monitoring system measures the radioactivity in the drywell and in doing so complies with Regulatory Guide 1.45, Revision 0, and GDC 30. The radioactivity in the drywell is from coolant and corrosion activation products plus fission products produced in the reactor and released through leaks.

This monitoring system has two redundant subsystems, each having two detectors, individually monitoring particulates, and noble gas activity. Additionally a charcoal sample cartridge is provided to trap halogen gases. The detectors are housed in divisionally separated sample

racks located in a reactor building sample room. The sample racks have incorporated blowers and flow controls to withdraw gas samples from the primary containment atmosphere via stainless steel sampling lines and vent back to the containment. The environment in which the local cabinets are located is maintained to limit upper temperature excursions that may occur in the reactor building during an accident. Associated radiation readout modules and recorders are mounted in the main control room along with alarm annunciators.

Required 120-V ac supply for Division 1 and 2 equipment in both the main control room and the reactor building sample room is provided on a divisional basis by the 120/240-V ac critical (Class 1E) instrumentation power system.

The two-channel detector assemblies are provided with lead shielding to minimize the effects of background radiation to ensure high sensitivity. The detectors are of the beta scintillation type and are provided with check sources to verify system operability. The particulate detector views a fixed filter collector on which airborne particulates are trapped. The noble gas detector views a fixed volume of gas.

Each radiation monitor has three trip circuits: one upscale Hi to alarm and close sample line valve, one upscale Alert, and one downscale for instrument inoperative. All alarms annunciate in the main control room. This monitoring subsystem provides no control function and is a diagnostic tool which enables the main control room operator to take appropriate action.

Arrangement details are shown in Figure 11.5-9.

11.5.2.2.3.2 Loss-of-Coolant Accident Tracking Radiation Monitoring Systems (Containment Drywell). The LOCA monitoring systems, CMS-RE-27E and CMS-RE-27F, monitor the drywell atmosphere from inside the drywell; while CMS-RE-27A and CMS-RE-27B monitor the drywell atmosphere through penetrations in the bioshield wall. These four monitors provide LOCA monitoring for abnormal radioactivity levels following an accident condition involving rupture of the reactor coolant boundary.

The in-containment LOCA monitors CMS-RE-27E and CMS-RE-27F, located at approximately 517 ft level Azimuth 291° and 515 ft level Azimuth 51.5° respectively, track long-term decreases in containment radioactivity that take place with decay and decontamination and comply with GDC 13 and 64. The in-containment LOCA monitors each contain a Victoreen Model 877-1 ionization chamber with a range of 1 to 10^7 R/hr. These two detectors are qualified and meet the criteria of Table II.F.1-3 of NUREG 0737, including calibration with a high level gamma source. The chambers will respond to low energy gamma radiation such as 81 keV from ^{133}Xe fulfilling the Regulatory Guide 1.97, Revision 2 criteria.

The LOCA monitors, CMS-RE-27A and CMS-RE-27B, are housed in the bioshield wall and are against the containment steel shell. These monitors will monitor the drywell radiation in conjunction with the in-containment LOCA monitors CMS-RE-27E and CMS-RE-27F. The

LOCA monitors, while not required for plant operation, are used to monitor the drywell in the post-LOCA situation.

The LOCA monitoring systems (CMS-RE-27A, CMS-RE-27B, CMS-RE-27E, and CMS-RE-27F) are redundant and separately supplied with power from Division 1 and 2 in both the main control room and remote locations.

Each ionization chamber is wired to a local monitor. Output from each local monitor is wired to a monitor located in the main control room. Radiation levels within the primary containment for each LOCA monitoring system are recorded in the main control room.

Each monitor has alarm circuitry and indication for high radiation and for instrument inoperative that annunciate in the main control room. This monitoring subsystem provides no control function and is a diagnostic tool which enables the main control room operator to take appropriate action.

Arrangement details are shown in Figure 11.5-10.

11.5.2.3 Sampling

The following sections present a detailed description of the radiological sampling procedures, frequencies and objectives for all plant process and effluent sampling. This sample program provides the means to show compliance with the ODCM for the process radiation monitoring system and radwaste system.

11.5.2.3.1 Process Sampling

Section 9.3.2 describes the design of sampling facilities provided for general sampling. The sample frequency, type of analyses, analytical sensitivity, and the purpose of the sample are summarized in Table 11.5-4 for each liquid process sample location and in Table 11.5-5 for each gas process sample location. The analytical procedures used in sample analysis are presented in Section 11.5.2.3.3. These samples serve to monitor radioactivity levels within various plant systems.

11.5.2.3.2 Effluent Sampling

Effluent sampling of all potentially radioactive liquid and gaseous effluent paths is conducted on a regular basis to verify the adequacy of effluent processing to meet the discharge limits to unrestricted areas. This effluent sampling program will be of such a comprehensive nature as to provide the information for the effluent measuring and reporting programs required by 10 CFR Part 50.36a in annual reports to the NRC. The frequency of the periodic sampling and analysis described herein is normal and will be increased if effluent levels approach

Technical Specifications limits. Tables 11.5-3 and 11.5-6 summarize the sample and analysis schedules which correspond to Regulatory Guide 1.21, Revision 1, guidance.

11.5.2.3.3 Analytical Procedures

Samples of process and effluent gases and liquids will be analyzed in the laboratory by the following techniques: gross beta counting, gross alpha counting, gamma spectrometry, liquid scintillation counting, and radiochemical separations.

Instrumentation which is available in the laboratory for the measurement of radioactivity includes a single channel analyzer, alpha counter, beta counter, liquid scintillation counter, and multichannel gamma spectrometer.

Samples for beta counting are evaporated to dryness on metal planchets prior to counting. Sample volume, counting geometry, and counting time are chosen to achieve the required measurement sensitivities. Correction factors are applied for sample-detector geometry, self-absorption and counter-resolving time.

Gross beta and gross alpha analyses of liquid effluent samples may be performed with an internal proportional counter. The samples are prepared for counting by evaporation onto metal planchets. Gross beta counting is also performed using a liquid scintillation counter.

Sample volume and counting times are chosen to achieve the required measurement sensitivity. When possible, sample volume is selected to maintain a sample residue thickness of less than 1 mg/cm^2. Correction factors are applied for self-absorption.

Gamma ray spectrometry is used for isotopic analyses of gaseous, airborne particulate and iodine, and liquid samples. High-resolution germanium solid-state detectors are available for this purpose. The detectors are calibrated against NIST traceable gamma ray standards for a variety of sample detector geometries. The gamma spectra are resolved using a software program developed for computerized spectrum resolution.

Gaseous tritium samples are collected by condensations or adsorption, and the resultant liquid is analyzed by liquid scintillation counting techniques.

Radiochemical separations are used for the routine analysis of ^{89}Sr and ^{90}Sr.

11.5.2.3.4 Inservice Inspection, Calibration, and Maintenance

During reactor operation, checks of system operability are made by observing channel behavior. During reactor operation at intervals specified in the Offsite Dose Calculation Manual (ODCM) and the Technical Specifications (TS), the response of each detector supplied with a remotely activated check source or LED source simulator will be recorded together with the instrument background count rate to ensure proper functioning of the

monitor. Any detector whose response is observed to be inconsistent with that expected for power operation and is not supplied with a remotely activated check source or LED source simulator will be checked with a portable source or by comparison of detector response to system activity. An exception to the portable source check requirement will be allowed for those detectors mounted in areas that are deemed hazardous to personnel.

The system has electronic testing and calibrating equipment, which permits channel testing without relocating or dismantling channel components. An internal trip test circuit, adjustable over the full range of the readout meter, is used for testing. Each channel is tested prior to performing a calibration check. Verification of valve operation, ventilation diversion, or other trip functions is done at this time if it can be done without jeopardizing the plant safety. The tests are performed in conformance with the ODCM test frequencies.

The continuous radiation monitors are calibrated to commercial radionuclide standards traceable to the NIST. Each continuous monitor is calibrated during the refueling outage or every 18 months as required by the ODCM, using standard sources with NIST traceability.

A calibration can also be performed by using liquid or gaseous radionuclide standards or by comparison analysis of particulate, iodine, liquid, or gaseous grab samples with laboratory instruments.

11.5.3 EFFLUENT MONITORING AND SAMPLING

The implementation of the requirements of GDC 64 concerning monitoring of gaseous effluent discharge paths for radioactivity is discussed in Section 11.5.2.2.1. Section 11.5.2.2.2 provides applicable discussions for liquid effluent radiation monitors.

11.5.4 PROCESS MONITORING AND SAMPLING

The implementation of the requirements of GDC 60 concerning automatic closure of isolation valves in gaseous and liquid effluent discharge paths is discussed in Sections 11.5.2.1 and 11.5.2.2. Section 11.5.2.2.1 provides a discussion of gaseous process radiation monitors. Section 11.5.2.2.2.2 provides a discussion of liquid process radiation monitors.

Table 11.5-1

Process and Effluent Radiation Monitoring System
(Gaseous and Airborne Monitors)

Monitor	Detector Location (Number of Channels)	Type	Efficiency/ Sensitivity	Range	Principal Radionuclides Measured	Expected Activity	Upscale Setpoints	
							Alarms	Trips
A. SAFETY-RELATED SYSTEMS								
Main steam line radiation monitors MS-RE-3A, 3B, 3C, and 3D	Adjacent to steam lines (4)	γ-ion chamber	3×10^{-10} amp/R/h	10^0 - 10^6 mR/h (6 dec. log)	Coolant activation gases	Steam line activity defined in Table 11.1-4	Above full power background, below trip	N/A
Reactor building exhaust plenum radiation monitors REA-RE-9A, 9B, 9C, and 9D	Inline (4)	GM[b]		10^{-2} - 10^2 mR/hr (4 dec. log)		Reactor bldg. activity defined in Table 11.3-7	Above background, below trip	Tech Specs
B. SYSTEMS REQUIRED FOR PLANT OPERATION								
Offgas pretreatment radiation monitor OG-RE-2	Offline, adjacent to sample chamber (1)	γ-ion chamber	3×10^{-10} amp/R/h	10^0 - 10^6 mR/h (6 dec. log)	Noble gas fission products	Offgas activity defined in Table 11.3-1	Above background	Tech Specs
Offgas posttreatment radiation monitors OG-RE-601A and OG-RE-601B	Offline (2)	β-scint, Part. Filter, Iodine Filter	4.08×10^5 cpm/μCi/ mL	10^0 - 10^7 cpm (2×10^{-5} to 24 μCi/cc) (7 dec. log)	^{133}Xe [a]	Offgas activity defined in Table 11.3-1	Above background	ODCM

Table 11.5-1

Process and Effluent Radiation Monitoring System
(Gaseous and Airborne Monitors) (Continued)

Monitor	Detector Location (Number of Channels)	Type	Efficiency/ Sensitivity	Range	Principal Radionuclides Measured	Expected Activity	Upscale Setpoints	
							Alarms	Trips
Charcoal bed vault radiation monitor OG-RE-11	Charcoal bed vault (1)	GM	N/A	10^{0} - 10^{6} mR/h (6 dec. log)	Noble gas	Charcoal bed inventory defined in Section 11.3	Above background	N/A
Mechanical vacuum pump discharge AR-RE-21	Inline (1)	GM		10^{-2} - 10^{6} mR/hr (4 dec. log)	^{133}Xe	Within monitor range	Above background	ODCM
Reactor building elevated discharge radiation monitor								
PRM-RE-11	Offline (1)	ß-scint	1.810E+08 cpm/(µCi/cc)	1E-07 µCi/cc to 1E-01 µCi/cc	^{133}Xe [a]	RB activity Table 11.3-7	ODCM	N/A
PRM-RE-12	Offline (1)	CdTe	2.506E+04 cpm/(µCi/cm)	1E-03 µCi/cc to 1E+02 µCi/cc	^{133}Xe [a]	LOCA Table 15.6-10 and 15.6-11	N/A	N/A
PRM-RE-13	Offline (1)	CdTe	1.105E+01 cpm/(µCi/cm)	1E00 µCi/cc to 1E+06 µCi/cc	^{133}Xe [a]	LOCA Table 15.6-10 and 15.6-11	N/A	N/A
Particulate filter	Offline	Filter	N/A	from 10^{-12} µCi/cm^{3}	γ emitters[c]	ODCM Table 6.2.2.1-1	N/A	N/A
Iodine filter	Offline	Charc. cart.	N/A	from 10^{-12} µCi/cm^{3}	γ emitters	ODCM Table 6.2.2.1-1	N/A	N/A

Table 11.5-1

Process and Effluent Radiation Monitoring System
(Gaseous and Airborne Monitors) (Continued)

Monitor	Detector Location (Number of Channels)	Type	Efficiency Sensitivity	Range	Principal Radionuclides Measured	Expected Activity	Upscale Setpoints	
							Alarms	Trips
Turbine bldg. vent. exhaust radiation	Offline (2)	Part. filter[c] iodine filter				Turbine bldg. activity defined in Table 11.3-7	ODCM	N/A
TEA-RE-13 Low Range		ß-scint	N/A	10^{-7} -10^{-1} μCi/cc (6 dec. log)	^{133}Xe[a]			
TEA-RE-13A High Range		ß-scint	N/A	10^{-3} - 10^{3} µCi/cc (6 dec. log)	^{133}Xe[a]	LOCA mixture of F.P. activity see Table 15.6-9		
Radwaste bldg. vent. exhaust radiation	Offline (2)					Radwaste bldg. activity defined in Table 11.3-7	ODCM	N/A
WEA-RE-14 Low Range		ß-scint	N/A	10^{-6} -10^{-1} μCi/cc (5 dec. log)	^{133}Xe[a]			
WEA-RE-14A High Range		ß-scint	N/A	10^{-3} - 10^{3} µCi/cc (6 dec. log)	^{133}Xe[a]	LOCA mixture of F.P. activity see Table 15.6-9		
Primary containment LOCA monitors	Adjacent to containment steel walls	γ-ion chambers	amp/R/h	R/h (6 dec. log)	Fission product gases	Within monitor range	Above background	N/A
CMS-RE-27A, 27B	Outside(2)		1 x 10^{-10} amp/R/h	10^{-2} x 10^{4} R/h (6 dec. log)				
[b] CMS-RE-27E, 27F	Inside (2)		7 x 10^{-11} amp/R/h	10^{0} - 10^{7} R/h (6 dec. log)				

Table 11.5-1

Process And Effluent Radiation Monitoring System
(Gaseous And Airborne Monitors) (Continued)

Monitor	Location (Number of Channels)	Type	Efficiency/ Sensitivity	Range	Principal Radionuclides Measured	Expected Activity	Upscale Setpoints	
							Alarms	Trips
Primary containment radiation monitor	Offline					Containment discussed in Section 12.2		
Particulate CMS-RE-12-1A CMS-RE-12-1B	(2)	Part. filter ß-scint	N/A	10^{-12} - 10^{-6} μCi/cc (6 dec. log)	Fission gas daughter and corrosion activation product [d]		Above background, below trip	Full scale
Gas CMS-RE-12-3A CMS-RE-12-3B	(2)	ß-scint	N/A	10^{-7} – 10^{-1} μCi/cc (6 dec. log)			Above background, below trip	Full scale

[a] Sensitivity based on this radionuclide.

[b] Not required for plant operation.

[c] Composite of particulate filters analyzed for ^{89}Sr and ^{90}Sr.

[d] For particulate, Cs-137 is the fission product radionuclide used for converting counts per minute to μCi/cc. For gas, Kr-85 is the fission product radionuclide used for converting counts per minute to μCi/cc. The monitors display units of μCi/cc.

Table 11.5-2

Process and Effluent Radiation Monitoring System (Liquid Monitors)

Monitor	Detector Location (Number of Channels)	Type	Range	Principal Radionuclides Measured	Expected Activity	Upscale Setpoints	
						Alarms	Trips
Residual heat removal standby service water radiation monitor SW-RE-4, 5	Offline (2)	γ-scint	10^{-1} - 10^{6} cps (7 dec. log)	^{137}Cs [a] ^{60}Co	Less than minimum detector sensitivity	Above background	Not applicable
Reactor building closed cooling water radiation monitor RCC-RE-7	Offline (1)	γ-scint	10^{-1} - 10^{6} cps (7 dec. log)	^{137}Cs [a] ^{60}Co	Less than minimum detector sensitivity	Above background	Not applicable
Radwaste effluent radiation monitor FDR-RE-6	Offline (1)	γ-scint	10^{-1} - 10^{6} cps (7 dec. log)	^{137}Cs [a] ^{60}Co	Section 11.2	Above background	ODCM[b]
Circulating water effluent radiation monitor CBD-RE-8	Inline (1)	γ-scint	10^{-1} -10^{6} cps (7 dec. log)	^{137}Cs [a] ^{60}Co	Less than minimum detector sensitivity	Above background	Section 11.5.2.2.2.4
Plant service water radiation monitor TSW-RE-5	Offline (1)	γ-scint	1.0E-08 to 1.0E-02 μCi/cc (6 dec. log)	^{137}Cs [a] ^{60}Co	Less than minimum detector sensitivity	Above background	Not applicable

[a] Sensitivity based on this radionuclide.

[b] The alarm point will be set, based upon the activity, radionuclides, and dilution factor so that the concentration in the discharge line is less than 10 CFR 20 Appendix B, Table II, Column 2 limits.

Table 11.5-3

Radiological Analysis Summary of Gaseous Effluent Samples

Grab Sample Sample Description	Frequency	Sensitivity Analysis	Sensitivity μCi/ml	Purpose
Reactor building elevated release exhaust	Weekly	Gamma spectrum[a] Radioiodines[b]	10^{-11} 10^{-10}	Effluent record
	Quarterly	^{89}Sr and ^{90}Sr[c]	10^{-11}	
	Monthly	Gross alpha Tritium	10^{-11} 10^{-11}	
Radwaste building exhaust	As above	As above		Effluent record
Turbine building exhaust	As above	As above		Effluent record

[a] On particulate filter

[b] On charcoal cartridge

[c] On composite of particulate filters

Table 11.5-4

Radiological Analysis Summary of Liquid Process Samples

Grab Sample Sample Description	Frequency	Sensitivity		Purpose
		Analysis	μCi/ml	
Reactor coolant	In accordance with Technical Specifications			
		^{131}I, ^{133}I	10^{-6}	Evaluate fuel cladding integrity
		Gamma spectrum	10^{-6}	Determine radionuclides present in system
Reactor water cleanup system	Periodically	Gamma spectrum	10^{-6}	Evaluate cleanup efficiency
Condensate storage tanks	Weekly	Gamma spectrum	10^{-6}	Tank inventory
Fuel pool filter - demineralizer inlet and outlet	Periodically	Gamma spectrum	10^{-6}	Evaluate system performance
Waste collector tank	Batch[a]	Gamma spectrum	10^{-6}	Evaluate system performance
Floor drain collector tank	Batch[a]	Gamma spectrum	10^{-6}	Evaluate system performance
Chemical waste tank	Batch[a]	Gamma spectrum	10^{-6}	Evaluate system performance
Detergent drain tank (2)	Batch[a]	Gamma spectrum	10^{-6}	Tank inventory

[a]Analysis performed on an infrequent basis as needed to evaluate equipment performance or unusual water chemistry.

Table 11.5-5

Radiological Analysis Summary of Gaseous Process Samples

Grab Sample Sample Description	Frequency	Sensitivity Analysis	Sensitivity μCi/ml	Purpose
Containment atmosphere (drywell)	Periodically	Gamma spectrum[a]	10^{-11}	Evaluate prior to discharge
	Prior to entry	Gamma spectrum	10^{-10}	Determine need for respiratory equipment
Offgas pretreatment monitor sample	Periodically	Gamma spectrum	10^{-6}	Determine offgas activity
Offgas posttreatment sample	Periodically	Gamma spectrum[a,b]	10^{-10}	Determine offgas system cleanup performance
		Gamma spectrum[c]	10^{-6}	

[a] On particulate filter

[b] On charcoal cartridge

[c] Noble gas

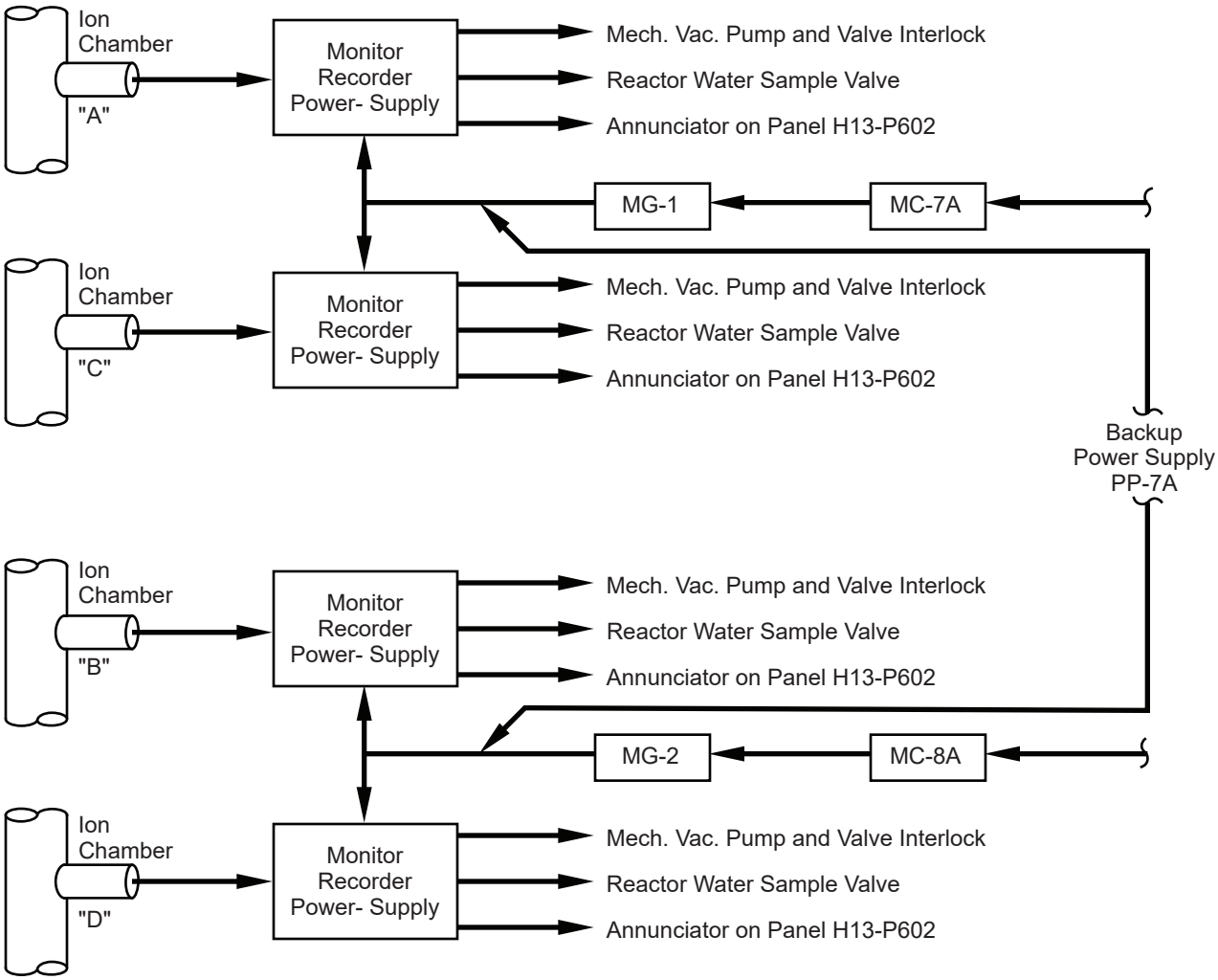
Table 11.5-6

Radiological Analysis Summary of Liquid Effluent Samples

Grab Sample Sample Description	Frequency	Sensitivity Analysis	Sensitivity μCi/ml	Purpose
Floor drain sample tank	Batch[a]	Gamma spectrum	10^{-6}	Effluent discharge record
Waste sample tanks (2)	Batch[a]	Gamma spectrum	10^{-6}	Effluent discharge record
Liquid radwaste effluent (composite of all tanks discharged)	Monthly[b]	Gross alpha Tritium	10^{-7} 10^{-5}	Effluent discharge record
	Quarterly[b]	$^{89}Sr/^{90}Sr$	10^{-8}	
Circulating water discharge line	Weekly grab of continuously collected proportional sample	Gross ß or Gamma Spectrum Tritium	10^{-6} 10^{-5}	Effluent discharge record (backup sample)

[a] If tank is to be discharged, analyses will be performed on each batch. If tank is not to be discharged, analyses will be performed periodically to evaluate equipment performance.

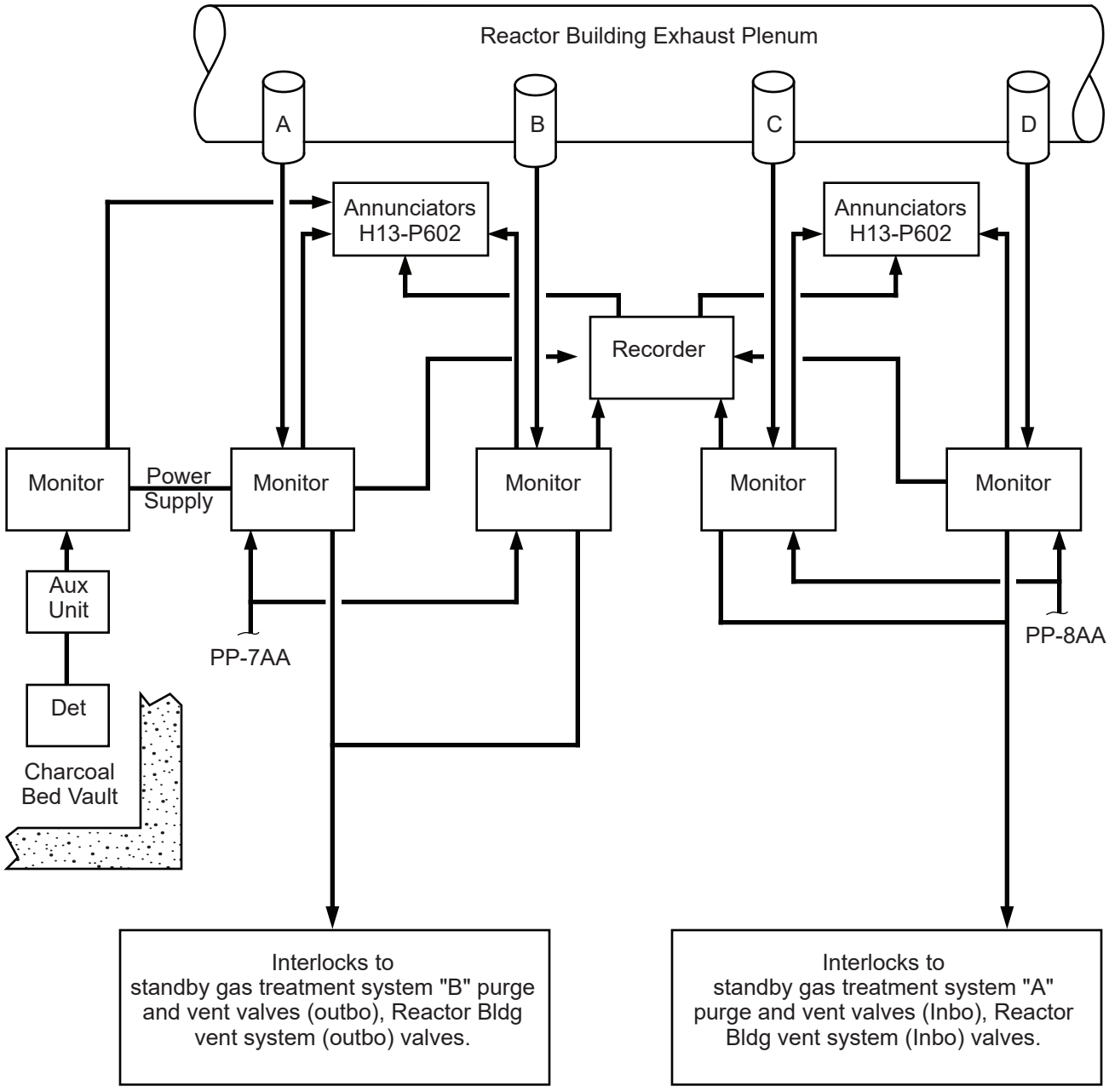
[b] If liquids were discharged during the previous month or quarter, as applicable.



Columbia Generating Station
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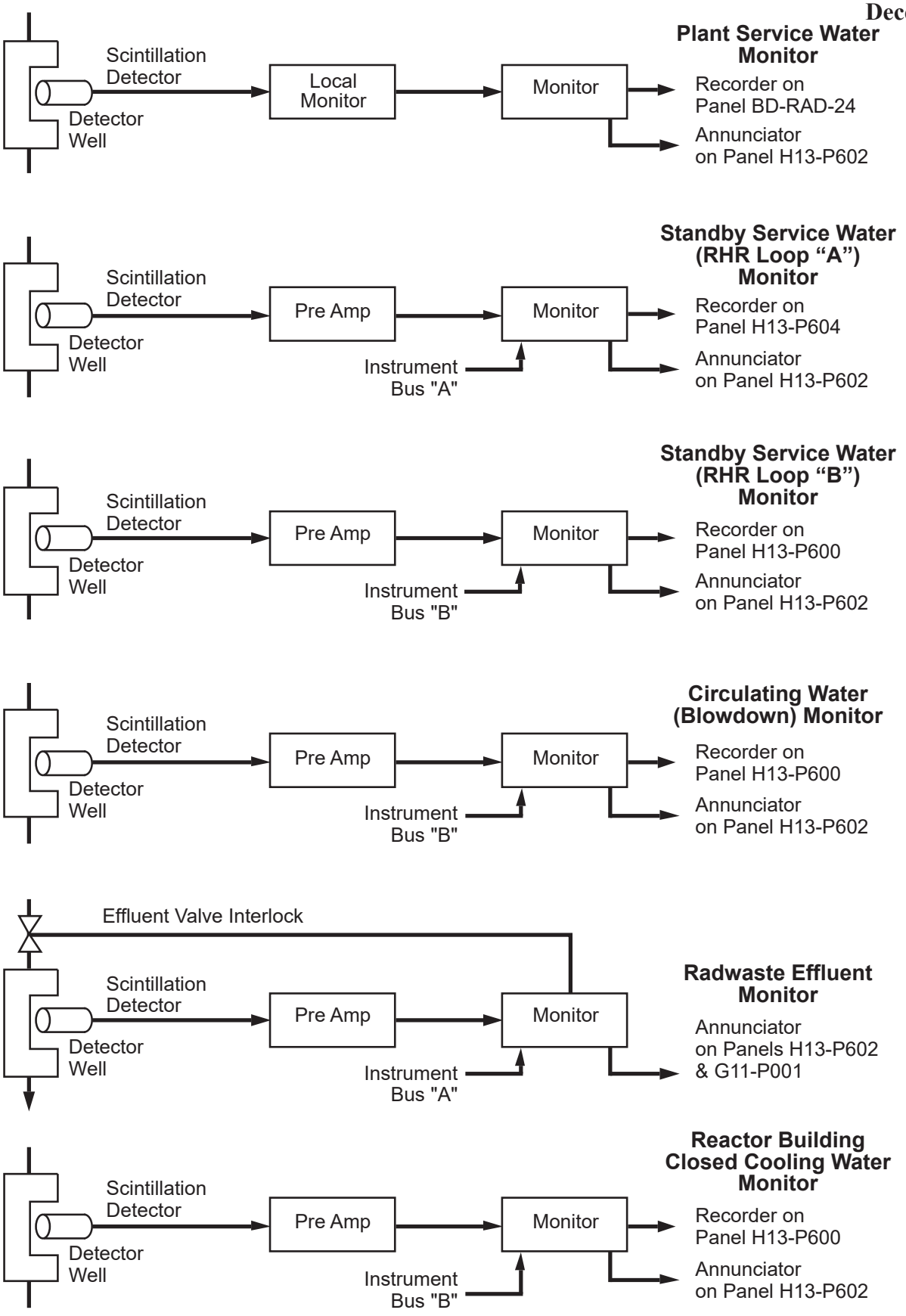
Main Steam Line Monitors

Draw. No. 900547.32	Rev.	Figure 11.5-1



Columbia Generating Station Final Safety Analysis Report	Reactor Building Exhaust Plenum Monitors and Charcoal Bed Vault Monitor		
	Draw. No. 900547.48	Rev.	Figure 11.5-2

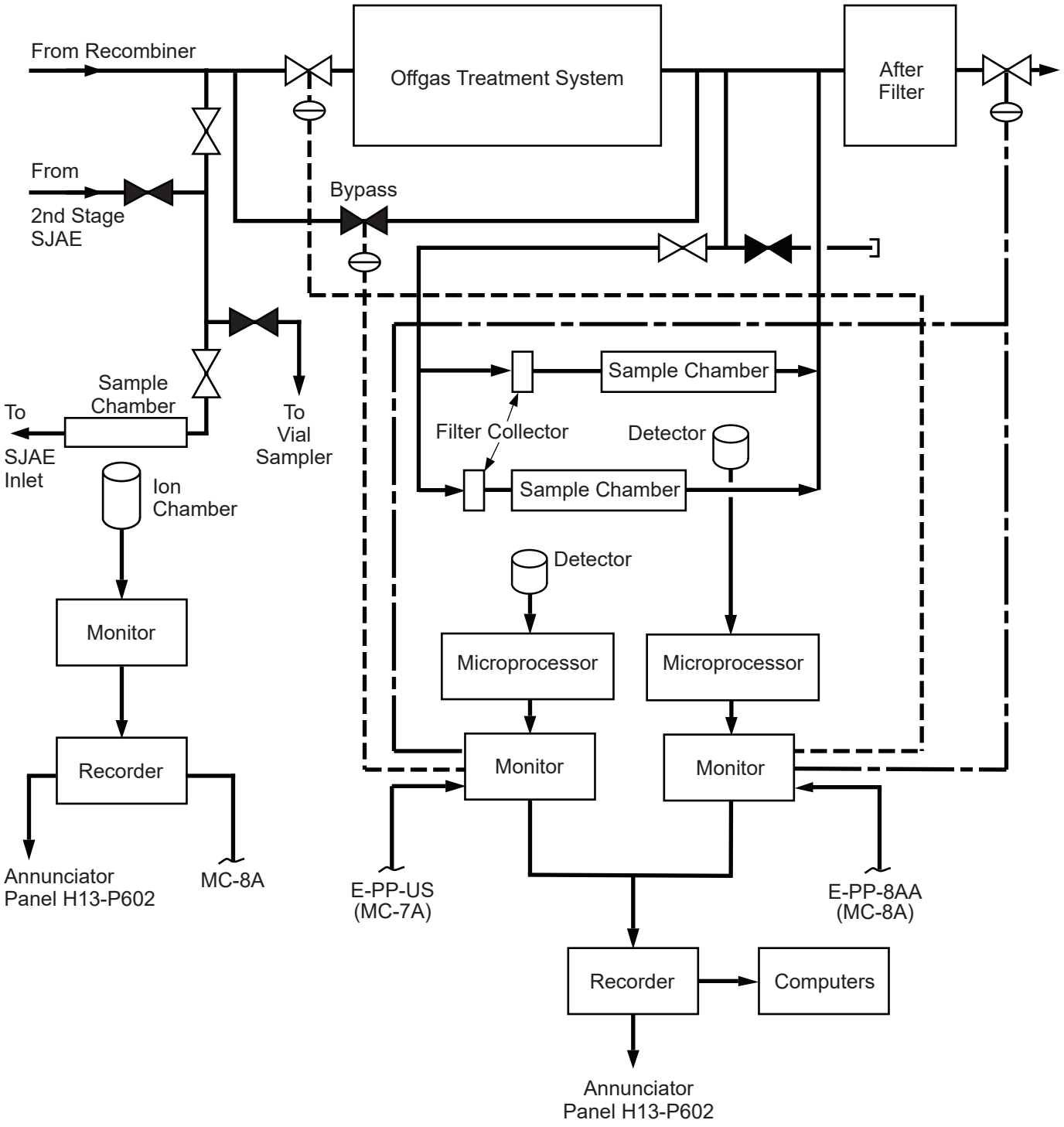
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Columbia Generating Station
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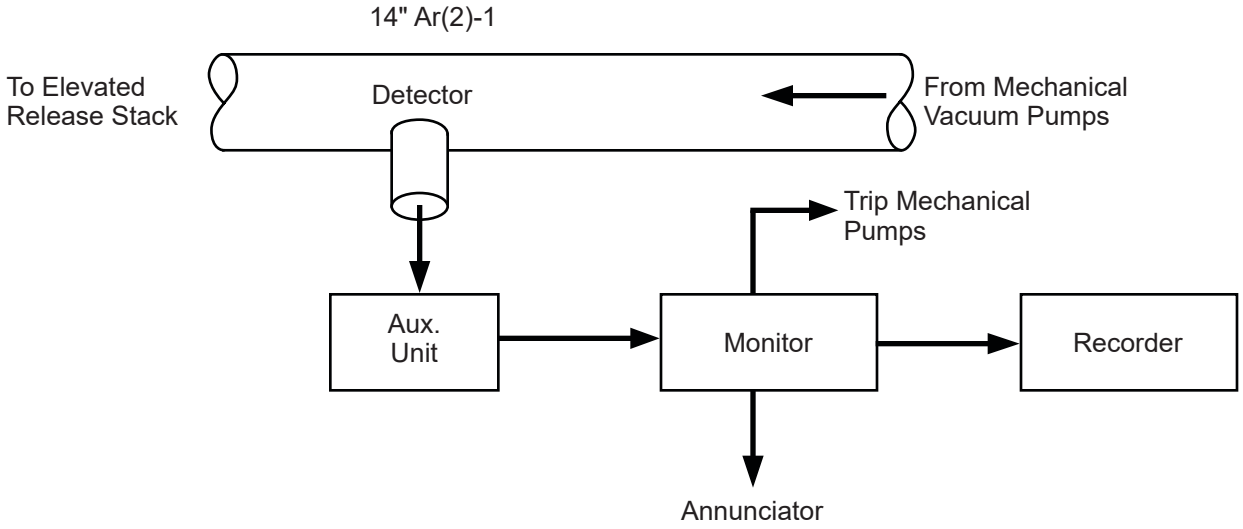
Process and Effluent Liquid Radiation Monitors

Draw. No. 900547.50	Rev.	Figure 11.5-4

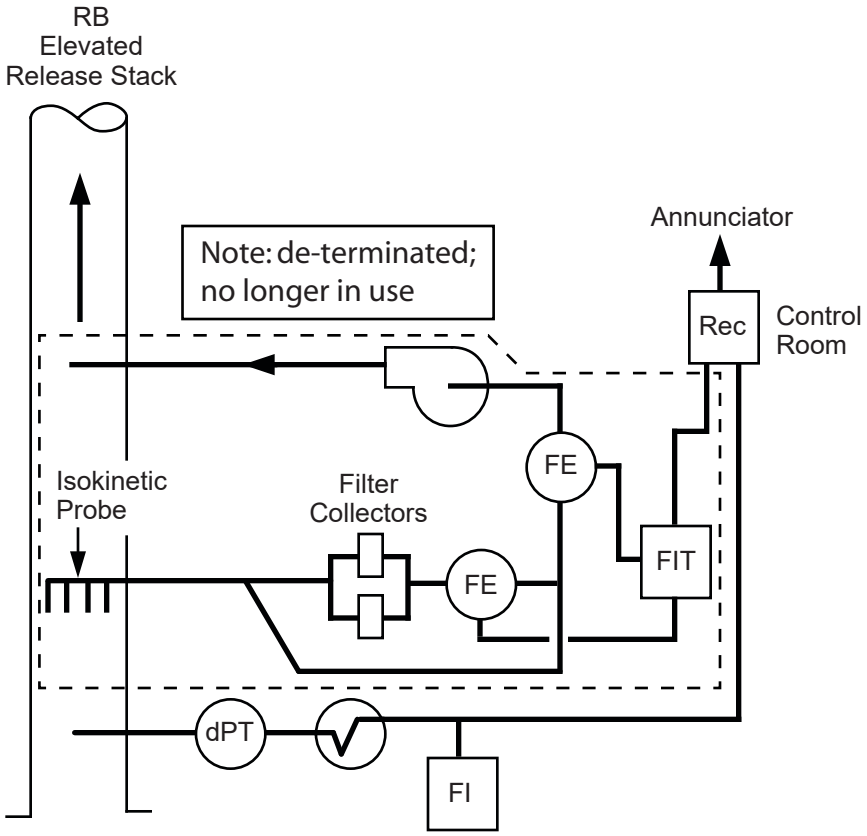


Columbia Generating Station Final Safety Analysis Report	Offgas Pretreatment and Posttreatment Radiation Monitors		
	Draw. No. 900547.51	Rev.	Figure 11.5-5

Mechanical Vacuum Pumps Exhaust Monitor

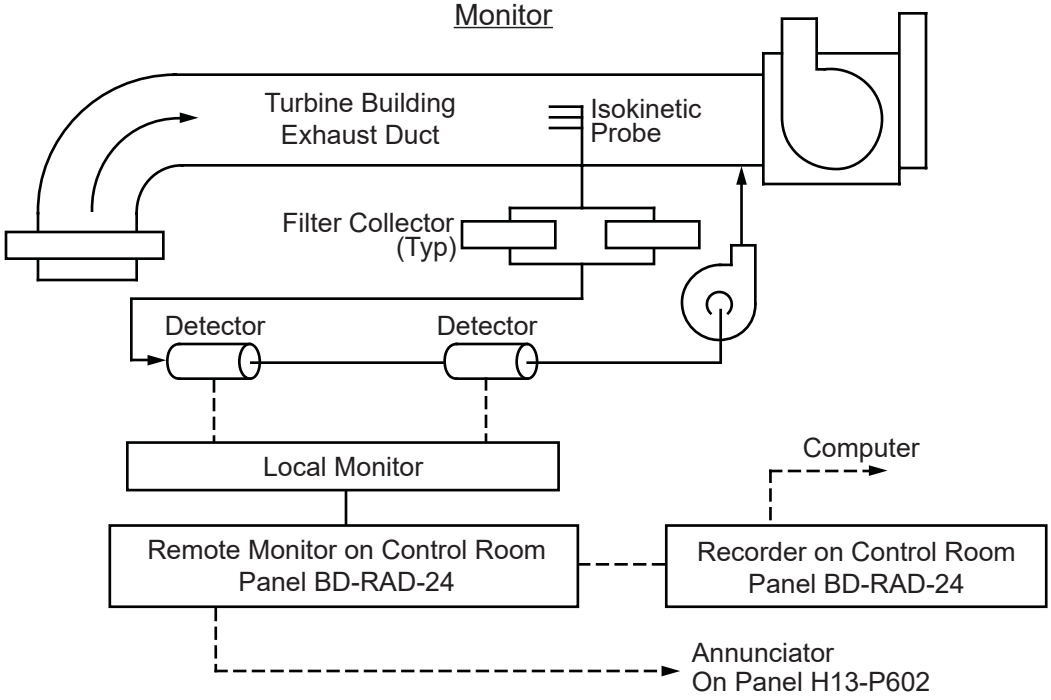


Reactor Building Ventilation Release Duct Monitor

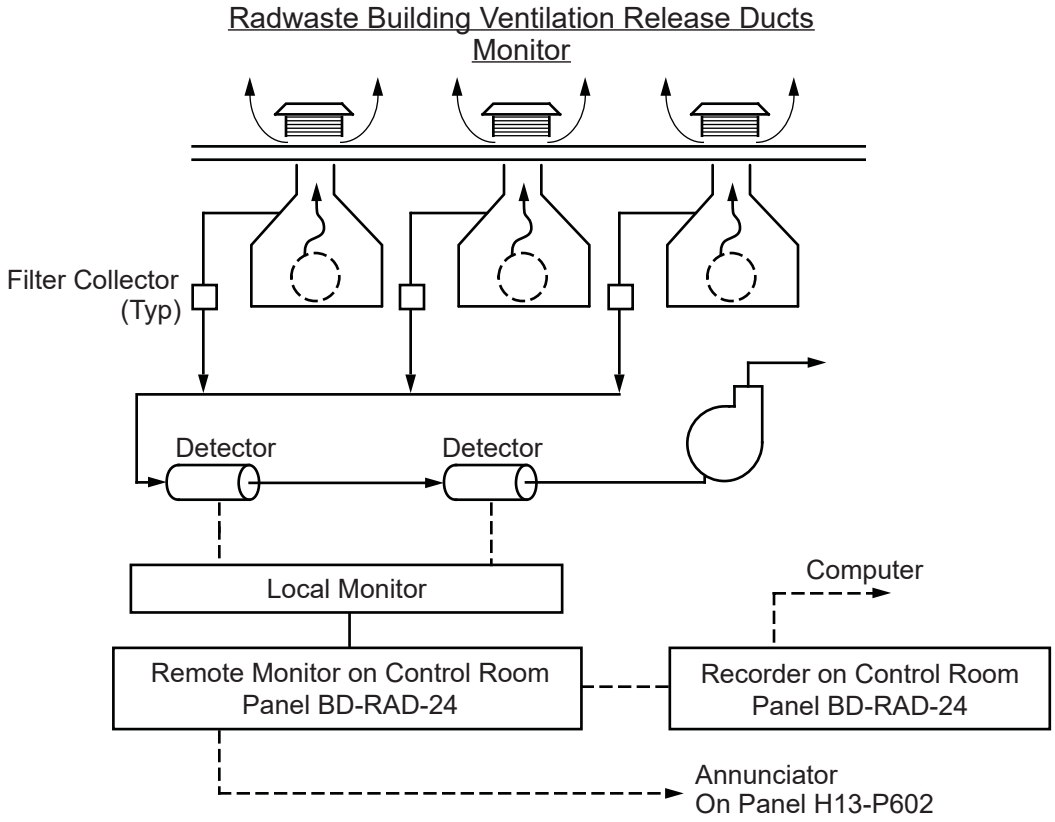


Columbia Generating Station Final Safety Analysis Report	Mechanical Vacuum Pumps Exhaust Monitor and Reactor Building Elevated Release Stack Monitor		
	Draw. No. 950021.32	Rev.	Figure 11.5-6

Turbine Building Ventilation Release Duct Monitor

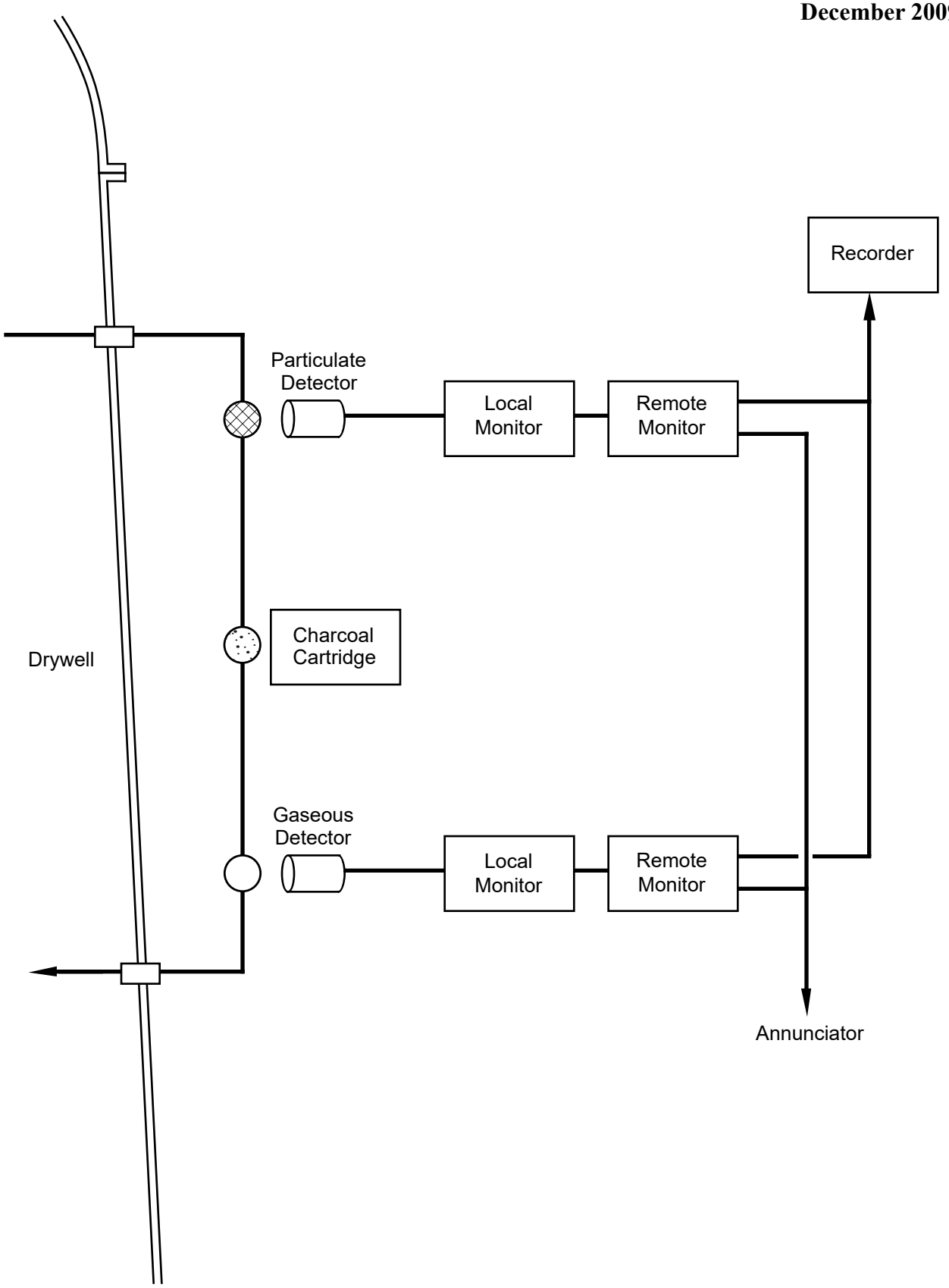


Radwaste Building Ventilation Release Ducts Monitor

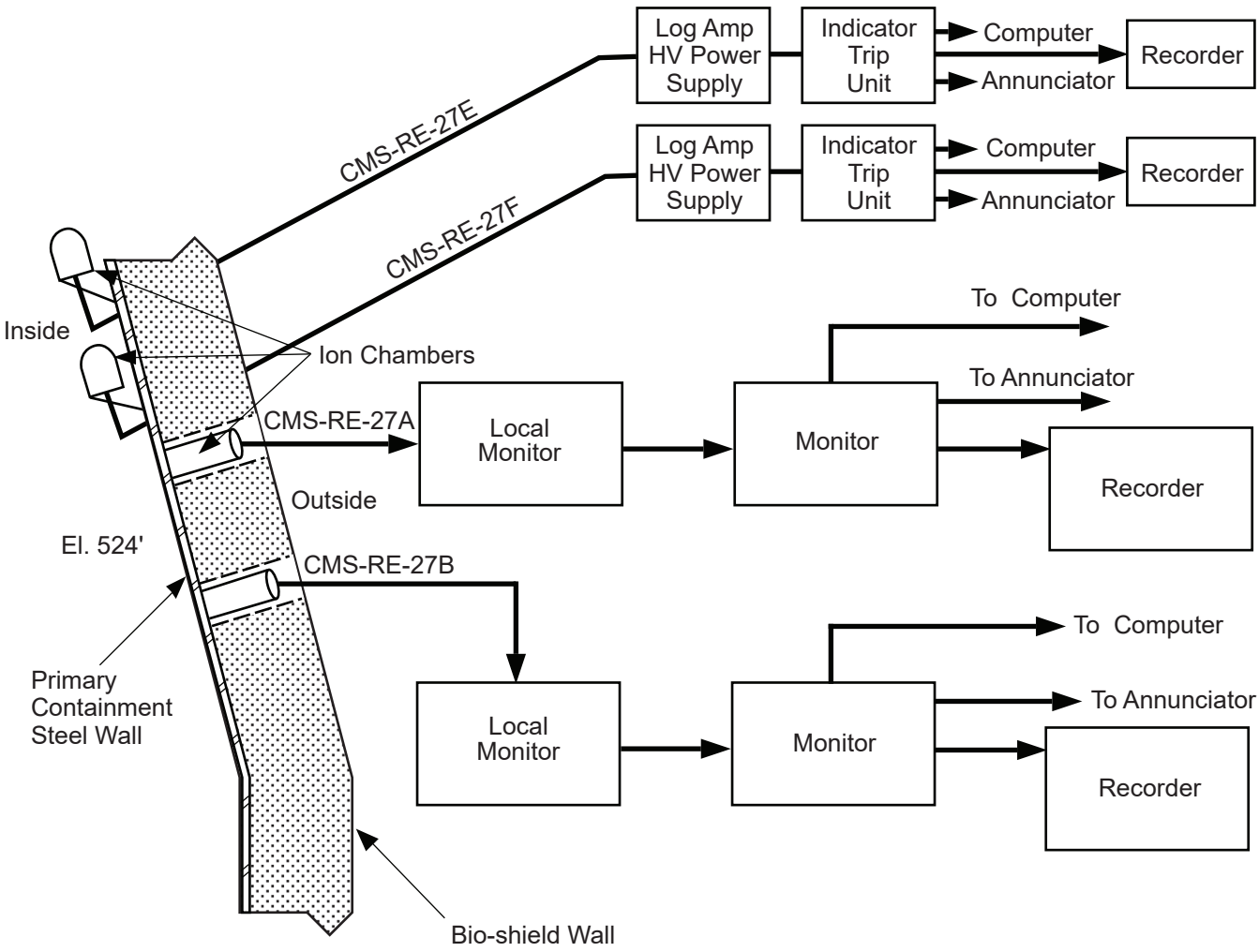


Columbia Generating Station Final Safety Analysis Report	Turbine and Radwaste Building Ventilation Release Duct Monitors		
	Draw. No. 960690.01	Rev. 1	Figure 11.5-7

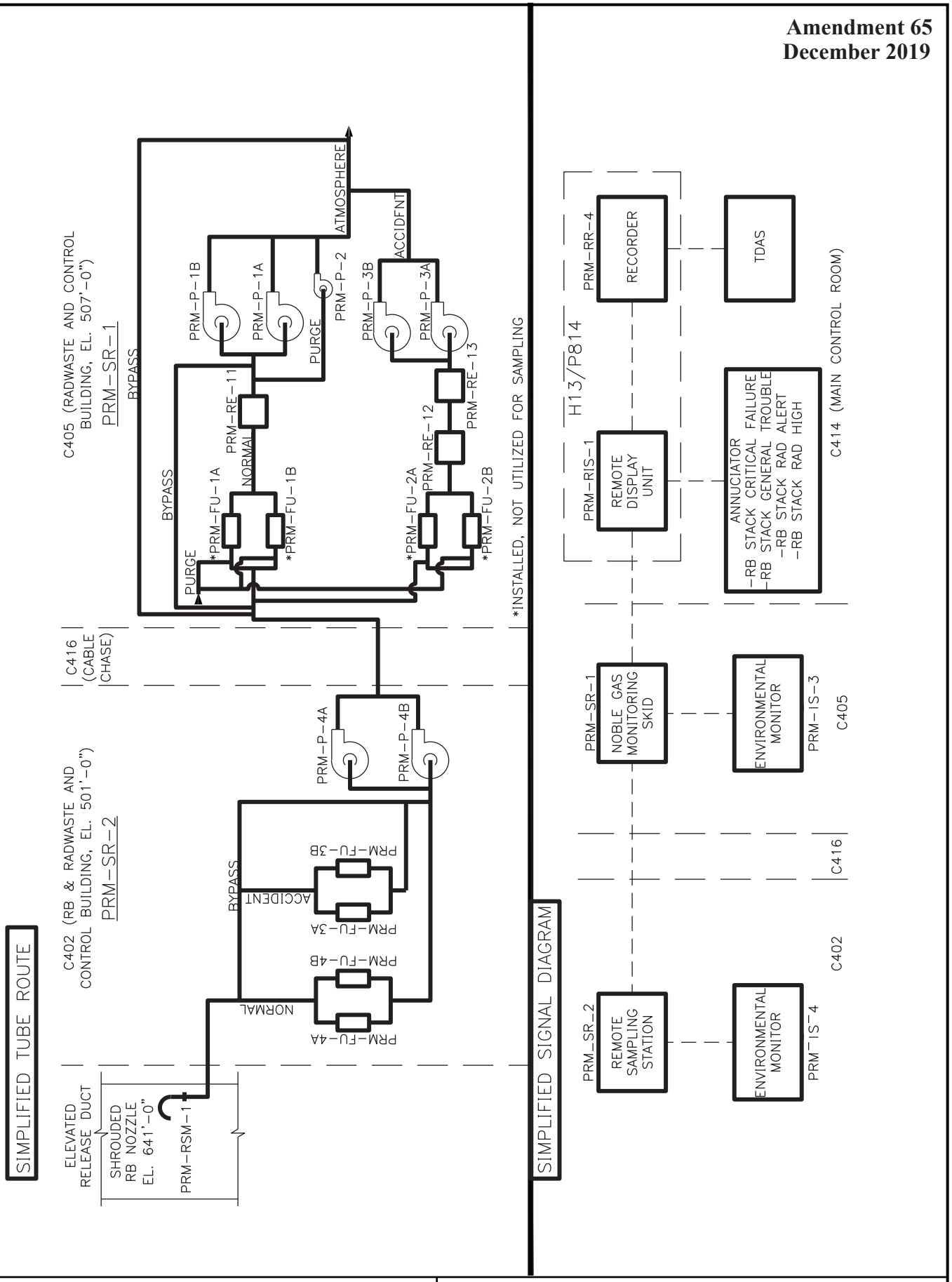
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Columbia Generating Station Final Safety Analysis Report	Primary Containment Leak Detection Monitor		
	Draw. No. 970187.21	Rev.	Figure 11.5-9



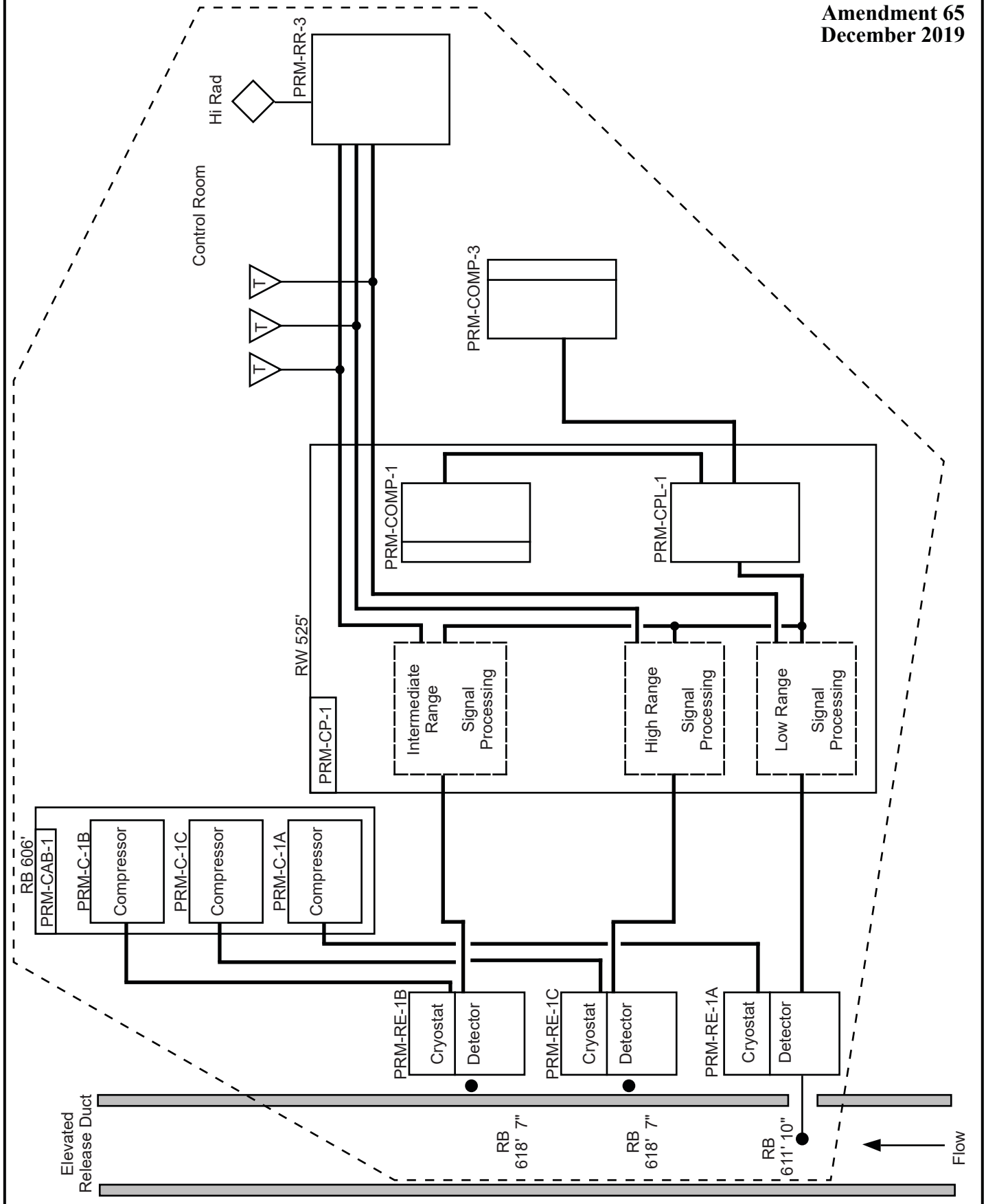
Columbia Generating Station Final Safety Analysis Report	Primary Containment and Elevated Release Stack LOCA Tracking		
	Draw. No. 950021.33	Rev.	Figure 11.5-10



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Elevated Release Stack Monitoring and Sampling System (ANSI N13.1-1999 Compliant)

Draw. No. 190165.01	Rev.	Figure 11.5-11



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Elevated Release Stack LOCA Monitoring
Note: de-terminated; no longer in use

Draw. No. 950021.29	Rev.	Figure 11.5-11A

11.6 POSTACCIDENT SAMPLING SYSTEM

11.6.1 DESIGN BASIS

Columbia Generating Station is using a General Electric postaccident sampling system (PASS) capable of sampling the primary containment and reactor building atmosphere and of obtaining liquid samples from the reactor, RHR loops and various reactor building sumps. This system is designed to obtain grab samples which may be analyzed onsite or transported to offsite facilities for more detailed analysis if necessary. The sample station is located in the radwaste building and is shielded to reduce radiation exposure rates to the operator. All remote-operated valves are controlled from this area. Lead pigs are provided for radiation protection when transporting samples either to onsite facilities or offsite.

All valves used are fully qualified for the environment in which they are located inside and outside reactor containment. Power for the postaccident sampling equipment is supplied from Division 1 and Division 2 critical power sources and will be available during accident conditions.

License Amendment No. 184 removed the requirement for PASS from the Technical Specifications effective January 27, 2003. While this amendment removes the Technical Specification requirements for PASS, it requires a commitment for contingency plans for obtaining and analyzing highly radioactive samples from the Reactor Coolant System, suppression pool, and containment atmosphere. Until further changes are made to the PASS hardware it will be operated as described in this FSAR Section. This operation meets the commitment for contingency plans for sampling.

11.6.2 SYSTEM DESCRIPTION

Gas samples will be obtained from locations in the drywell, the suppression pool atmosphere, and from the secondary containment atmosphere. The sample system is designed to operate at pressures ranging from subatmospheric to maximum design pressures of the primary and secondary containment. Heat-traced sample lines are used outside the primary containment to prevent precipitation of moisture and resultant loss of particulates and iodines in the sample lines. The gas samples may be passed through a particulate filter and silver zeolite cartridge for determination of particulate activity and iodine activity by subsequent analysis of the samples on a gamma spectrometer system. Alternatively, the sample flow bypasses the particulate/iodine sampler, is chilled to remove moisture, and a 15-ml grab sample can be taken for determination of gaseous radioactivity and for gas composition by gas chromatography. This size sample vial has been adopted for all gas samples to be consistent with present offgas sample vial counting factors.

Reactor coolant samples will be obtained from two points in the jet pump pressure instrument system when the reactor is at pressure. The jet pump pressure system has been determined to

be an optimum sample point for accident conditions. The pressure taps are well protected from damage and debris. If the recirculation pumps are secured, the water level will be raised about 18 in. above normal. This provides natural circulation of the bulk coolant past the taps. Also, the pressure taps are located sufficiently low to permit sampling at a reactor water level even below the lower core support plate.

A single sample line is also connected to both loops in the RHR system. This provides a means of obtaining a reactor coolant sample when the reactor is depressurized and at least one of the RHR loops is operated in the shutdown cooling mode. Similarly, a suppression pool liquid sample can be obtained from the RHR loop lined up in the suppression pool cooling mode. Samples from the five drain sumps in the reactor building are also available.

The sample system isolation valves are controlled from the local control panel. The sample system is designed for a purge flow of 1 gpm, which is sufficient to maintain turbulent flow in the sample line. Purge flow is returned to the suppression pool. The high flush flow also serves to alleviate cross-contamination of the samples when switching from one sample point to another.

All liquid samples are taken into septum bottles mounted on sampling needles. The sample station is basically a bypass loop on the sample purge line. In the normal lineup, the sample flows through a conductivity cell (readable range 0.1 to 1000 μS/cm) and then through a ball valve bored out to 0.10-ml volume. Flow through the sample panel is established, the valve is rotated 90°, and a syringe is used to flush the sample plus a measured volume of diluent (generally 10 ml) through the valve and into the sample bottle. This provides a dilution of 100:1 to the sample. Alternatively, the valve sampling sequence can be repeated 10 times to provide a 1-ml sample diluted 10:1. The sample is transported to the laboratory for further dilution and subsequent analysis. Alternatively, the sample flow can be diverted through a 70-ml bomb to obtain a large pressurized volume. This 70-ml volume can be circulated and depressurized into a known volume gas expansion chamber. The pressure change in this chamber will be used to calculate the total dissolved gases in the reactor coolant. A grab sample of these gases may be taken through a septum port for subsequent analysis.
Ten milliliter aliquots of this degassed liquid can also be taken for on or offsite chemical analyses requiring a relatively large sample. A radiation monitor in the liquid sample enclosure monitors liquid flow from the sample station to provide immediate assessment of the sample activity level. This monitor also provides information as to the effectiveness of the demineralized water flushing of the sample system following sample operation. The control instrumentation is installed in two 2 ft x 2 ft x 6 ft high standard cabinet control panels. One panel contains the conductivity and radiation level readouts. Another control panel contains the flow, pressure and temperature indicators, and the various control valves and switches.

A graphic display panel, installed directly below the main control panel, shows the status of the pumps and valves at all times. The panel also indicates the relative position of the pressure

gauges and other items of concern to the operator. The use of this panel will improve operator comprehension and assist in trouble-shooting operation.

Appropriate sample handling tools, a gas sampler vial positioner and gas vial cask are available to the operator at the sampling station. The gas vial is installed and removed by use of the vial positioner through the front of the gas sampler. The vial is then manually placed down in the cask with the positioner which allows the vial to be maintained about 3 ft from the individual performing the operation.

The small-volume (10 ml) liquid sample is remotely obtained through the bottom of the sample station by use of the small-volume cask and cask positioner. The cask positioner holds the cask and positions the cask directly under the liquid sampler. The sample vial is manually raised within the cask to engage the hypodermic needles. When the sample vial has been filled, the bottle is manually withdrawn into the cask. The sample vial is always contained within lead shielding during this operation. The cask is then lowered and sealed prior to transport to the laboratory.

A large-volume cask and cask positioner is available for transporting large liquid samples. A 21-ml bottle is contained within a lead shielded cask. This sample bottle is raised from its location in the cask to the sample station needles for bottle filling. The sample station will only deliver 10 ml to this sample bottle. When filled, the bottle is withdrawn into the cask. The sample bottle is always shielded by 5 to 6 in. of lead when in position under the sample station and during the fill and withdraw cycles, thus reducing operator exposure.

The cask is transported to the required position under the sample station by a dolly cask positioner. When in position this cask is hydraulically elevated approximately 1.5 in. by a small hand pump for contact with the sample station shielding under the liquid sample enclosure floor. The sample bottle is raised, held, and lowered by a simple push/pull cable. The cask is sealed by a threaded top plug that inserts above the sample bottle. The weight of this large-volume cask is approximately 700 lb.

The particulate filters and iodine cartridges are removed via a drawer arrangement. The quantity of activity which is accumulated on the cartridges is controlled by a combination of flow orificing and time sequence control of the flow valve opening. In addition, the deposition of iodine is monitored during sampling using a radiation detector installed adjacent to the cartridge. These samples will hence be limited to activity levels which will normally not require shielded sample carriers to transport the samples to the laboratory.

Based on information developed by General Electric, Energy Northwest has developed plant-specific procedures for the determination of the extent of core damage under accident conditions using inplant radiation and hydrogen monitors. This meets the commitment for ability to determine fuel damage, postaccident.